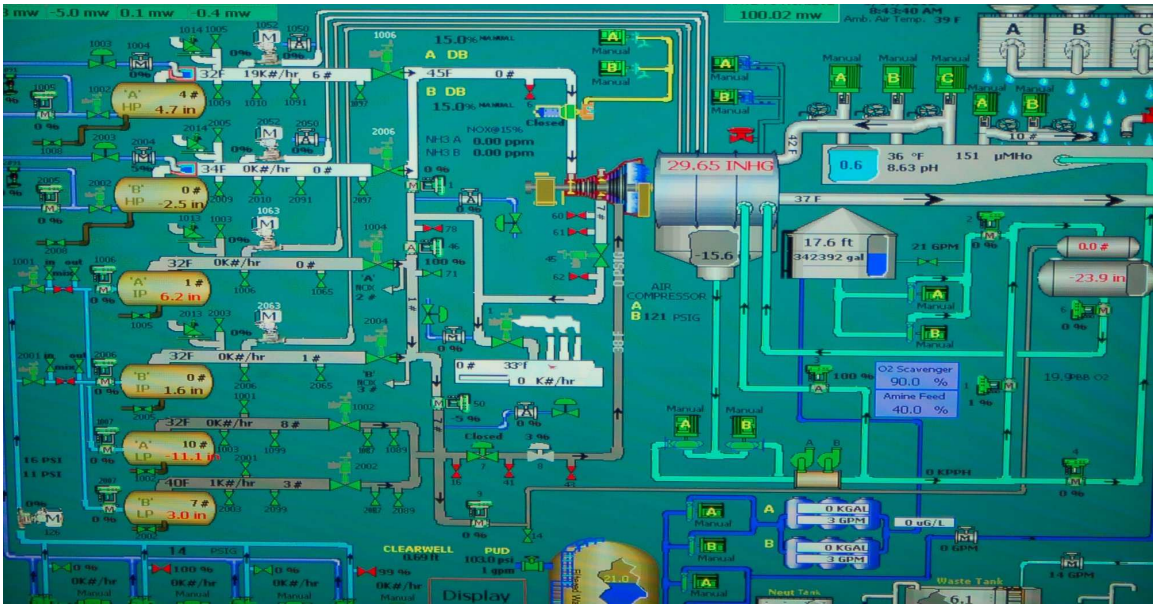
Lessons In Industrial Instrumentation



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[1] Version numbers ending in odd digits are developmental (e.g. 0.7, 1.23, 4.5), with only the latest revision made accessible to the public. Version numbers ending in even digits (e.g. 0.6, 1.0, 2.14) are considered "public-release" and will be archived. Version numbers beginning with zero (e.g. 0.1, 0.2, etc.) represent early editions that were substantially incomplete.

Chapter 32

Process safety and instrumentation

This chapter discusses instrumentation issues related to industrial process safety. Instrumentation safety may be broadly divided into two categories: how instruments themselves may pose a safety hazard (electrical signals possibly igniting hazardous atmospheres), and how instruments and control systems may be configured to detect unsafe process conditions and automatically shut an unsafe process down.

In either case, the intent of this chapter is to help define and teach how to mitigate hazards encountered in certain instrumented processes. I purposely use the word "mitigate" rather than "eliminate" because the complete elimination of all risk is an impossibility. Despite our best efforts and intentions, no one can absolutely eliminate all dangers from industrial processes[1]. What we can do, though, is *significantly* reduce those risks to the point they begin to approach the low level of "background" risks we all face in daily life, and that is no small achievement.

An important philosophy to follow in the safe design is something called *defense-in-depth*. This is the principle of using multiple layers[2] of protection, in case one or more of those layers fail. Applying defense-in-depth to process design means regarding each and every safety tool and technique as part of a multi-faceted strategy, rather than as a set of mutually-exclusive alternatives.

To give a brief example of defense-in-depth applied to overpressure protection in a fluid processing system, that system might defend against excessive fluid pressure using all of the following techniques:

- A pressure-control system with an operator-adjusted setpoint
- High-pressure alarms to force operator attention
- A safety shutdown system triggered by abnormally high pressure
- Temperature control systems (both regulatory and safety shutdown) to prevent excessive temperature from helping to create excessive fluid pressure

[1] For that matter, it is impossible to eliminate all danger from *life in general*. Every thing you do (or don't do) involves some level of risk. The question really should be, "how much risk is there in a given action, and how much risk am I willing to tolerate?" To illustrate, there does exist a non-zero probability that something you will read in this book is so shocking it will cause you to suffer a heart attack. However, the odds of you walking away from this book and never reading it again over concern of epiphany-induced cardiac arrest are just as slim.

[2] Also humorously referred to as the "belt *and* suspenders" school of engineering.

- Pressure-relief valves which automatically open to vent high pressure
- Pressure vessels built with "frangible[3]" tops designed to burst in the safest manner possible
- Locating the process far away from anything (or anyone) that might be harmed by an overpressure event

Any one of these techniques will work to reduce the risk posed by excessive fluid pressure in the system, but all of them used together will provide greater risk reduction than any one used alone.

32.1 Classified areas and electrical safety measures

Any physical location in an industrial facility harboring the potential of explosion due to the presence of flammable process matter suspended in the air is called a *hazardous* or *classified* location. In this context, the label "hazardous" specifically refers to the hazard of explosion, not of other health or safety hazards[4].

[3] Frangible roofs are a common design applied to liquid storage tanks harboring the potential for overpressure, such as sulfuric acid storage tanks which may generate accumulations of explosive hydrogen gas. Having the roof seam rupture from overpressure is a far less destructive event than having a side seam or floor seam rupture and consequently spill large volumes of acid. This technique of mitigating overpressure risk does not work to reduce pressure in the system, but it does reduce the risk of damage caused by overpressure in the system.

[4] Chemical corrosiveness, biohazardous substances, poisonous materials, and radiation are all examples of other types of industrial hazards not covered by the label "hazardous" in this context. This is not to understate the danger of these other hazards, but merely to focus our attention on the specific hazard of explosions and how to build instrument systems that will not trigger explosions due to electrical spark.

32.1.1 Classified area taxonomy

In the United States, the National Electrical Code (NEC) published by the National Fire Protection Association (NFPA) defines different categories of "classified" industrial areas and prescribes safe electrical system design practices for those areas. Article 500 of the NEC categorizes classified areas into a system of *Classes* and *Divisions*. Articles 505 and 506[5] of the NEC provide alternative categorizations for classified areas based on *Zones* that is more closely aligned with European safety standards.

The Class and Division taxonomy defines classified areas in terms of hazard type and hazard probability. Each "Class" contains (or may contain) different types of potentially explosive substances: Class I is for gases or vapors, Class II is for combustible dusts, and Class III is for flammable fibers. The three-fold class designation is roughly scaled on the size of the flammable particles, with Class I being the smallest (gas or vapor molecules) and Class III being the largest (fibers of solid matter). Each "Division" ranks a classified area according to the likelihood of explosive gases, dusts, or fibers being present. Division 1 areas are those where explosive concentrations can or do exist under normal operating conditions. Division 2 areas are those where explosive concentrations only exist infrequently or under abnormal conditions[6].

The "Zone" method of area classifications defined in Article 505 of the National Electrical Code applies to Class I (explosive gas or vapor) applications, but the three-fold Zone ranks (0, 1, and 2) are analogous to Divisions in their rating of explosive concentration probabilities. Zone 0 defines areas where explosive concentrations are continually present or normally present for long periods of time. Zone 1 defines areas where those concentrations may be present under normal operating conditions, but not as frequently as Zone 0. Zone 2 defines areas where explosive concentrations are unlikely under normal operating conditions, and when present do not exist for substantial periods of time. This three-fold Zone taxonomy may be thought of as expansion on the two-fold Division system, where Zones 0 and 1 are sub-categories of Division 1 areas, and Zone 2 is nearly equivalent to a Division 2 area[7]. A similar three-zone taxonomy for Class II and Class III applications is defined in Article 506 of the National Electrical Code, the zone ranks for these dust and fiber hazards numbered 20, 21, and 22 (and having analogous meanings to zones 0, 1, and 2 for Class I applications).

An example of a classified area common to most peoples' experience is a vehicle refueling station. Being a (potentially) explosive *vapor*, the hazard in question here is deemed Class I. The Division rating varies with proximity to the fume source. For an upward-discharging vent pipe from an underground gasoline storage tank, the area is rated as Division 1 within 900 millimeters (3 feet) from the vent hole. Between 3 feet and 5 feet away from the vent, the area is rated as Division 2. In relation to an outdoor fuel pump (dispenser), the space internal to the pump enclosure is rated Division 1, and any space up to 18 inches from grade level and up to 20 feet away (horizontally) from the pump is rated Division 2.

[5] Article 506 is a new addition to the NEC as of 2008. Prior to that, the only "zone"-based categories were those specified in Article 505.

[6] The final authority on Class and Division definitions is the National Electrical Code itself. The definitions presented here, especially with regard to Divisions, may not be precise enough for many applications. Article 500 of the NEC is quite specific for each Class and Division combination, and should be referred to for detailed information in any particular application.

[7] Once again, the final authority on this is the National Electrical Code, in this case Article 505. My descriptions of Zones and Divisions are for general information only, and may not be specific or detailed enough for many applications.

Within Class I and Class II (but not Class III), the National Electrical Code further sub-divides hazards according to explosive properties called *Groups*. Each group is defined either according to a substance type, or according to specific ignition criteria. Ignition criteria listed in the National Electrical Code (Article 500) include the *maximum experimental safe gap* (MESG) and the *minimum ignition current ratio* (MICR). The MESG is based on a test where two hollow hemispheres separated by a small gap enclose both an explosive air/fuel mixture and an ignition source. Tests are performed with this apparatus to determine the maximum gap width between the hemispheres that will not permit the excursion of flame from an explosion within the hemispheres triggered by the ignition source. The MICR is the ratio of electrical ignition current for an explosive air/fuel mixture compared to an optimum mixture of methane and air. The smaller of either these two values, the more dangerous the explosive substance is.

Class I substances are grouped according to their respective MESG and MICR values, with typical gas types given for each group:

Group	**Typical substance**	**Safe gap**	**Ignition current**
A	Acetylene		
B	Hydrogen	MESG $\leq$ 0.45 mm	MICR $\leq$ 0.40
C	Ethylene	0.45 mm $<$ MESG $\leq$ 0.75 mm	0.40 $<$ MICR $\leq$ 0.80
D	Propane	0.75 mm $<$ MESG	0.80 $<$ MICR

Class II substances are grouped according to material type:

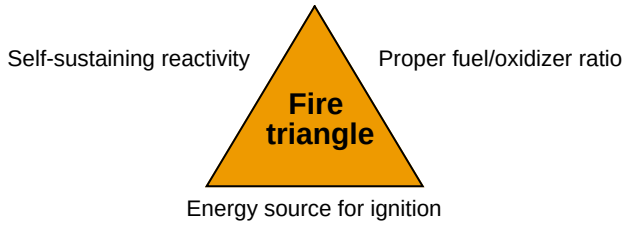
Group	**Substances**
E	Metal dusts
F	Carbon-based dusts
G	Other dusts (wood, grain, flour, plastic, etc.)

Just to make things confusing, the Class/Zone system described in NEC Article 505 uses a completely different lettering order to describe gas and vapor groups (at the time of this writing there is no grouping of dust or fiber types for the zone system described in Article 506 of the NEC):

Group	**Typical substance(s)**	**Safe gap**	**Ignition current**
IIC	Acetylene, Hydrogen	MESG $\leq$ 0.50 mm	MICR $\leq$ 0.45
IIB	Ethylene	0.50 mm $<$ MESG $\leq$ 0.90 mm	0.45 $<$ MICR $\leq$ 0.80
IIA	Acetone, Propane	0.90 mm $<$ MESG	0.80 $<$ MICR

32.1.2 Explosive limits

In order to have combustion (an explosion being a particularly aggressive form of combustion), certain basic criteria must be satisfied: a proper *oxidizer/fuel ratio*, sufficient *energy* for ignition, and the *potential for a self-sustaining chemical reaction* (i.e. the absence of any chemical inhibitors). We may show these criteria in the form of a *fire triangle*[8], the concept being that removing any of these three critical elements renders a fire (or explosion) impossible:



The fire triangle serves as a qualitative guide for *preventing* fires and explosions, but it does not give sufficient information to tell us if the necessary conditions exist to *support* a fire or explosion. In order for a fire or explosion to occur, we need to have an adequate mixture of fuel and oxidizer in the correct proportions, and a source of ignition energy exceeding a certain minimum threshold.

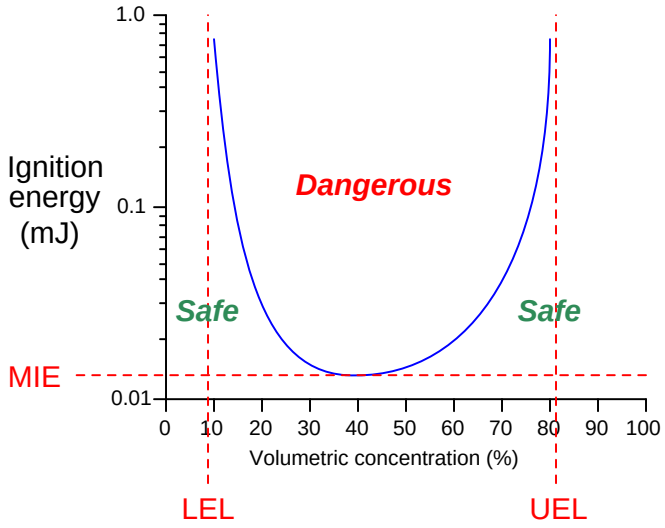
Suppose we had a laboratory test chamber filled with a mixture of acetone vapor (70% by volume) and air at room temperature, with an electrical spark gap providing convenient ignition. No matter how energetic the spark, this mixture would not explode, because there is too *rich* a mixture of acetone (i.e. too much acetone mixed with not enough air). Every time the spark gap discharges, its energy would surely cause some acetone molecules to combust with available oxygen molecules. However, since the air is so dilute in this rich acetone mixture, those scarce oxygen molecules are depleted fast enough that the flame temperature quickly falls off and is no longer hot enough to trigger the remaining oxygen molecules to combust with the plentiful acetone molecules.

The same problem occurs if the acetone/air mixture is too *lean* (not enough acetone and too much air). This is what would happen if we diluted the acetone vapors to a volumetric concentration of only 0.5% inside the test chamber: any spark at the gap would indeed cause some acetone molecules to combust, but there would be too few available to support expansive combustion across the rest of the chamber.

We could also have an acetone/air mixture in the chamber ideal for combustion (about 9.5% acetone by volume) and still not have an explosion if the spark's energy were insufficient. Most combustion reactions require a certain minimum level of *activation energy* to overcome the potential barrier before molecular bonding between fuel atoms and oxidizer atoms occurs. Stated differently, many combustion reactions are not *spontaneous* at room temperature and at atmospheric pressure – they need a bit of "help" to initiate.

[8] Traditionally, the three elements of a "fire triangle" were fuel, oxidizer, and ignition source. However, this model fails to account for fuels not requiring oxygen as well as cases where a chemical inhibitor prevents a self-sustaining reaction even in the presence of fuel, oxidizer, and ignition source.

All the necessary conditions for an explosion (assuming no chemical inhibitors are present) may be quantified and plotted as an *ignition curve* for any particular fuel and oxidizer combination. This next graph shows an ignition curve for an hypothetical fuel gas mixed with air:



Note how any point in the chart lying *above* the curve is "dangerous," while any point *below* the curve is "safe." The three critical values on this graph are the *Lower Explosive Limit* (LEL), the *Upper Explosive Limit* (UEL), and the *Minimum Ignition Energy* (MIE). These critical values differ for every type of fuel and oxidizer combination, change with ambient temperature and pressure, and may be rendered irrelevant in the presence of a catalyst (a chemical substance that works to promote a reaction without itself being consumed by the reaction). Most ignition curves are published with the assumed conditions of air as the oxidizer, at room temperature and atmospheric pressure, with no catalyst(s) present.

Some substances are so reactive that their minimum ignition energy (MIE) levels are well below the thermal energy of ambient air temperatures. Such fuels will *auto-ignite* the moment they come into contact with air, which effectively means one cannot prevent a fire or explosion by eliminating sources of flame or sparks. When dealing with such substances, the only means for preventing fires and explosions lies with maintaining fuel/air ratios outside of the danger zone (i.e. below the LEL or above the UEL), or by using a chemical inhibitor to prevent a self-sustaining reaction.

The greater the difference in LEL and UEL values, the greater "explosive potential" a fuel gas or vapor presents (all other factors being equal), because it means the fuel may explode over a wider range of mixture conditions. It is instructive to research the LEL and UEL values for many common substances, just to see how "explosive" they are relative to each other:

Substance	**LEL** (% volume)	**UEL** (% volume)
Acetylene	2.5%	100%
Acetone	2.5%	12.8%
Butane	1.5%	8.5%
Carbon disulfide	1.3%	50%
Carbon monoxide	12.5%	74%
Ether	1.9%	36%
Ethylene oxide	2.6%	100%
Gasoline	1.4%	7.6%
Kerosene	0.7%	5%
Hydrazine	2.9%	98%
Hydrogen	4.0%	74.2%
Methane	4.4%	17%
Propane	2.1%	9.5%

Note how both acetylene and ethylene oxide have UEL values of 100%. This means it is possible for these gases to explode *even when there is no oxidizer present*. Some other chemical substances exhibit this same property (n-propyl nitrate being another example), where the lack of an oxidizer does not prevent an explosion. With these substances in high concentration, our only practical hope of avoiding explosion is to eliminate the possibility of an ignition source in its presence. Some substances have UEL values so high that the elimination of oxidizers is only an uncertain guard against combustion: hydrazine being one example with a UEL of 98%, and diborane being another example with a UEL of 88%.

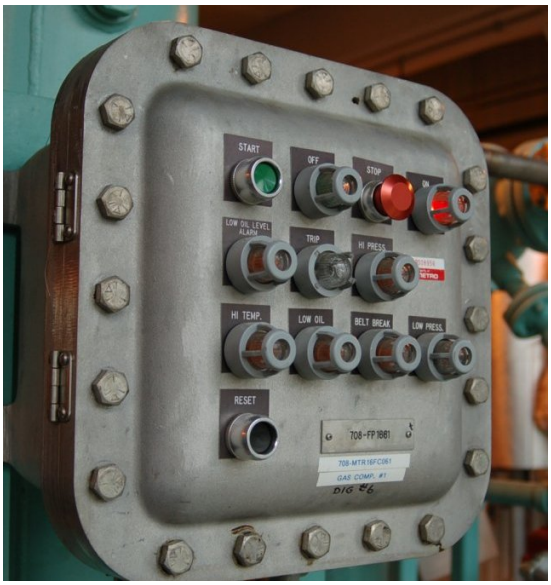
32.1.3 Protective measures

Different strategies exist to help prevent electrical devices from triggering fires or explosions in classified areas. These strategies may be broadly divided four ways:

- **Contain the explosion:** enclose the device inside a very strong box that contains any explosion generated by the device so as to not trigger a larger explosion outside the box. This strategy may be viewed as eliminating the "ignition" component of the fire triangle, from the perspective of the atmosphere outside the explosion-proof enclosure (ensuring the explosion inside the enclosure does not ignite a larger explosion outside).
- **Shield the device:** enclose the electrical device inside a suitable box or shelter, then purge that enclosure with clean air (or a pure gas) that prevents an explosive mixture from forming inside the enclosure. This strategy works by eliminating the "proper fuel/oxidizer ratio" component of the fire triangle: by eliminating fuel (if purged by air), or by eliminating oxidizer (if purged by fuel gas), or by eliminating both (if purged by an inert gas).
- **Encapsulated design:** manufacture the device so that it is self-enclosing. In other words, build the device in such a way that any spark-producing elements are sealed air-tight within the device from any explosive atmosphere. This strategy works by eliminating the "ignition" component of the fire triangle (from the perspective of outside the device) or by eliminating the "proper fuel/oxidizer ratio" component (from the perspective of inside the device).
- **Limit total circuit energy:** design the circuit such that there is insufficient energy to trigger an explosion, even in the event of an electrical fault. This strategy works by eliminating the "ignition" component of the fire triangle.

It should be noted that any one of these strategies, correctly and thoroughly applied, is sufficient to mitigate the risk of fire and explosion. For this reason you will seldom see more than one of these strategies simultaneously applied (e.g. an explosion-proof enclosure housing a circuit with insufficient energy to trigger an explosion).

A common example of the first strategy is to use extremely rugged metal *explosion-proof* (NEMA 7 or NEMA 8) enclosures instead of the more common sheet-metal or fiberglass enclosures to house electrical equipment. Two photographs of explosion-proof electrical enclosures reveal their unusually rugged construction:





Note the abundance of bolts securing the covers of these enclosures! This is necessary in order to withstand the enormous forces generated by the pressure of an explosion developing inside the enclosure. Note also how most of the bolts have been removed from the door of the right-hand enclosure. This is an unsafe and very unfortunate occurrence at many industrial facilities, where technicians leave just a few bolts securing the cover of an explosion-proof enclosure because it is so time-consuming to remove all of them to gain access inside the enclosure for maintenance work. Such practices negate the safety of the explosion-proof enclosure, rendering it just as dangerous as a sheet metal enclosure in a classified area.

Explosion-proof enclosures are designed in such a way that high-pressure gases resulting from an explosion within the enclosure must pass through small gaps (either holes in vent devices, and/or the gap formed by a bulging door forced away from the enclosure box) en route to exiting the enclosure. As hot gases pass through these tight metal gaps, they are forced to cool to the point where they will not ignite explosive gases outside the enclosure, thus preventing the original explosion inside the enclosure from triggering a far more violent event. This is the same phenomenon measured in determinations of MESG (Maximum Experimental Safe Gap) for an explosive air/fuel mixture. With an explosion-proof enclosure, all gaps are designed to be less than the MESG for the mixtures in question.

A similar strategy involves the use of a non-flammable *purge gas* pressurizing an ordinary electrical enclosure such that explosive atmospheres are prevented from entering the enclosure. Ordinary compressed air may be used as the purge gas, so long as provisions are made to ensure

the air compressor supplying the compressed air is in a non-classified area where explosive gases will never be drawn into the compressed air system.

Devices may be encapsulated in such a way that explosive atmospheres cannot penetrate the device to reach anything generating sufficient spark or heat. *Hermetically sealed* devices are an example of this protective strategy, where the structure of the device has been made completely fluid-tight by fusion joints of its casing. Mercury tilt-switches are good examples of such electrical devices, where a small quantity of liquid mercury is hermetically sealed inside a glass tube. No outside gases, vapors, dusts, or fibers can ever reach the spark generated when the mercury comes into contact (or breaks contact with) the electrodes:



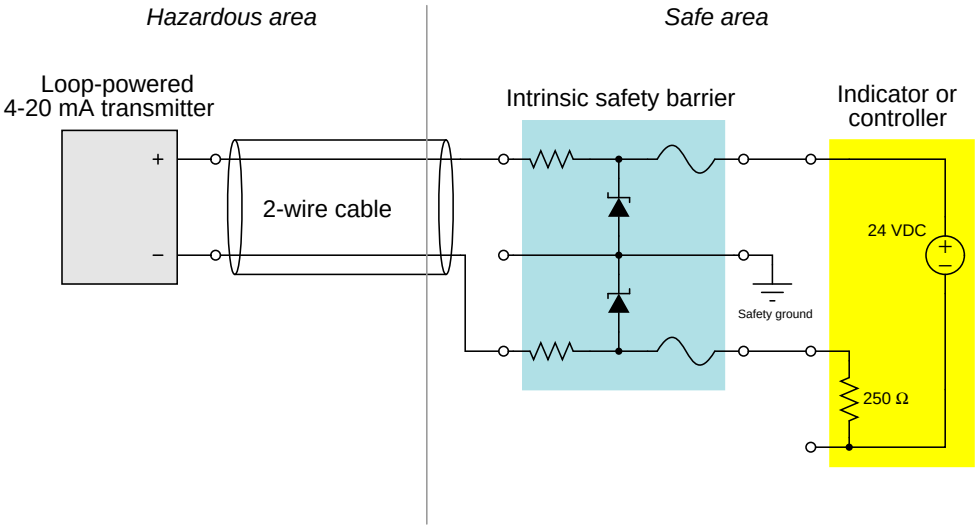


The ultimate method for ensuring instrument circuit safety in classified areas is to intentionally limit the amount of energy available within a circuit such that it *cannot* generate enough heat or spark to ignite an explosive atmosphere, even in the event of an electrical fault within the circuit. Article 504 of the National Electrical Code specifies standards for this method. Any system meeting these requirements is called an *intrinsically safe* or *I.S.* system. The word "intrinsic" implies that the safety is a natural property of the circuit, since it lacks even the ability to produce an explosion-triggering spark[9].

One way to underscore the meaning of intrinsic safety is to contrast it against a different concept that has the appearance of similarity. Article 500 of the National Electrical Code defines *nonincendive equipment* as devices incapable of igniting a hazardous atmosphere *under normal operating conditions*. However, the standard for nonincendive devices or circuits does not guarantee what will happen under *abnormal* conditions, such as an open- or short-circuit in the wiring. So, a "nonincendive" circuit may very well pose an explosion hazard, whereas an "intrinsically safe" circuit will not because the intrinsically safe circuit simply does not possess enough energy to trigger an explosion under *any* electrical fault condition. As a result, nonincendive circuits are not approved in Class I or Class II Division 1 locations whereas intrinsically safe circuits are approved for all hazardous locations.

[9]To illustrate this concept in a different context, consider my own personal history of automobiles. For many years I drove an ugly and inexpensive truck which I joked had "intrinsic theft protection:" it was so ugly, no one would ever want to steal it. Due to this "intrinsic" property of my vehicle, I had no need to invest in an alarm system or any other protective measure to deter theft. Similarly, the components of an intrinsically safe system need not be located in explosion-proof or purged enclosures because the intrinsic energy limitation of the system is protection enough.

Most modern 4 to 20 mA analog signal instruments may be used as part of intrinsically safe circuits so long as they are connected to control equipment through suitable *safety barrier* interfaces, the purpose of which is to limit the amount of voltage and current available at the field device to low enough levels that an explosion-triggering spark is impossible even under fault conditions (e.g. a short-circuit in the field instrument or wiring). A simple intrinsic safety barrier circuit made from passive components is shown in the following diagram[10]:



In normal operation, the 4-20 mA field instrument possesses insufficient terminal voltage and insufficient loop current to pose any threat of hazardous atmosphere ignition. However, the normally modest voltage and current values within a healthy 4-20 mA loop circuit are enough for that circuit to be considered *intrinsically safe*. In order to be intrinsically safe, the circuit's voltage and current levels must be limited even in the event of device or wiring faults. This is the purpose of the intrinsic safety barrier circuit: to serve as a safeguard in the event of unforseen wiring and/or component faults so that there is no possible way for enough voltage or current to develop to trigger an explosion.

If a short-circuit develops in the field instrument, the series resistance of the barrier circuit will limit fault current to a value low enough not to pose a threat in the hazardous area. If something fails in the receiving instrument to cause a much greater power supply voltage to develop at its terminals, the zener diode inside the barrier will break down and provide a shunt path for fault current that bypasses the field instrument (and may possibly blow the fuse in the barrier). Thus, the intrinsic safety barrier circuit provides protection against overcurrent *and* overvoltage faults, so that neither type of fault will result in enough electrical energy available at the field device to ignite an explosive atmosphere.

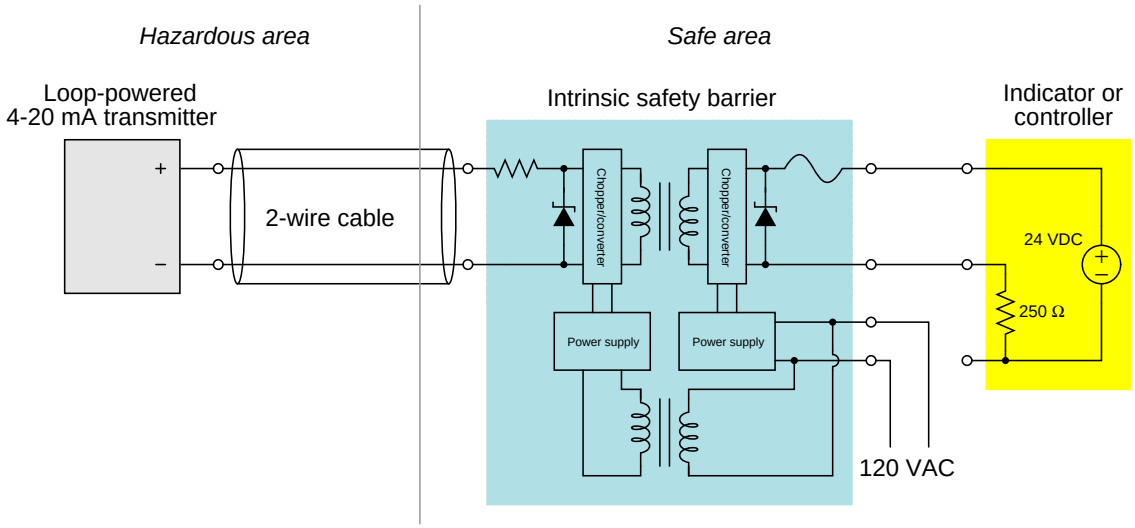
[10] Real passive barriers often used redundant zener diodes connected in parallel to ensure protection against excessive voltage even in the event of a zener diode failing open.

A photograph of an MTL-brand intrinsic safety barrier is shown here. A schematic diagram on the side of this barrier shows its internal circuitry:



Note that a barrier device such as this *must* be present in the 4-20 mA analog circuit in order for the circuit to be intrinsically safe. The "intrinsic" safety rating of the circuit depends on this barrier, not on the integrity of the field device or of the receiving device. Without this barrier in place, the instrument circuit is not intrinsically safe, even though the *normal* operating voltage and current parameters of the field and receiving devices are well within the parameters of safety for classified areas. It is the barrier and the barrier alone which guarantees those voltage and current levels will remain within safe limits in the event of *abnormal* circuit conditions such as a field wiring short, field device fault, or a faulty loop power supply.

More sophisticated *active* barrier devices are manufactured which provide electrical isolation from ground in the instrument wiring, thus eliminating the need for a safety ground connection at the barrier device.



In the example shown here, transformers[11] are used to electrically isolate the analog current signal so that there is no path for DC fault current between the field instrument and the receiving instrument, ground or no ground.

Safety barrier circuits fundamentally limit the amount of power deliverable to a field device from a power supply located in the safe area. Barrier circuits cannot, however, ensure safety for field devices capable of *generating* their own electrical energy. In order for such devices to be considered intrinsically safe, their natural abilities for generating voltage, current, and power must fall below limits defined in NEC article 504. Sensors such as pH electrodes, thermocouples, and photovoltaic light detectors are examples of such field devices, and are called *simple apparatus* by the NEC. The qualifications for a generating device to be a "simple apparatus" is that it cannot generate more than 1.5 volts of voltage, and more than 100 milliamps of current, and more than 25 milliwatts of power. If a device's ability to generate electricity exceeds these limits, the device is not a "simple apparatus" and therefore its circuit is not intrinsically safe.

An example of a generating field device exceeding these limits is a tachogenerator: a small DC generator used to measure the speed of rotating equipment by outputting a DC voltage proportional to speed (typically over a 0-10 volt range). An alternative to a tachogenerator for measuring machine speed is an *optical encoder*, using a slotted wheel to chop a light beam (from an LED), generating a pulsed electrical signal of sufficiently low intensity to qualify as a simple apparatus.

[11] Of course, transformers cannot be used to pass DC signals of any kind, which is why chopper/converter circuits are used before and after the signal transformer to convert each DC current signal into a form of chopped (AC) signal that *can* be fed through the transformer. This way, the *information* carried by each 4-20 mA DC current signal passes through the barrier, but electrical fault current cannot.

Passive (non-generating) field devices may also be classified as "simple apparatus" if they do not dissipate more than 1.3 watts of power. Examples of passive, simple apparatus include switches, LED indicator lamps, and RTD (Resistive Temperature Detector) sensors. Even devices with internal inductance and/or capacitance may be deemed "simple apparatus" if their stored energy capacity is insufficient to pose a hazard.

In addition to the use of barrier devices to create an intrinsically safe circuit, the National Electrical Code (NEC) article 504 specifies certain wiring practices different from normal control circuits. The conductors of an intrinsically safe circuit (i.e. conductors on the "field" side of a barrier) must be separated from the conductors of the non-intrinsically safe circuit (i.e. conductors on the "supply" side of the barrier) by at least 50 millimeters, which is approximately 2 inches. Conductors must be secured prior to terminals in such a way that they cannot come into contact with non-intrinsically safe conductors if the terminal becomes loose. Also, the color *light blue* may be used to identify intrinsically safe conductors, raceways, cable trays, and junction boxes so long as that color is not used for any other wiring in the system.

32.2 Concepts of probability

While the term "probability" may evoke images of imprecision, probability is in fact an exact mathematical concept: *the ratio a specific outcome to total possible outcomes* where 1 (100%) represents certainty and 0 (0%) represents impossibility. A probability value between 1 and 0 describes an outcome that occurs some of the time but not all of the time. Reliability – which is the expression of how likely a device or a system is to function as intended – is based on the mathematics of probability. Therefore, a rudimentary understanding of probability mathematics is necessary to grasp what reliability means.

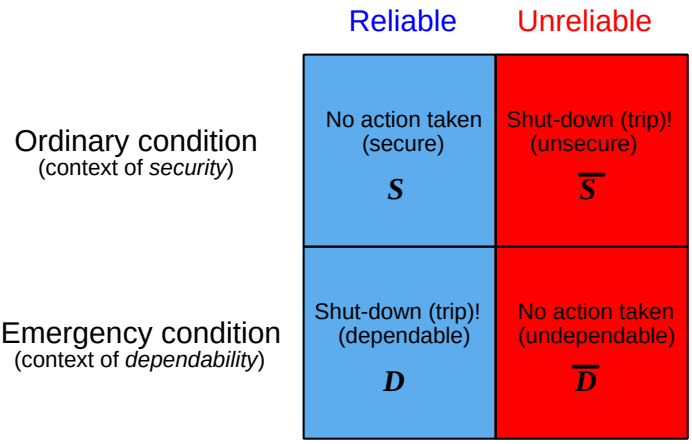
Before we delve too deeply into discussions of reliability, some definition of terms is in order. We have defined "reliability" to mean the probability of a device or system functioning as designed, which is a good general definition but sometimes not specific enough for our needs. There are usually a variety of different ways in which a device or system can fail, and these different failure modes usually have different probability values. Let's take for example a fire alarm system triggered by a manual pushbutton switch: the intended function of such a system is to activate an alarm whenever the switch is pressed. If we wish to express the reliability of this system, we must first carefully define what we mean by "failure". One way in which this simple fire alarm system could fail is if it remained silent when the pushbutton switch was pressed (i.e. *not* alerting people when it should have). Another, completely different, way this simple system could fail is by accidently sounding the alarm when no one pressed the switch (i.e. alerting people when it had no reason to, otherwise known as a "false alarm"). If we discuss the "reliability" of this fire alarm system, we may need to differentiate between these two different kinds of unreliable system behaviors.

The electrical power industry has an interest in ensuring the safe delivery of electrical power to loads, both ensuring maximum service to customers while simultaneously shutting power off as quickly as possible in the event of dangerous system faults. A complex system of fuses, circuit breakers, and protective relays work together to ensure the flow of power remains uninterrupted as long as safely possible. These protective devices must shut off power when they sense dangerous conditions, but they must also refrain from needlessly shutting off power when there is no danger. Like our fire alarm system which must alert people when needed yet not sound false alarms, electrical protective systems serve two different needs. In order to avoid confusion when quantifying the reliability of electrical protective systems to function as designed, the power industry consistently uses the following terms:

- **Dependability**: The probability a protective system will shut off power when needed
- **Security**: The probability a protective system will allow power to flow when there is no danger
- **Reliability**: A combination of dependability and security

For the sake of clarity I will use these same terms when discussing the reliability of any instrument or control systems. "Dependability" is how reliably a device or system will take appropriate action when it is actively called to do so – in other words, the degree to which we may *depend* on this device or system to do its job when activated. "Security" is how reliably a device or system refrains from taking action when no action should be taken – in other words, the degree to which we may feel *secure* it won't needlessly trigger a system shutdown or generate a false alarm. If there is no need to differentiate, the term "reliability" will be used as a general description of how probable a device or system will do what it was designed to do.

The following matrix should help clarify the meanings of these three terms, defined in terms of what the protective component or system does under various conditions:



In summary: a protective function that does not trip when it doesn't need to is *secure*; a protective function that trips when it needs to is *dependable*; a protective system that does both is *reliable*.

The Boolean variables used to symbolize dependability (D), security (S), undependability ($\overline{D}$), and unsecurity ($\overline{S}$) tell us something about the relationships between those four quantities. A bar appearing over a Boolean variable represents the *complement* of that variable. For example, security (S) and unsecurity ($\overline{S}$) are complementary to each other: if we happen to know the probability that a device or system will be secure, then we may calculate with assurance the probability that it is unsecure. A fire alarm system that is 99.3% secure (i.e. 99.3% of the time it generates no false alarms) must generate false alarms the other 0.7% of the time in order to account for all possible system responses 100% of the time no fires occur. If that same fire alarm system is 99.8% dependable (i.e. it alerts people to the presence of a real fire 99.8% of the time), then we may conclude it will fail to report 0.02% of real fire incidents in order to account for all possible responses during 100% of fire incidents.

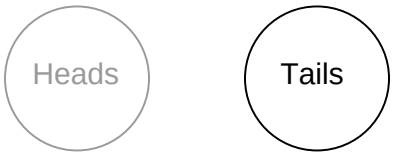
However, it should be clearly understood that there is no such simple relationship between security (S) and dependability (D) because these two measures refer to completely different conditions and (potentially) different modes of failure. The specific faults causing a fire alarm system to generate a false alarm (an example of an *unsecure* outcome, $\overline{S}$) are quite different from the faults disabling that same fire alarm system in the event of a real fire (an example of an *undependable* outcome, $\overline{D}$). Through the application of redundant components and clever system design we may augment dependability and/or security (sometimes improving one at the expense of the other), but it should be understood that these are really two fundamentally different probability measures and as such are not necessarily related.

32.2.1 Mathematical probability

Probability may be defined as a ratio of specific outcomes to total (possible) outcomes. If you were to flip a coin, there are really only two possibilities[12] for how that coin may land: face-up ("heads") or face-down ("tails"). The probability of a coin falling "tails" is thus one-half ($\frac{1}{2}$), since "tails" is one specific outcome out of two total possibilities. Calculating the probability (P) is a matter of setting up a ratio of outcomes:

$$P(\text{“tails”}) = \frac{\text{“tails”}}{\text{“heads”} + \text{“tails”}} = \frac{1}{2} = 0.5$$

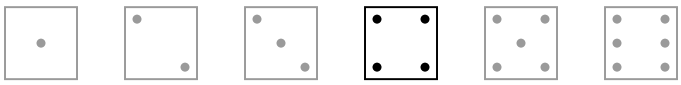
This may be shown graphically by displaying all possible outcomes for the coin's landing ("heads" or "tails"), with the one specific outcome we're interested in ("tails") highlighted for emphasis:



The probability of the coin landing "heads" is of course exactly the same, because "heads" is also *one* specific outcome out of *two* total possibilities.

If we were to roll a six-sided die, the probability of that die landing on any particular side (let's arbitrarily choose the "four" side) is one out of six, because we're looking at one specific outcome out of six total possibilities:

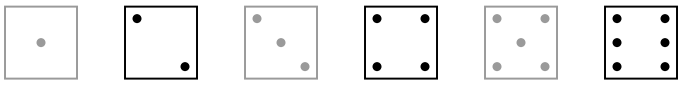
$$P(\text{“four”}) = \frac{\text{“four”}}{\text{“one”} + \text{“two”} + \text{“three”} + \text{“four”} + \text{“five”} + \text{“six”}} = \frac{1}{6} = 0.1\overline{66}$$



[12] To be honest, the coin could also land on its edge, which is a third possibility. However, that third possibility is so remote as to be negligible in the presence of the other two. Strictly speaking, $P(\text{“heads”}) + P(\text{“tails”}) + P(\text{“edge”}) = 1$.

If we were to roll the same six-sided die, the probability of that die landing on an even-numbered side (2, 4, or 6) is three out of six, because we're looking at three specific outcomes out of six total possibilities:

$$P(\text{even}) = \frac{\text{“two”} + \text{“four”} + \text{“six”}}{\text{“one”} + \text{“two”} + \text{“three”} + \text{“four”} + \text{“five”} + \text{“six”}} = \frac{3}{6} = 0.5$$



As a ratio of specific outcomes to total possible outcomes, the probability of any event will always be a number ranging in value from 0 to 1, inclusive. This value may be expressed as a fraction ($\frac{1}{2}$), as a per unit value (0.5), as a percentage (50%), or as a verbal statement (e.g. “three out of six”). A probability value of zero (0) means a specific event is impossible, while a probability of one (1) means a specific event is guaranteed to occur.

Probability values realistically apply only to large samples. A coin tossed ten times may very well fail to land “heads” exactly five times and land “tails” exactly five times. For that matter, it may fail to land on each side exactly 500000 times out of a million tosses. However, so long as the coin and the coin-tossing method are *fair* (i.e. not biased in any way), the experimental results will approach[13] the ideal probability value as the number of trials approaches infinity. Ideal probability values become less and less certain as the number of trials decreases, and are completely useless for singular (non-repeating) events.

A familiar application of probability values is the forecasting of meteorological events such as rainfall. When a weather forecast service provides a rainfall prediction of 65% for a particular day, it means that out of a large number of days sampled in the past having similar measured conditions (cloud cover, barometric pressure, temperature and dew point, etc.), 65% of those days experienced rainfall. This past history gives us some idea of how likely rainfall will be for any present situation, based on similarity of measured conditions.

Like all probability values, forecasts of rainfall are more meaningful with greater samples. If we wish to know how many days with measured conditions similar to those of the forecast day will experience rainfall over the *next ten years*, the forecast probability value of 65% will be quite accurate. However, if we wish to know whether or not rain will fall on any particular (single) day having those same conditions, the value of 65% tells us very little. So it is with all measurements of probability: precise for large samples, ambiguous for small samples, and virtually meaningless for singular conditions[14].

In the field of instrumentation – and more specifically the field of *safety* instrumented systems – probability is useful for the mitigation of hazards based on equipment failures where the probability of failure for specific pieces of equipment is known from mass production of that equipment and years of data gathered describing the reliability of the equipment. If we have data showing the probabilities

[13] In his excellent book, *Reliability Theory and Practice*, Igor Bazovsky describes the relationship between true probability (P) calculated from ideal values and estimated probability ($\hat{P}$) calculated from experimental trials as a limit function: $P = \lim_{N \to \infty} \hat{P}$, where N is the number of trials.

[14] Most people can recall instances where a weather forecast proved to be completely false: a prediction for rainfall resulting in a completely dry day, or vice-versa. In such cases, one is tempted to blame the weather service for poor forecasting, but in reality it has more to do with the nature of probability, specifically the meaninglessness of probability calculations in predicting singular events.

of failure for different pieces of equipment, we may use this data to calculate the probability of failure for the system as a whole. Furthermore, we may apply certain mathematical laws of probability to calculate system reliability for different equipment configurations, and therefore minimize the probability of system failure by optimizing those configurations.

As with weather predictions, predictions of system reliability (or conversely, of system failure) become more accurate as the sample size grows larger. Given an accurate probabilistic model of system reliability, a system (or a set of systems) with enough individual components, and a sufficiently long time-frame, an organization may accurately predict the number of system failures and the cost of those failures (or alternatively, the cost of minimizing those failures through preventive maintenance). However, no probabilistic model can accurately predict which component in a large system will fail at any specific point in time.

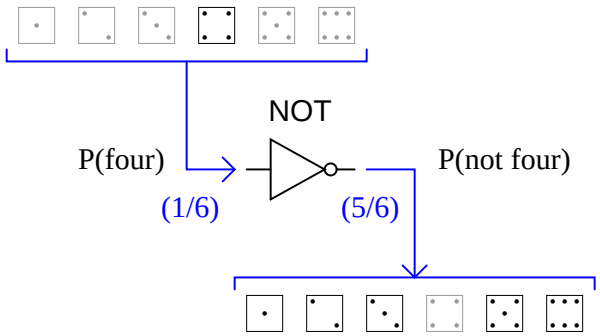
The ultimate purpose, then, in probability calculations for process systems and automation is to optimize the safety and availability of large systems over many years of time. Calculations of reliability, while useful to the technician in understanding the nature of system failures and how to minimize them, are actually more valuable (more meaningful) at the enterprise level.

32.2.2 Laws of probability

Probability mathematics bears an interesting similarity to Boolean algebra in that probability values (like Boolean values) range between zero (0) and one (1). The difference, of course, is that while Boolean variables may *only* have values equal to zero or one, probability variables range continuously between those limits. Given this similarity, we may apply standard Boolean operations such as `NOT`, `AND`, and `OR` to probabilities. These Boolean operations lead us to our first "laws" of probability for combination events.

The logical "NOT" function

For instance, if we know the probability of rolling a "four" on a six-sided die is $\frac{1}{6}$, then we may safely say the probability of *not* rolling a "four" is $\frac{5}{6}$, the complement of $\frac{1}{6}$. The common "inverter" logic symbol is shown here representing the complementation function, turning a probability of rolling a "four" into the probability of *not* rolling a "four":



Symbolically, we may express this as a sum of probabilities equal to one:

$$P(\text{total}) = P(\text{“one”}) + P(\text{“two”}) + P(\text{“three”}) + P(\text{“four”}) + P(\text{“five”}) + P(\text{“six”}) = 1$$

$$P(\text{total}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$$

$$P(\text{total}) = P(\text{“four”}) + P(\text{not “four”}) = \frac{1}{6} + \frac{5}{6} = 1$$

$$P(\text{“four”}) = 1 - P(\text{not “four”}) = 1 - \frac{5}{6} = \frac{1}{6}$$

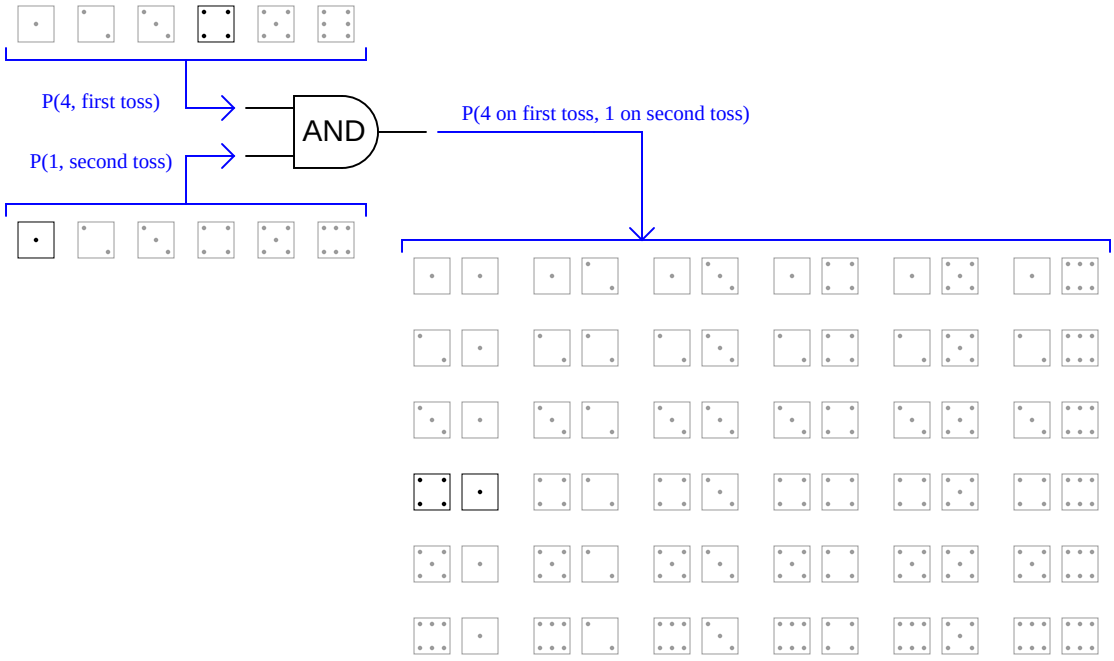
We may state this as a general "law" of complementation for any event (A):

$$P(A) = 1 - \overline{P}(A)$$

Complements of probability values find frequent use in reliability engineering. If we know the probability value for the failure of a component (i.e. how likely it is to fail when called upon to function – termed the *Probability of Failure on Demand*, or *PFD* – which is a measure of that component's *undependability*), then we know the *dependability* value (i.e. how likely it is to function on demand) will be the mathematical complement. To illustrate, consider a device with a PFD value of $\frac{1}{100000}$. Such a device could be said to have a dependability value of $\frac{99999}{100000}$, or 99.999%, since $1 - \frac{1}{100000} = \frac{99999}{100000}$.

The logical "AND" function

The AND function regards probabilities of two or more coincidental events (i.e. where the outcome of interest only happens if two or more events happen together, or in a specific sequence). Another example using a die is the probability of rolling a "four" on the first toss, then rolling a "one" on the second toss. It should be intuitively obvious that the probability of rolling this specific combination of values will be less (i.e. less likely) than rolling either of those values in a single toss. The shaded field of possibilities (36 in all) demonstrate the unlikelihood of this sequential combination of values compared to the unlikelihood of either value on either toss:



As you can see, there is only one outcome matching the specific criteria out of 36 total possible outcomes. This yields a probability value of one-in-thirty six ($\frac{1}{36}$) for the specified combination, which is the *product* of the individual probabilities. This, then, is our second law of probability:

$$P(\text{A } \textit{and } \text{B}) = P(\text{A}) \times P(\text{B})$$

A practical application of this would be the calculation of failure probability for a double-block valve assembly, designed to positively stop the flow of a dangerous process fluid. Double-block valves are used to provide increased assurance of shut-off, since the shutting of *either* block valve is sufficient in itself to stop fluid flow. The probability of failure for a double-block valve assembly – "failure" defined as not being able to stop fluid flow when needed – is the product of each valve's un-dependability (i.e. probability of failing in the open position when commanded to shut off):



With these two valves in service, the probability of neither valve successfully shutting off flow (i.e. *both* valve 1 *and* valve 2 failing on demand; remaining open when they should shut) is the product of their individual failure probabilities.

$$P(\text{assembly fail}) = P(\text{valve 1 fail open}) \times P(\text{valve 2 fail open})$$

$$P(\text{assembly fail}) = 0.0002 \times 0.0003$$

$$P(\text{assembly fail}) = 0.00000006 = 6 \times 10^{-8}$$

An extremely important assumption in performing such an `AND` calculation is that the probabilities of failure for each valve are completely unrelated. For instance, if the failure probabilities of both valve 1 and valve 2 were largely based on the possibility of a certain residue accumulating inside the valve mechanism (causing the mechanism to freeze in the open position), and *both* valves were equally susceptible to this residue accumulation, there would be virtually no advantage to having double block valves. If said residue were to accumulate in the piping, it would affect both valves practically the same. Thus, the failure of one valve due to this effect would virtually ensure the failure of the other valve as well. *The probability of simultaneous or sequential events being the product of the individual events' probabilities is true if and only if the events in question are completely independent.*

We may illustrate the same caveat with the sequential rolling of a die. Our previous calculation showed the probability of rolling a "four" on the first toss and a "one" on the second toss to be $\frac{1}{6} \times \frac{1}{6}$, or $\frac{1}{36}$. However, if the person throwing the die is extremely consistent in their throwing technique and the way they orient the die after each throw, such that rolling a "four" on one toss makes it very likely to roll a "one" on the next toss, the sequential events of a "four" followed by a "one" would be far more likely than if the two events were completely random and independent. The probability calculation of $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$ holds true only if all the throws' results are completely unrelated to each other.

Another, similar application of the Boolean AND function to probability is the calculation of system reliability (R) based on the individual reliability values of components necessary for the system's function. If we know the reliability values for several essential[15] system components, and we also know those reliability values are based on independent (unrelated) failure modes, the overall system reliability will be the product (Boolean AND) of those component reliabilities. This mathematical expression is known as *Lusser's product law of reliabilities*:

$$R_{system} = R_1 \times R_2 \times R_3 \times \cdots \times R_n$$

As simple as this law is, it is surprisingly unintuitive. Lusser's Law tells us that any system depending on the performance of several essential components will be *less* reliable than the least-reliable of those components. This is akin to saying that a chain will be *weaker* than its weakest link!

To give an illustrative example, suppose a complex system depended on the reliable operation of six key components in order to function, with the individual reliabilities of those six components being 91%, 92%, 96%, 95%, 93%, and 92%, respectively. Given individual component reliabilities all greater than 90%, one might be inclined to think the overall reliability would be quite good. However, following Lusser's Law we find the reliability of this system (as a whole) is only 65.3% because $0.91 \times 0.92 \times 0.96 \times 0.95 \times 0.93 \times 0.92 = 0.653$.

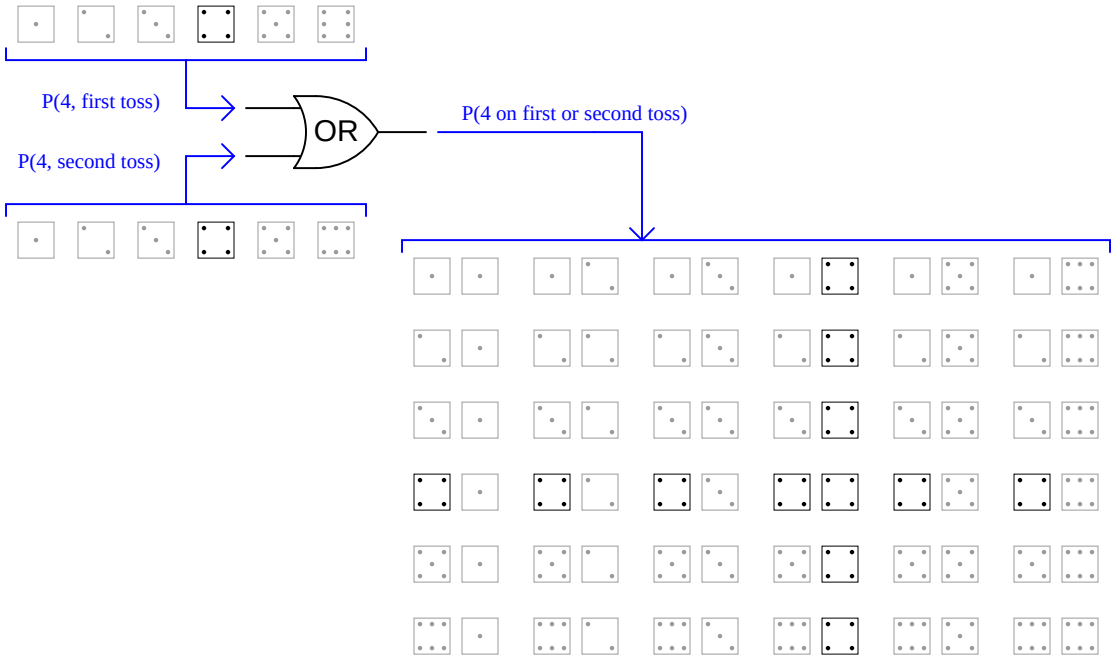
In his excellent text *Reliability Theory and Practice*, author Igor Bazovsky recounts the German V1 missile project during World War Two, and how early assumptions of system reliability were grossly inaccurate[16]. Once these faulty assumptions of reliability were corrected, development of the V1 missile resulted in greatly increased reliability until a system reliability of 75% (three out of four) was achieved.

[15] Here, "essential" means the system will fail if *any* of these identified components fails. Thus, Lusser's Law implies a logical "AND" relationship between the components' reliability values and the overall system reliability.

[16] According to Bazovsky (pp. 275-276), the first reliability principle adopted by the design team was that the system could be no more reliable than its least-reliable (weakest) component. While this is technically true, the mistake was to assume that the system would be *as reliable* as its weakest component (i.e. the "chain" would be exactly as strong as its weakest link). This proved to be too optimistic, as the system would still fail due to the failure of "stronger" components even when the "weaker" components happened to survive. After noting the influence of "stronger" components' unreliabilities on overall system reliability, engineers somehow reached the bizarre conclusion that system reliability was equal to the mathematical *average* of the components' reliabilities. Not surprisingly, this proved even less accurate than the "weakest link" principle. Finally, the designers were assisted by the mathematician Erich Pieruschka, who helped formulate Lusser's Law.

The logical "OR" function

The OR function regards probabilities of two or more redundant events (i.e. where the outcome of interest happens if any one of the events happen). Another example using a die is the probability of rolling a "four" on the first toss *or* on the second toss. It should be intuitively obvious that the probability of rolling a "four" on either toss will be more probable (i.e. more likely) than rolling a "four" on a single toss. The shaded field of possibilities (36 in all) demonstrate the likelihood of this either/or result compared to the likelihood of either value on either toss:

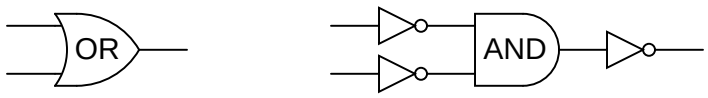


As you can see, there are eleven outcomes matching the specific criteria out of 36 total possible outcomes (the outcome with two "four" rolls counts as a single trial matching the stated criteria, just as all the other trials containing only one "four" roll count as single trials). This yields a probability value of eleven-in-thirty six ($\frac{11}{36}$) for the specified combination. This result may defy your intuition, if you assumed the OR function would be the simple *sum* of individual probabilities ($\frac{1}{6} + \frac{1}{6} = \frac{2}{6}$ or $\frac{1}{3}$), as opposed to the AND function's *product* of probabilities ($\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$). In truth, there is an application of the OR function where the probability is the simple sum, but that will come later in this presentation.

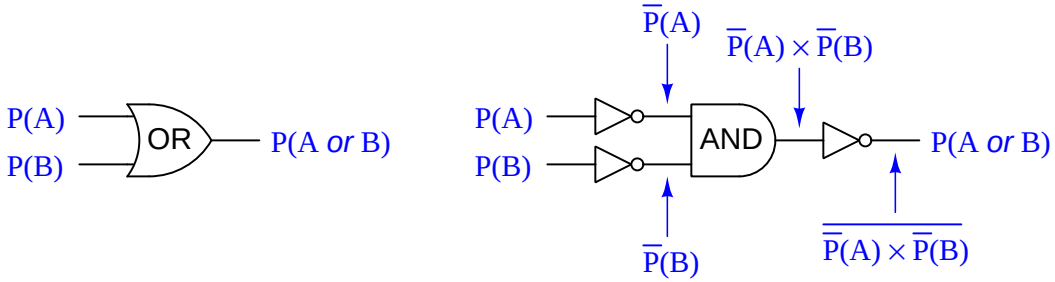
As with the logical "AND" function, the logical "OR" function assumes the events in question are independent from each other. That is to say, the events lack a common cause, and are not contingent upon one another in any way.

For now, a way to understand why we get a probability value of $\frac{11}{36}$ for our OR function with two $\frac{1}{6}$ input probabilities is to derive the OR function from other functions whose probability laws we already know with certainty. From Boolean algebra, DeMorgan's Theorem tells us an OR function is equivalent to an AND function with all inputs and outputs inverted ($A + B = \overline{\overline{A}\,\overline{B}}$):

(Equivalent logic functions)



We already know the complement (inversion) of a probability is the value of that probability subtracted from one ($\overline{P} = 1 - P$). This gives us a way to symbolically express the DeMorgan's Theorem definition of an OR function in terms of an AND function with three inversions:



Knowing that $\overline{P}(A) = 1 - P(A)$ and $\overline{P}(B) = 1 - P(B)$, we may substitute these inversions into the triple-inverted AND function to arrive at an expression for the OR function in simple terms of $P(A)$ and $P(B)$:

$$P(A \text{ or } B) = \overline{\overline{P}(A) \times \overline{P}(B)}$$

$$P(A \text{ or } B) = \overline{(1 - P(A))(1 - P(B))}$$

$$P(A \text{ or } B) = 1 - [(1 - P(A))(1 - P(B))]$$

Distributing terms on the right side of the equation:

$$P(A \text{ or } B) = 1 - [1 - P(B) - P(A) + P(A)P(B)]$$

$$P(A \text{ or } B) = P(B) + P(A) - P(A)P(B)$$

This, then, is our third law of probability:

$$P(\text{A } \textit{or} \text{ B}) = P(B) + P(A) - P(A) \times P(B)$$

Inserting our example probabilities of $\frac{1}{6}$ for both $P(A)$ and $P(B)$, we obtain the following probability for the `OR` function:

$$P(A \text{ or } B) = \frac{1}{6} + \frac{1}{6} - \left(\frac{1}{6}\right)\left(\frac{1}{6}\right)$$

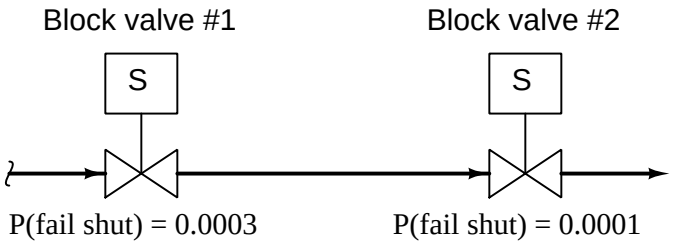
$$P(A \text{ or } B) = \frac{2}{6} - \left(\frac{1}{36}\right)$$

$$P(A \text{ or } B) = \frac{12}{36} - \frac{1}{36}$$

$$P(A \text{ or } B) = \frac{11}{36}$$

This confirms our previous conclusion of there being an $\frac{11}{36}$ probability of rolling a "four" on the first or second rolls of a die.

We may return to our example of a double-block valve assembly for a practical application of `OR` probability. When illustrating the `AND` probability function, we focused on the probability of both block valves failing to shut off when needed, since both valve 1 *and* valve 2 would have to fail open in order for the double-block assembly to fail in shutting off flow. Now, we will focus on the probability of *either* block valve failing to open when needed. While the `AND` scenario was an exploration of the system's un-dependability (i.e. the probability it might fail to stop a dangerous condition), this scenario is an exploration of the system's *un-security* (i.e. the probability it might fail to resume normal operation).



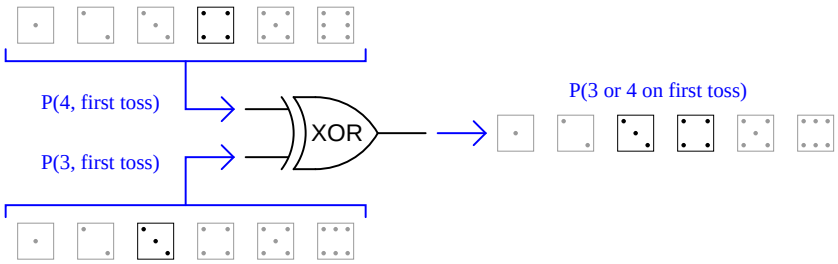
Each block valve is designed to be able to shut off flow independently, so that the flow of (potentially) dangerous process fluid will be halted if *either or both* valves shut off. The probability that process fluid flow may be impeded by the failure of either valve to open is thus a simple (non-exclusive) `OR` function:

$$P(\text{assembly fail}) = P(\text{valve 1 fail shut}) + P(\text{valve 2 fail shut}) - P(\text{valve 1 fail shut}) \times P(\text{valve 2 fail shut})$$

$$P(\text{assembly fail}) = 0.0003 + 0.0001 - (0.0003 \times 0.0001)$$

$$P(\text{assembly fail}) = 0.0003997 = 3.9997 \times 10^{-4}$$

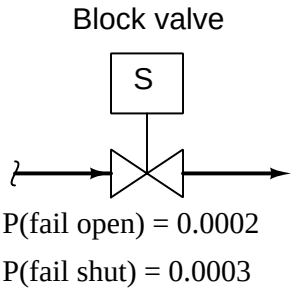
A similar application of the OR function is seen when we are dealing with *exclusive* events. For instance, we could calculate the probability of rolling either a "three" or a "four" in a single toss of a die. Unlike the previous example where we had two opportunities to roll a "four," and two sequential rolls of "four" counted as a single successful trial, here we know with certainty that the die cannot land on "three" *and* "four" in the same roll. Therefore, the exclusive OR probability (XOR) is much simpler to determine than a regular OR function:



This is the only type of scenario where the function probability is the simple sum of the input probabilities. In cases where the input probabilities are mutually exclusive (i.e. they *cannot* occur simultaneously or in a specific sequence), the probability of one *or* the other happening is the sum of the individual probabilities. This leads us to our fourth probability law:

$$P(\text{A } \textit{exclusively or} \text{ B}) = P(A) + P(B)$$

A practical example of the exclusive-or (`XOR`) probability function may be found in the failure analysis of a single block valve. If we consider the probability this valve may fail in either condition (stuck open or stuck shut), and we have data on the probabilities of the valve failing open and failing shut, we may use the `XOR` function to model the system's general unreliability[17]. We know that the exclusive-or function is the appropriate one to use here because the two "input" scenarios (failing open versus failing shut) *absolutely cannot* occur at the same time:



$$P(\text{valve fail}) = P(\text{valve fail open}) + P(\text{valve fail shut})$$

$$P(\text{valve fail}) = 0.0002 + 0.0003$$

$$P(\text{valve fail}) = 0.0005 = 5 \times 10^{-4}$$

If the intended safety function of this block valve is to shut off the flow of fluid if a dangerous condition is detected, then the probability of this valve's failure to shut when needed is a measure of its *undependability*. Conversely, the probability of this valve's failure to open under normal (safe) operating conditions is a measure of its *unsecurity*. The `XOR` of the valve's undependability and its unsecurity therefore represents its *unreliability*. The complement of this value will be the valve's *reliability*: $1 - 0.0005 = 0.9995$. This reliability value tells us we can expect the valve to operate as it's called to 99.95% of the time, and we should expect 5 failures out of every 10,000 calls for action.

[17] Here we have an example where dependability and security are lumped together into one "reliability" quantity.

Summary of probability laws

The complement (inversion) of a probability:

$$P(A) = 1 - \overline{P}(A)$$

The probability of coincidental events (where both must happen either simultaneously or in specific sequence) for the result of interest to occur:

$$P(\text{A } \textit{and } \text{B}) = P(\text{A}) \times P(\text{B})$$

The probability of redundant events (where either or both may happen) for the result of interest to occur:

$$P(\text{A } \textit{or } \text{B}) = P(B) + P(A) - P(A) \times P(B)$$

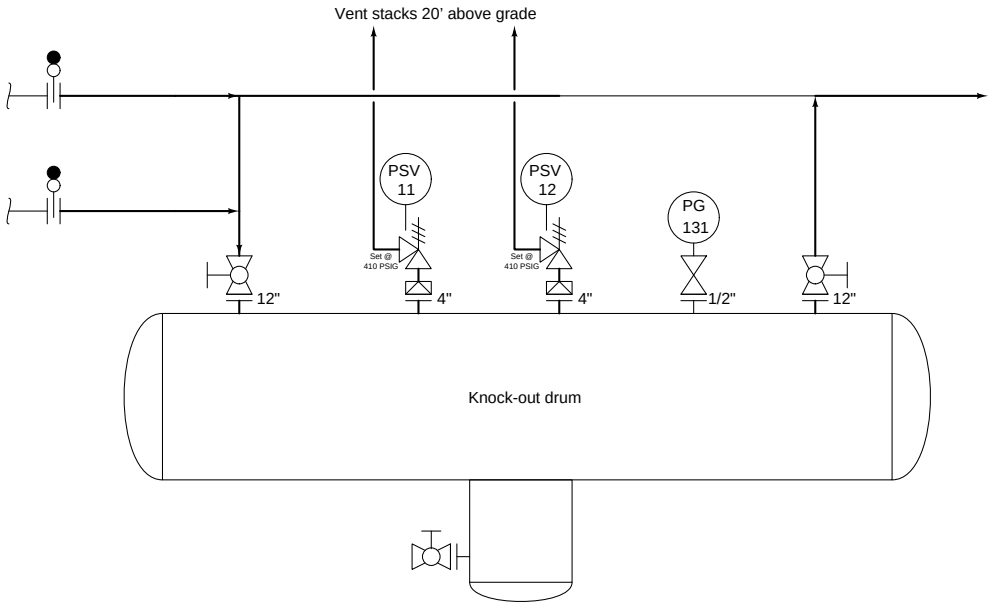
The probability of exclusively redundant events (where either may happen, but not simultaneously or in specific sequence) for the result of interest to occur:

$$P(\text{A } \textit{exclusively or } \text{B } \textit{exclusively}) = P(A) + P(B)$$

32.2.3 Applying probability laws to real systems

The relatively simple concepts of `AND` and `OR` Boolean functions become surprisingly complicated when applying them to real-life measures of component reliability, mainly because reliability is measured in multiple ways. As we have already seen, *dependability* (D) and *security* (S) are related concepts in that they both describe the probability of a system or system component functioning properly, but defy simple correlation because they imply different failure modes. "Dependability" for any safety-related system or component is the probability that it will perform its safety function when called upon in an emergency. "Security" by contrast is the probability that the system or component in question will maintain normal operation when there is no emergency.

To illustrate, we will examine the overpressure protection features of a "knock-out drum" used to collect small amounts of liquid entrained in a gas stream. This particular vessel is equipped with two pressure-safety valves (PSV-11 and PSV-12) designed to open and vent gas to atmosphere in the event of an overpressure condition (over 410 PSIG):



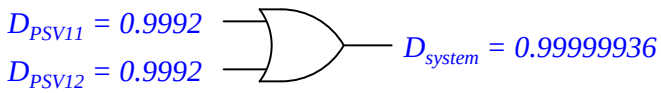
Suppose each of these pressure safety valves has a dependability (D) rating of 0.9992, which means each one has a 99.92% chance of opening up to relieve excess pressure when a high-pressure condition exists. Let us also suppose each of these PSVs has a security (S)[18] rating of 0.995, which means each one has a 99.5% chance of remaining in the shut condition when no overpressure condition exists. Furthermore, assume each of the two pressure safety valves individually has a high enough flow capacity to adequately vent the vessel during an overpressure condition.

[18] An easy way to remember what each of these terms mean in the context of a protective system is to associate D (Dependability) with a *dangerous* scenario and S (Security) with a *safe* scenario: D expresses what the system or component will do when a dangerous condition presents itself to the protective system and it needs to act; S expresses what the system or component will do when conditions are safe and there is no need to act.

How might we calculate the overall dependability and security ratings of this dual-PSV overpressure protection system? Clearly, we must use Boolean functions to combine the two valves' D ratings into a D_{system} rating, and likewise with the two valves' S ratings, but which logical function should we use to calculate each measure of reliability? The choice between `AND` and `OR` functions may not be obvious at first inspection.

One way to analyze logical functions is in terms of what state (0 or 1) at any input will *guarantee* a certain output state. For an `AND` function, any 0 state in guarantees a 0 state out. For an `OR` function, any 1 state in guarantees a 1 state out. These facts are useful when selecting logical functions for a variety of purposes, and they will serve us well in this application of probability values too.

A useful problem-solving technique for this application is called *limiting cases*, where we take some quantity to its extreme limits in an effort to simply the problem at hand. To begin, we will assume that one of the two pressure safety valves in this system has a D rating of 1, which means it is perfectly reliable when called to open by a high-pressure condition. A D rating of 1 is a "limiting case" of the pressure safety valve's dependability: a perfectly dependable PSV. If this were true, would it guarantee the whole overpressure protection system is dependable, or not? Since we know each valve is sized large enough to protect the vessel on its own (without need of the second PSV opening), then the answer to this question is "yes": a single PSV with a D rating of 1 guarantees a D_{system} rating of 1. All we need is for one of these PSVs to vent when it senses a high-pressure condition to protect the vessel from overpressure damage. Therefore, the proper Boolean function to calculate D_{system} from the valves' individual D ratings is the `OR` function, because given the choices of `AND` and `OR` only the `OR` function guarantees a certain output state with any "1" input. Calculating system dependability using both valves' D ratings:

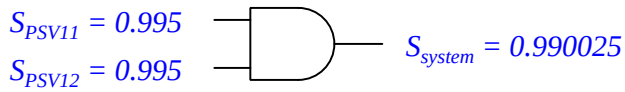


The numerical results shown here should make sense: in an overpressure protection system where we only need one of the two valves to vent gas during an overpressure condition, having two valves increases the probability that the vessel will be adequately protected.

Now we will apply this same problem-solving strategy to the system's *security* (S). Taking the high limiting-case value of either PSV's S rating, we ask ourselves the question "Does any one perfectly secure PSV ($S = 1$) make the system secure?" In other words, if one of these valves was guaranteed not to vent when no overpressure condition exists, would that mean the entire system was guaranteed not to vent when it didn't need to? The answer here is "no", since the presence of *two* pressure safety valves increases the chance of unnecessary leakage. This tells us we cannot use the `OR` function for security, because a perfectly secure PSV ($S = 1$) does *not* guarantee a perfectly secure system.

At this point we may conclude that the proper Boolean function for system security in this application is the `AND`, by process of elimination. However, we may also consider a different limiting-case scenario to verify this conclusion. Let us suppose one of the pressure safety valves failed in

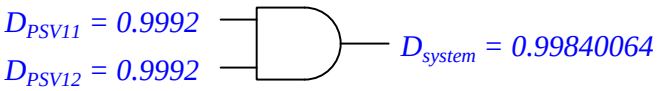
such a way that it had *zero* security, meaning there was no chance at all it would remain shut when no overpressure condition existed (i.e. a security rating of $S = 0$ means it is guaranteed to vent when it shouldn't). Would one PSV in this state guarantee a certain system security state? We see here that this is true: any one PSV with an S rating of zero means the system as a whole has a zero S rating as well, because all it takes is one PSV to unnecessarily vent to make the system as a whole unnecessarily vent. Since we know the Boolean `AND` function guarantees a zero output for any zero input, this is the function we should use when calculating system security. Calculating system security using both valves' S ratings::



These numerical results should make sense as well: in an overpressure protection system where a leak in one valve is enough to constitute a problem, the presence of multiple valves is a liability and therefore reduces the over-all security.

It is worth noting that a simple change in parameters may strongly impact our reliability calculations. In this scenario we were told each pressure safety valve was sized large enough to adequately vent the vessel on its own, without the help of the other PSV, in the event of an overpressure condition. What if the PSVs were undersized, and *both* of them would be required to vent in order to protect the vessel from overpressure damage? How would this alteration impact our reliability calculations?

It should be obvious that this change will have no effect whatsoever on the system's security, because it still takes just one PSV to leak in order to make the whole system unsecure. However, dependability will definitely be affected by this change because now a single PSV with a $D = 1$ rating is not enough to guarantee a protected system. With undersized PSVs, *both* valves must be dependable in order to guarantee dependable overpressure protection. Conversely, if only one of the PSVs fails in such a way as to be completely undependable ($D = 0$, meaning the valve is guaranteed to fail in the shut condition when faced with high pressure), it makes the whole system undependable because the other valve on its own is not enough to adequately vent the excess gas. From this analysis we can see that the proper Boolean function for dependability will now be `AND`, because any zero into an `AND` function guarantees a 0 output. Re-calculating dependability for undersized PSVs:



32.3 Practical measures of reliability

In reliability engineering, it is important to be able to quantity the reliability (or conversely, the probability of failure) for common components, and for systems comprised of those components. As such, special terms and mathematical models have been developed to describe probability as it applies to component and system reliability.

32.3.1 Failure rate and MTBF

Perhaps the first and most fundamental measure of (un)reliability is the *failure rate* of a component or system of components, symbolized by the Greek letter lambda (λ). The definition of "failure rate" for a group of components undergoing reliability tests is the instantaneous rate of failures per number of surviving components:

$$\lambda = \frac{\frac{dN_f}{dt}}{N_s} \qquad \text{or} \qquad \lambda = \frac{dN_f}{dt}\frac{1}{N_s}$$

Where,

λ = Failure rate

N_f = Number of components failed during testing period

N_s = Number of components surviving during testing period

t = Time

The unit of measurement for failure rate (λ) is inverted time units (e.g. "per hour" or "per year"). An alternative expression for failure rate sometimes seen in reliability literature is the acronym *FIT* ("Failures In Time"), in units of 10^{-9} failures per hour. Using a unit with a built-in multiplier such as 10^{-9} makes it easier for human beings to manage the very small λ values normally associated with high-reliability industrial components and systems.

Failure rate may also be applied to discrete-switching (on/off) components and systems of discrete-switching components on the basis of the number of on/off cycles rather than clock time. In such cases, we define failure rate in terms of cycles (c) instead of in terms of minutes, hours, or any other measure of time (t):

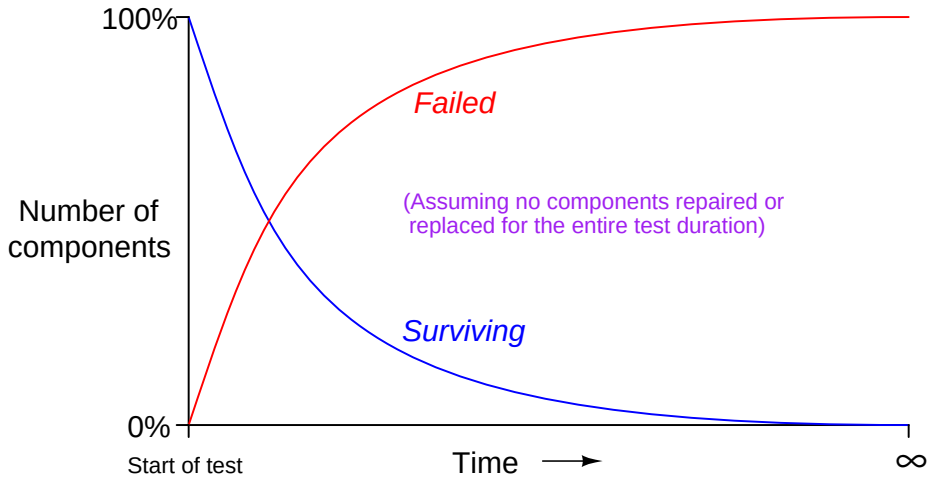
$$\lambda = \frac{\frac{dN_f}{dc}}{N_s} \qquad \text{or} \qquad \lambda = \frac{dN_f}{dc}\frac{1}{N_s}$$

One of the conceptual difficulties inherent to the definition of lambda (λ) is that it is fundamentally a *rate* of failure over time. This is why the calculus notation $\frac{dN_f}{dt}$ is used to define lambda: a "derivative" in calculus always expresses a rate of change. However, a failure *rate* is not the same thing as the number of devices failed in a test, nor is it the same thing as the probability of failure for one or more of those devices. Failure rate (λ) has more in common with the *time constant* of an resistor-capacitor circuit (τ) than anything else.

An illustrative example is helpful here: if we were to test a large batch of identical components for proper operation over some extended period of time with no maintenance or other intervention, the number of failed components in that batch would gradually accumulate while the number of surviving components in the batch would gradually decline. The reason for this is obvious: every component that fails remains failed (with no repair), leaving one fewer surviving component to function. If we limit the duration of this test to a time-span much shorter than the expected lifetime of the components, any failures that occur during the test must be due to random causes ("Acts of God") rather than component wear-out.

This scenario is analogous to another random process: rolling a large set of dice, counting any "1" roll as a "fail" and any other rolled number as a "survive." Imagine rolling the whole batch of dice at once, setting aside any dice landing on "1" aside (counting them as "failed" components in the batch), then only rolling the *remaining* dice the next time. If we maintain this protocol – setting aside "failed" dice after each roll and only continuing to roll "surviving" dice the next time – we will find ourselves rolling fewer and fewer "surviving" dice in each successive roll of the batch. Even though each of the six-sided die has a fixed failure probability of $\frac{1}{6}$, the population of "failed" dice keeps growing over time while the population of "surviving" dice keeps dwindling over time.

Not only does the number of surviving components in such a test dwindle over time, but that number dwindles at an ever-decreasing rate. Likewise with the number of failures: the number of components failing (dice coming up "1") is greatest at first, but then tapers off after the population of surviving components gets smaller and smaller. Plotted over time, the graph looks something like this:



Rapid changes in the failed and surviving component populations occurs at the start of the test when there is the greatest number of functioning components "in play." As components fail due to random events, the smaller and smaller number of surviving components results in a slower approach for both curves, simply because there are fewer surviving components remaining to fail.

These curves are precisely identical to those seen in RC (resistor-capacitor) charging circuits, with voltage and current tracing complementary paths: one climbing to 100% and the other falling to 0%, but both of them doing so at ever-decreasing rates. Despite the asymptotic approach of both curves, however, we can describe their approaches in an RC circuit with a constant value τ, otherwise known as the *time constant* for the RC circuit. Failure rate (λ) plays a similar role in describing the failed/surviving curves of a batch of tested components:

$$N_{surviving} = N_o e^{-\lambda t} \qquad N_{failed} = N_o \left(1 - e^{-\lambda t}\right)$$

Where,

$N_{surviving}$ = Number of components surviving at time t
N_{failed} = Number of components failed at time t
N_o = Total number of components in test batch
e = Euler's constant (≈ 2.71828)
λ = Failure rate (assumed to be a constant during the useful life period)

Following these formulae, we see that 63.2% of the components will fail (36.8% will survive) when $\lambda t = 1$ (i.e. after one "time constant" has elapsed).

Unfortunately, this definition for lambda doesn't make much intuitive sense. There is a way, however, to model failure rate in a way that not only makes more immediate sense, but is also more realistic to industrial applications. Imagine a different testing protocol where we maintain a constant sample quantity of components over the entire testing period by immediately replacing each failed device with a working substitute as soon as it fails. Now, the number of functioning devices under test will remain constant rather than declining as components fail. Imagine counting the number of "fails" (dice falling on a "1") for each batch roll, and then rolling *all* the dice in each successive trial rather than setting aside the "failed" dice and only rolling those remaining. If we did this, we would expect a constant fraction $\left(\frac{1}{6}\right)$ of the six-sided dice to "fail" with each and every roll. The number of failures per roll divided by the total number of dice would be the failure rate (lambda, λ) for these dice. We do not see a curve over time because we do not let the failed components remain failed, and thus we see a constant number of failures with each period of time (with each group-roll).

We may mathematically express this using a different formula:

$$\lambda = \frac{\frac{N_f}{t}}{N_o} \qquad \text{or} \qquad \lambda = \frac{N_f}{t}\frac{1}{N_o}$$

Where,

λ = Failure rate

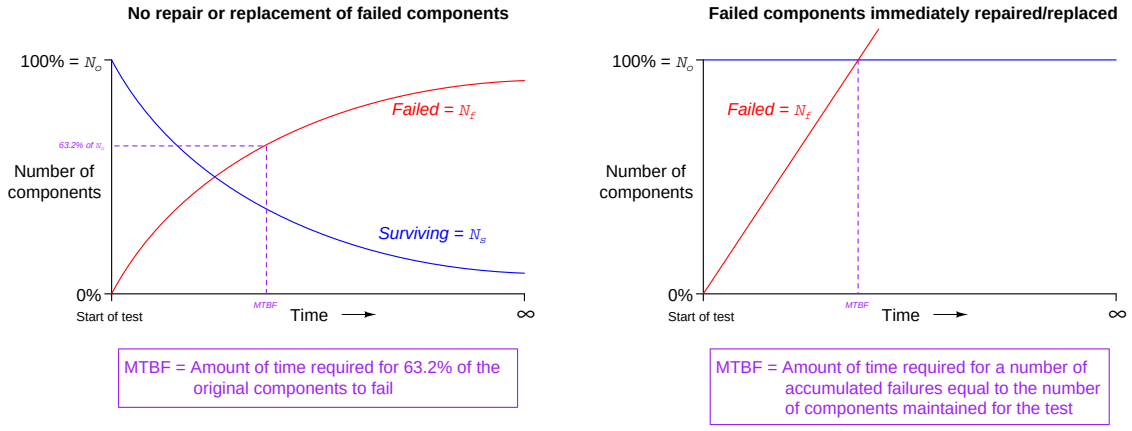
N_f = Number of components failed during testing period

N_o = Number of components under test (maintained constant) during testing period by immediate replacement of failed components

t = Time

An alternative way of expressing the failure rate for a component or system is the reciprocal of lambda ($\frac{1}{\lambda}$), otherwise known as *Mean Time Between Failures* (MTBF). If the component or system in question is repairable, the expression *Mean Time To Failure* (MTTF) is often used instead[19]. Whereas failure rate (λ) is measured in reciprocal units of time (e.g. "per hour" or "per year"), MTBF is simply expressed in units of time (e.g. "hours" or "years").

For non-maintained tests where the number of failed components accumulates over time (and the number of survivors dwindles), MTBF is precisely equivalent to "time constant" in an RC circuit: MTBF is the amount of time it will take for 63.2% of the components to fail due to random causes, leaving 36.8% of the component surviving. For maintained tests where the number of functioning components remains constant due to swift repairs or replacement of failed components, MTBF (or MTTF) is the amount of time it will take for the total number of tested components to fail[20].



It should be noted that these definitions for lambda and MTBF are idealized, and do not necessarily represent all the complexity we see in real-life applications. The task of calculating

[19] Since most high-quality industrial devices and systems are repairable for most faults, MTBF and MTTF are interchangeable terms.

[20] This does not mean the amount of time for *all* components to fail, but rather the amount of time to log a total number of failures equal to the total number of components tested. Some of those failures may be multiple for single components, while some other components in the batch might never fail within the MTBF time.

lambda or MTBF for any real component sample can be quite complex, involving statistical techniques well beyond the scope of instrument technician work.

Simple calculation example: transistor failure rate

Problem: Suppose a semiconductor manufacturer creates a microprocessor "chip" containing 2500000 transistors, each of which is virtually identical to the next in terms of ruggedness and exposure to degrading factors such as heat. The architecture of this microprocessor is such that there is enough redundancy to allow continued operation despite the failure of some of its transistors. This integrated circuit is continuously tested for a period of 1000 days (24000 hours), after which the circuit is examined to count the number of failed transistors. This testing period is well within the useful life of the microprocessor chip, so we know none of the failures will be due to wear-out, but rather to random causes.

Supposing several tests are run on identical chips, with an average of 3.4 transistors failing per 1000-day test. Calculate the failure rate (λ) and the MTBF for these transistors.

Solution: The testing scenario is one where failed components are not replaced, which means both the number of failed transistors and the number of surviving transistors changes over time like voltage and current in an RC charging circuit. Thus, we must calculate lambda by solving for it in the exponential formula.

Using the appropriate formula, relating number of failed components to the total number of components:

$$N_{failed} = N_o \left(1 - e^{-\lambda t}\right)$$

$$3.4 = 2500000 \left(1 - e^{-24000\lambda}\right)$$

$$1.36 \times 10^{-6} = 1 - e^{-24000\lambda}$$

$$e^{-24000\lambda} = 1 - 1.36 \times 10^{-6}$$

$$-24000\lambda = \ln(1 - 1.36 \times 10^{-6})$$

$$-24000\lambda = -1.360000925 \times 10^{-6}$$

$$\lambda = 5.66667 \times 10^{-11} \text{ per hour} = 0.0566667 \text{ FIT}$$

Failure rate may be expressed in units of "per hour," "Failures In Time" (FIT, which means failures per 10^9 hours), or "per year" (pa).

$$\text{MTBF} = \frac{1}{\lambda} = 1.7647 \times 10^{10} \text{ hours} = 2.0145 \times 10^{6} \text{ years}$$

Recall that Mean Time Between Failures (MTBF) is essentially the "time constant" for this decaying collection of transistors inside each microprocessor chip.

Simple calculation example: control valve failure rate

Problem: Suppose a control valve manufacturer produces a large number of valves, which are then sold to customers and used in comparable process applications. After a period of 5 years, data is collected on the number of failures these valves experienced. Five years is well within the useful life of these control valves, so we know none of the failures will be due to wear-out, but rather to random causes.

Supposing customers report an average of 15 failures for every 200 control valves in service over the 5-year period, calculate the failure rate (λ) and the MTTF for these control valves.

Solution: The testing scenario is one where failures are repaired in a short amount of time, since these are working valves being maintained in a real process environment. Thus, we may calculate lambda as a simple fraction of failed components to total components.

Using the appropriate formula, relating number of failed components to the total number of components:

$$\lambda = \frac{N_f}{t}\frac{1}{N_o}$$

$$\lambda = \frac{15}{5 \text{ yr}}\frac{1}{200}$$

$$\lambda = \frac{3}{200 \text{ yr}}$$

$$\lambda = 0.015 \text{ per year (pa)} = 1.7123 \times 10^{-6} \text{ per hour}$$

With this value for lambda being so much larger than the microprocessor's transistors, it is not necessary to use a unit such as FIT to conveniently represent it.

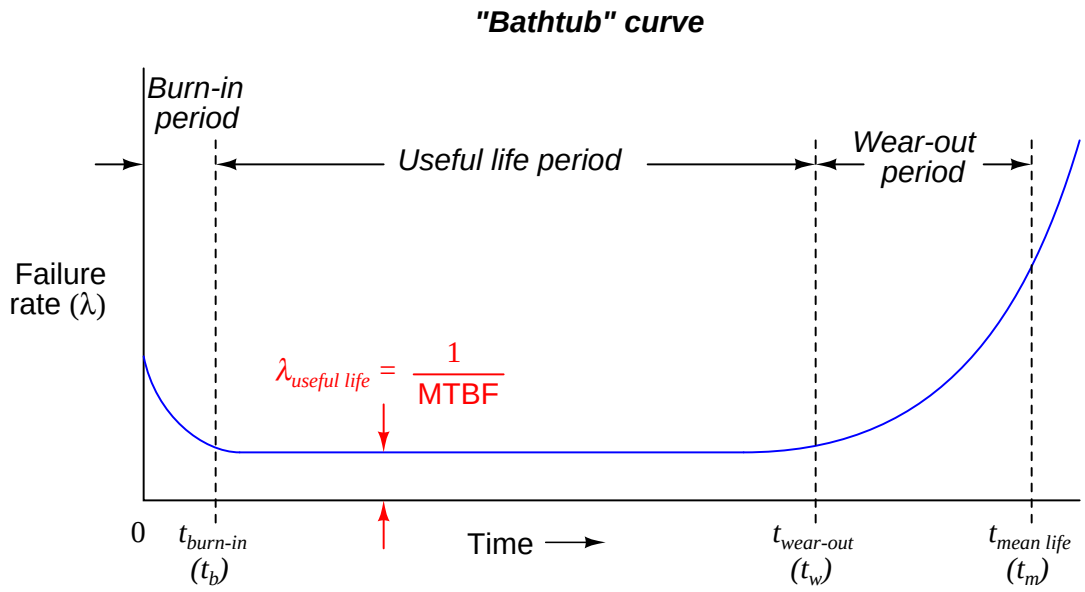
$$\text{MTTF} = \frac{1}{\lambda} = 66.667 \text{ years} = 584000 \text{ hours}$$

Recall that Mean Time To Failure (MTTF) is the amount of time it would take[21] to log a number of failures equal to the total number of valves in service, given the observed rate of failure due to random causes. Note that MTTF is largely synonymous with MTBF. The only technical difference between MTBF and MTTF is that MTTF more specifically relates to situations where components are repairable, which is the scenario we have here with well-maintained control valves.

[21] The typically large values we see for MTBF and MTTF can be misleading, as they represent a *theoretical* time based on the failure rate seen over relatively short testing times where all components are "young." In reality, the wear-out time of a component will be less than its MTBF. In the case of these control valves, they would likely all "die" of old age and wear long before reaching an age of 66.667 years!

32.3.2 The "bathtub" curve

Failure rate tends to be constant during a component's useful lifespan where the major cause of failure is random events ("Acts of God"). However, lambda does not remain constant over the entire life of the component or system. A common graphical expression of failure rate is the so-called *bathtub curve* showing the typical failure rate profile over time from initial manufacture (brand-new) to wear-out:



This curve profiles the failure rate of a large sample of components (or a large sample of systems) as they age. Failure rate begins at a relatively high value starting at time zero due to defects in manufacture. Failure rate drops off rapidly during a period of time called the *burn-in period* where defective components experience an early death. After the burn-in period, failure rate remains relatively constant over the useful life of the components, and this is where we typically define and apply the failure rate (λ). Any failures occurring during this "useful life" period are due to random mishaps ("Acts of God"). Toward the end of the components' working lives when the components enter the *wear-out period*, failure rate begins to rise until all components eventually fail. The *mean (average) life* of a component (t_m) is the time required for one-half of the components surviving up until the wear-out time (t_w) to fail, the other half failing after the mean life time.

Several important features are evident in this "bathtub" curve. First, component reliability is greatest between the times of burn-in and wear-out. For this reason, many manufacturers of high-reliability components and systems perform their own burn-in testing prior to sale, so that the customers are purchasing products that have already passed the burn-in phase of their lives. To express this using colloquial terms, we may think of "burnt-in" components as those having already passed through their "growing pains," and are now "mature" enough to face demanding applications.

Another important measure of reliability is the *mean life*. This is an expression of a component's (or system's) operating lifespan. At first this may sound synonymous with MTBF, but it is not.

MTBF – and by extension lambda, since MTBF is the reciprocal of failure rate – is an expression of susceptibility to random ("chance") failures. Both MTBF and λ_{useful} are quite independent of mean life[22]. In practice, values for MTBF often greatly exceed values for mean life.

To cite a practical example, the Rosemount model 3051C differential pressure transmitter has a suggested useful lifetime of 50 years (based on the expected service life of tantalum electrolytic capacitors used in its circuitry), while its demonstrated MTBF is 136 years. The larger value of 136 years is a projection based on the failure rate of large samples of these transmitters when they are all "young," which is why one should never confuse MTBF for service life. In reality, components within the instrument will begin to suffer accelerated failure rates as they reach their end of useful lifetime, as the instrument approaches the right-hand end of the "bathtub" curve.

When determining the length of time any component should be allowed to function in a high-reliability system, the mean life (or even better, the *wear-out* time) should be used as a guide, not the MTBF. This is not to suggest the MTBF is a useless figure – far from it. MTBF simply serves a different purpose, and that is to predict the rate of random failures *during* the useful life span of a large number of components or systems, whereas mean life predicts the service life period where the component's failure rate remains relatively constant.

[22] One could even imagine some theoretical component immune to wear-out, but still having finite values for failure rate and MTBF. Remember, λ_{useful} and MTBF refer to *chance* failures, not the normal failures associated with age and extended use.

32.3.3 Reliability

Reliability (R) is the probability a component or system will perform as designed. Like all probability figures, reliability ranges in value from 0 to 1, inclusive. Given the tendency of manufactured devices to fail over time, reliability decreases with time. During the useful life of a component or system, reliability is related to failure rate by a simple exponential function:

$$R = e^{-\lambda t}$$

Where,

R = Reliability as a function of time (sometimes shown as $R(t)$)
e = Euler's constant (≈ 2.71828)
λ = Failure rate (assumed to be a constant during the useful life period)
t = Time

Knowing that failure rate is the mathematical reciprocal of mean time between failures (MTBF), we may re-write this equation in terms of MTBF as a "time constant" (τ) for random failures during the useful life period:

$$R = e^{\frac{-t}{MTBF}} \qquad \text{or} \qquad R = e^{\frac{-t}{\tau}}$$

This inverse-exponential function mathematically explains the scenario described earlier where we tested a large batch of components, counting the number of failed components and the number of surviving components over time. Like the dice experiment where we set aside each "failed" die and then rolled only the remaining "survivors" for the next trial in the test, we end up with a diminishing number of "survivors" as the test proceeds.

The same exponential function for calculating reliability applies to single components as well. Imagine a single component functioning within its useful life period, subject only to random failures. The longer this component is relied upon, the more time it has to succumb to random faults, and therefore the less likely it is to function perfectly over the duration of its test. To illustrate by example, a pressure transmitter installed and used for a period of 1 year has a greater chance of functioning perfectly over that service time than an identical pressure transmitter pressed into service for 5 years, simply because the one operating for 5 years has five times more opportunity to fail. In other words, the reliability of a component over a specified time is a function of time, and not just the failure rate (λ).

Using dice once again to illustrate, it is as if we rolled a single six-sided die over and over, waiting for it to "fail" (roll a "1"). The more times we roll this single die, the more likely it will eventually "fail" (eventually roll a "1"). With each roll, the probability of failure is $\frac{1}{6}$, and the probability of survival is $\frac{5}{6}$. Since survival over multiple rolls necessitates surviving the first roll *and* and next roll *and* the next roll, all the way to the last surviving roll, the probability function we should apply here is the "AND" (multiplication) of survival probability. Therefore, the survival probability after a single roll is $\frac{5}{6}$, while the survival probability for two successive rolls is $\left(\frac{5}{6}\right)^2$, the survival probability for three successive rolls is $\left(\frac{5}{6}\right)^3$, and so on.

The following table shows the probabilities of "failure" and "survival" for this die with an increasing number of rolls:

Number of rolls	**Probability of failure** (1)	**Probability of survival** (2, 3, 4, 5, 6)
1	1 / 6 = 0.16667	5 / 6 = 0.83333
2	11 / 36 = 0.30556	25 / 36 = 0.69444
3	91 / 216 = 0.42129	125 / 216 = 0.57870
4	671 / 1296 = 0.51775	625 / 1296 = 0.48225
n	$1 - \left(\frac{5}{6}\right)^n$	$\left(\frac{5}{6}\right)^n$

A practical example of this equation in use would be the reliability calculation for a Rosemount model 1151 analog differential pressure transmitter (with a demonstrated MTBF value of 226 years as published by Rosemount) over a service life of 5 years following burn-in:

$$R = e^{\frac{-5}{226}}$$

$$R = 0.9781 = 97.81\%$$

Another way to interpret this reliability value is in terms of a large batch of transmitters. If three hundred Rosemount model 1151 transmitters were continuously used for five years following burn-in (assuming no replacement of failed units), we would expect approximately 293 of them to still be working (i.e. 6.564 random-cause failures) during that five-year period:

$$N_{surviving} = N_o e^{\frac{-t}{MTBF}}$$

$$\text{Number of surviving transmitters} = (300)\left(e^{\frac{-5}{226}}\right) = 293.436$$

$$N_{failed} = N_o\left(1 - e^{\frac{-t}{MTBF}}\right)$$

$$\text{Number of failed transmitters} = 300\left(1 - e^{\frac{-5}{226}}\right) = 6.564$$

It should be noted that the calculation will be linear rather than inverse-exponential if we assume immediate replacement of failed transmitters (maintaining the total number of functioning units at 300). If this is the case, the number of random-cause failures is simply $\frac{1}{226}$ per year, or 0.02212 per transmitter over a 5-year period. For a collection of 300 (maintained) Rosemount model 1151 transmitters, this would equate to 6.637 failed units over the 5-year testing span:

$$\text{Number of failed transmitters} = (300)\left(\frac{5}{226}\right) = 6.637$$

32.3.4 Probability of failure on demand (PFD)

Reliability, as previously defined, is the probability a component or system will perform as designed. Like all probability values, reliability is expressed a number ranging between 0 and 1, inclusive. A reliability value of zero (0) means the component or system is totally unreliable (i.e. it is guaranteed to fail). Conversely, a reliability value of one (1) means the component or system is completely reliable (i.e. guaranteed to properly function). In the context of dependability (i.e. the probability that a safety component or system will function *when called upon to act*), the unreliability of that component or system is referred to as *PFD*, an acronym standing for *Probability of Failure on Demand*. Like dependability, this is also a probability value ranging from 0 to 1, inclusive. A PFD value of zero (0) means there is no probability of failure (i.e. it is 100% dependable – guaranteed to properly perform when needed), while a PFD value of one (1) means it is completely undependable (i.e. guaranteed to fail when activated). Thus:

$$\text{Dependability} + \text{PFD} = 1$$

$$\text{PFD} = 1 - \text{Dependability}$$

$$\text{Dependability} = 1 - \text{PFD}$$

Obviously, a system designed for high dependability should exhibit a small PFD value (very nearly 0). Just how low the PFD needs to be is a function of how critical the component or system is to the fulfillment of our human needs.

The degree to which a system must be dependable in order to fulfill our modern expectations is often surprisingly high. Suppose someone were to tell you the reliability of seatbelts in a particular automobile was 99.9 percent (0.999). This sounds rather good, doesn't it? However, when you actually consider the fact that this degree of probability would mean an average of one failed seatbelt for every 1000 collisions, the results are seen to be rather poor (at least to modern American standards of expectation). If the dependability of seatbelts is 0.999, then the PFD is 0.001:

$$\text{PFD} = 1 - \text{Dependability}$$

$$\text{PFD} = 1 - 0.999$$

$$\text{PFD} = 0.001$$

Let's suppose an automobile manufacturer sets a goal of only 1 failed seatbelt in any of its cars during a 1 million unit production run, assuming each and every one of these cars were to crash. Assuming four seatbelts per car, this equates to a $\frac{1}{4000000}$ PFD. The necessary dependability of this manufacturer's seatbelts must therefore be:

$$\text{Dependability} = 1 - \text{PFD} = 1 - \frac{1}{4000000} = 0.99999975$$

Thus, the dependability of these seatbelts must be 99.999975% in order to fulfill the goal of only 1 (potential) seatbelt failure out of 4 million seatbelts produced.

A common order-of-magnitude expression of desired reliability is the number of "9" digits in the reliability value. A reliability value of 99.9% would be expressed as "three nine's" and a reliability value of 99.99% as "four nine's." Expressed thusly, the seatbelt dependability must be "six nine's" in order to achieve the automobile manufacturer's goal.

32.4 High-reliability systems

As discussed at the beginning of this chapter, instrumentation safety may be broadly divided into two categories: the safety hazards posed by malfunctioning instruments, and special instrument systems designed to reduce safety hazards of industrial processes. This section regards the first category.

All methods of reliability improvement incur some extra cost on the operation, whether it be capital expense (initial purchase/installation cost) or continuing expense (labor or consumables). The choice to improve system reliability is therefore very much an economic one. One of the human challenges associated with reliability improvement is continually justifying this cost over time. Ironically, the more successful a reliability improvement program has been, the less important that program seems. The manager of an operation suffering from reliability problems does not need to be convinced of the economic benefit of reliability improvement as much as the manager of a trouble-free facility. Furthermore, the people most aware of the benefits of reliability improvement are usually those tasked with reliability-improving duties (such as preventive maintenance), while the people least aware of the same benefits are usually those managing budgets. If ever a disagreement erupts between the two camps, pleas for continued financial support of reliability improvement programs may be seen as nothing more than self-interest, further escalating tensions[23].

A variety of methods exist to improve the reliability of systems. The following subsections investigate several of them.

[23] Preventive maintenance is not the only example of such a dynamic. Modern society is filled with monetarily expensive programs and institutions existing for the ultimate purpose of avoiding *greater* costs, monetary and otherwise. Public education, health care, and national militaries are just a few that come to my mind. Not only is it a challenge to continue justifying the expense of a well-functioning cost-avoidance program, but it is also a challenge to detect and remove unnecessary expenses (waste) within that program. To extend the preventive maintenance example, an appeal by maintenance personnel to continue (or further) the maintenance budget may happen to be legitimate, but a certain degree of self-interest will always be present in the argument. Just because preventive maintenance is actually necessary to avoid greater expense due to failure, does not mean *all* preventive maintenance demands are economically justified! Proper funding of any such program depends on the financiers being fair in their judgment *and* the executors being honest in their requests. So long as both parties are human, this territory will remain contentious.

32.4.1 Design and selection for reliability

Many workable designs may exist for electronic and mechanical systems alike, but not all are equal in terms of reliability. A major factor in machine reliability, for example, is *balance*. A well-balanced machine will operate with little vibration, whereas an ill-balanced machine will tend to shake itself (and other devices mechanically coupled to it) apart over time[24].

Electronic circuit reliability is strongly influenced by design as well as by component choice. An historical example of reliability-driven design is found in the Foxboro SPEC 200 analog control system. The reliability of the SPEC 200 control system is legendary, with a proven record of minimal failures over many years of industrial use. According to Foxboro technical literature, several design guidelines were developed following application experience with Foxboro electronic field instruments (most notably the "E" and "H" model lines), among them the following:

- All critical switches should spend most of their time in the *closed* state
- Avoid the use of carbon composition resistors – use wirewound or film-type resistors instead
- Avoid the use of plastic-cased semiconductors – use glass-cased or hermetically sealed instead
- Avoid the use of electrolytic capacitors wherever possible – use polycarbonate or tantalum instead

Each of these design guidelines is based on minimization of component failure. Having switches spend most of their lives in the closed state means their contact surfaces will be less exposed to air and therefore less susceptible to corrosion over time (leading to an "open" fault). Wirewound resistors are better able to tolerate vibration and physical abuse than brittle carbon-composition designs. Glass-cased and hermetically-sealed semiconductors are better at sealing out moisture than plastic-cased semiconductors. Electrolytic capacitors are famously unreliable compared to other capacitor types such as polycarbonate, and so their avoidance is wise.

In addition to high-quality component characteristics and excellent design practices, components used in these lines of Foxboro instruments were "burned in" prior to circuit board assembly, thus avoiding many "early failures" due to components "burning in" during actual service.

[24] Sustained vibrations can do really strange things to equipment. It is not uncommon to see threaded fasteners undone slowly over time by vibrations, as well as cracks forming in what appear to be extremely strong supporting elements such as beams, pipes, etc. Vibration is almost never good for mechanical (or electrical!) equipment, so it should be eliminated wherever reliability is a concern.

32.4.2 Preventive maintenance

The term *preventive maintenance* refers to the maintenance (repair or replacement) of components prior to their inevitable failure in a system. In order to intelligently schedule the replacement of critical system components, some knowledge of those components' useful lifetimes is necessary. On the standard "bathtub curve," this corresponds with the *wear-out time* or $t_{wear-out}$.

In many industrial operations, preventive maintenance schedules (if they exist at all) are based on past history of component lifetimes, and the operational expenses incurred due to failure of those components. Preventive maintenance represents an up-front cost, paid in exchange for the avoidance of larger costs later in time.

A common example of preventive maintenance and its cost savings is the periodic replacement of lubricating oil and oil filters for automobile engines. Automobile manufacturers provide specifications for the replacement of oil and filters based on testing of their engines, and assumptions made regarding the driving habits of their customers. Some manufacturers even provide dual maintenance schedules, one for "normal" driving and another for "heavy" or "performance" driving to account for accelerated wear. As trivial as an oil change might seem to the average driver, regular maintenance to an automobile's lubrication system is absolutely critical not only to long service life, but also to optimum performance. Certainly, the consequences of not performing this preventive maintenance task on an automobile's engine will be costly[25].

Another example of preventive maintenance for increased system reliability is the regular replacement of light bulbs in traffic signal arrays. For rather obvious reasons, the proper function of traffic signal lights is critical for smooth traffic flow and public safety. It would not be a satisfactory state of affairs to replace traffic signal light bulbs only when they failed, as is common with the replacement of most light bulbs. In order to achieve high reliability, these bulbs must be replaced in advance of their expected wear-out times[26]. The cost of performing this maintenance is undeniable, but then so is the (greater) cost of congested traffic and accidents caused by burned-out traffic light bulbs.

An example of preventive maintenance in industrial instrumentation is the installation and service of *dryer* mechanisms for compressed air, used to power pneumatic instruments and valve actuators. Compressed air is a very useful medium for transferring (and storing) mechanical energy, but problems will develop within pneumatic instruments if water is allowed to collect within air distribution systems. Corrosion, blockages, and hydraulic "locking" are all potential consequences of "wet" instrument air. Consequently, instrument compressed air systems are usually installed separate from utility compressed air systems (used for operating general-purpose pneumatic tools and equipment actuators), using different types of pipe (plastic, copper, or stainless steel rather than black iron or galvanized iron) to avoid corrosion and using *air dryer* mechanisms near the compressor to absorb and expel moisture. These air dryers typically use a beaded *desiccant* material to absorb

[25] On an anecdotal note, a friend of mine once destroyed his car's engine, having never performed an oil or filter change on it since the day he purchased it. His poor car expired after only 70000 miles of driving – a mere fraction of its normal service life with regular maintenance. Given the type of car it was, he could have easily expected 200000 miles of service between engine rebuilds had he performed the recommended maintenance on it.

[26] Another friend of mine used to work as a traffic signal technician in a major American city. Since the light bulbs they replaced still had some service life remaining, they decided to donate the bulbs to a charity organization where the used bulbs would be freely given to low-income citizens. Incidentally, this same friend also instructed me on the proper method of inserting a new bulb into a socket: twisting the bulb just enough to maintain some spring tension on the base, rather than twisting the bulb until it will not turn farther (as most people do). Maintaining some natural spring tension on the metal leaf within the socket helps extend the socket's useful life as well!

water vapor from the compressed air, and then this desiccant material is periodically purged of its retained water. After some time of operation, though, the desiccant must be physically removed and replaced with fresh desiccant.

32.4.3 Component de-rating

Some[27] control system components exhibit an inverse relationship between service load (how "hard" the component is used) and service life (how long it will last). In such cases, a way to increase service life is to *de-rate* that component: operate it at a load reduced from its design rating.

For example, a variable-frequency motor drive (VFD) takes AC power at a fixed frequency and voltage and converts it into AC power of varying frequency and voltage to drive an induction motor at different speeds and torques. These electronic devices dissipate some heat owing mostly to the imperfect (slightly resistive) "on" states of power transistors. Temperature is a wear factor for semiconductor devices, with greater temperatures leading to reduced service lives. A VFD operating at high temperature, therefore, will fail sooner than a VFD operating at low temperature, all other factors being equal. One way to reduce the operating temperature of a VFD is to over-size it for the application. If the motor to be driven requires 2 horsepower of electrical power at full load, and increased reliability is demanded of the drive, then perhaps a 5 horsepower VFD (programmed with reduced trip settings appropriate to the smaller motor) could be chosen to drive the motor.

In addition to extending service life, de-rating also has the ability to amplify the mean time between failure (MTBF) of load-sensitive components. Recall that MTBF is the reciprocal of failure rate during the low area of the "bathtub curve," representing failures due to random causes. This is distinct from wear-out, which is an increase in failure rate due to irreversible wear and aging. The main reason a component will exhibit a greater MTBF value as a consequence of de-rating is that the component will be better able to absorb transient overloads, which is a typical cause of failure during the operational life of system components.

Consider the example of a pressure sensor in a process known to exhibit transient pressure surges. A sensor chosen such that the typical process operating pressure spans most of its range will have little overpressure capacity. Perhaps just a few over-pressure events will cause this sensor to fail well before its rated service life. A de-rated pressure sensor (with a pressure-sensing range covering much greater pressures than what are normally encountered in this process), by comparison, will have more pressure capacity to withstand random surges, and therefore exhibit less probability of random failure.

The costs associated with component de-rating include initial investment (usually greater, owing to the greater capacity and more robust construction compared to a "normally" rated component) and reduced sensitivity. The latter factor is an important one to consider if the component is expected to provide high accuracy as well as high reliability. In the example of the de-rated pressure sensor, accuracy will likely suffer because the full pressure range of the sensor is not being used for normal process pressure measurements. If the instrument is digital, resolution will certainly suffer as a result of de-rating the instrument's measurement range. Alternative methods of reliability improvement (including more frequent preventive maintenance) may be a better solution than de-rating in such cases.

[27] Many components do not exhibit any relationship between load and lifespan. An electronic PID controller, for example, will last just as long controlling an "easy" self-regulating process as it will controlling a "difficult" unstable ("runaway") process. The same might not be said for the other components of those loops, however! If the control valve in the self-regulating process rarely changes position, but the control valve in the runaway process continually moves in an effort to stabilize it at setpoint, the less active control valve will most likely enjoy a longer service life.

32.4.4 Redundant components

The MTBF of any system dependent upon certain critical components may be extended by duplicating those components in parallel fashion, such that the failure of only one does not compromise the system as a whole. This is called *redundancy*. A common example of component redundancy in instrumentation and control systems is the redundancy offered by distributed control systems (DCSs), where processors, network cables, and even I/O (input/output) channels may be equipped with "hot standby" duplicates ready to assume functionality in the event the primary component fails.

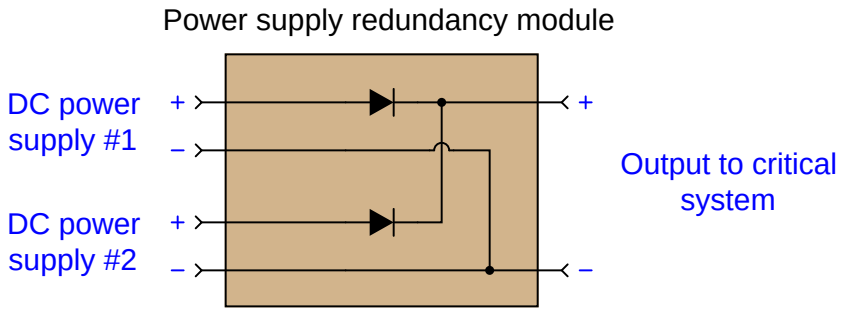
Redundancy tends to extend the MTBF of a system without necessarily extending its service life. A DCS, for example, equipped with redundant microprocessor control modules in its rack, will exhibit a greater MTBF because a random microprocessor fault will be covered by the presence of the spare ("hot standby") microprocessor module. However, given the fact that both microprocessors are continually powered, and therefore tend to "wear" at the same rate, their operating lives will not be additive. In other words, two microprocessors will not function twice as long before wear-out than one microprocessor.

The extension of MTBF resulting from redundancy holds true only if the random failures are truly independent events – that is, not associated by a common cause. To use the example of a DCS rack with redundant microprocessor control modules again, the susceptibility of that rack to a random microprocessor fault will be reduced by the presence of redundant microprocessors *only* if the faults in question are unrelated to each other, affecting the two microprocessors separately. There may exist common-cause fault mechanisms capable of disabling *both* microprocessor modules as easily as it could disable one, in which case the redundancy adds no value at all. Examples of such common-cause faults include power surges (because a surge strong enough to kill one module will likely kill the other at the same time) and a computer virus infection (because a virus able to attack one will be able to attack the other just as easily, and at the same time).

A simple example of component redundancy in an industrial instrumentation system is dual DC power supplies feeding through a diode module. The following photograph shows a typical example, in this case a pair of Allen-Bradley AC-to-DC power supplies for a DeviceNet digital network:



If either of the two AC-to-DC power supplies happens to fail with a low output voltage, the other power supply is able to carry the load by passing its power through the diode redundancy module[28]:



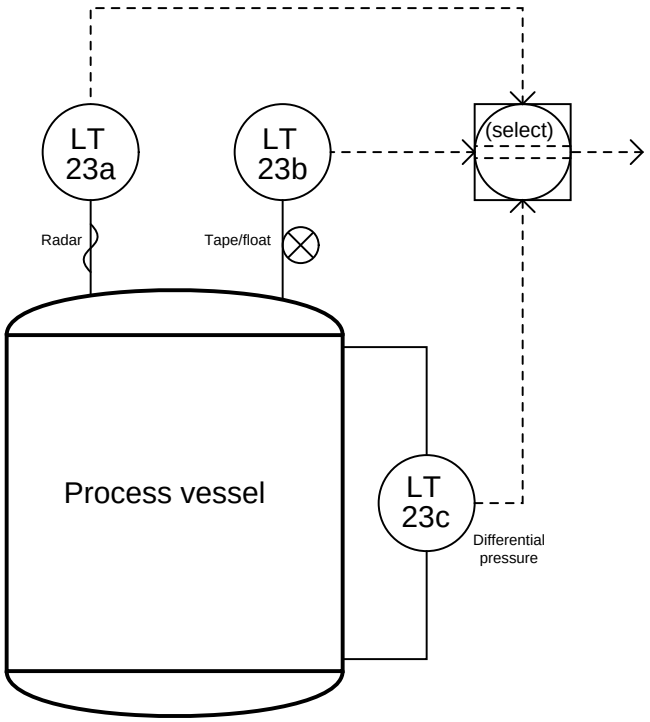
[28]This redundancy module has its own MTBF value, and so by including it in the system we are adding one more component that can fail. However, the MTBF rate of a simple diode network greatly exceeds that of an entire AC-to-DC power supply, and so we find ourselves at a greater level of reliability using this diode redundancy module than if we did not (and only had one power supply).

In order for redundant components to actually increase system MTBF, the potential for common-cause failures must be addressed. For example, consider the effects of powering redundant AC-to-DC power supplies from the exact same AC line. Redundant power supplies would increase system reliability in the face of a random power supply failure, but this redundancy would do *nothing at all* to improve system reliability in the event of the common AC power line failing! In order to enjoy the fullest benefit of redundancy in this example, we must source each AC-to-DC power supply from a different (unrelated) AC line.

Another example of redundancy in industrial instrumentation is the use of multiple transmitters to sense the same process variable, the notion being that the critical process variable will still be monitored even in the event of a transmitter failure. Thus, installing redundant transmitters should increase the MTBF of the system's sensing ability.

Here again, we must address common-cause failures in order to reap the full benefits of redundancy. If three liquid level transmitters are installed to measure the exact same liquid level, their combined signals represent an increase in measurement system MTBF *only* for independent faults. A failure mechanism common to all three transmitters will leave the system just as vulnerable to random failure as a single transmitter. In order to achieve optimum MTBF in redundant sensor arrays, the sensors must be immune to common faults.

In this example, three different types of level transmitter monitor the level of liquid inside a vessel, their signals processed by a *selector* function programmed inside a DCS:



Here, level transmitter 23a is a guided-wave radar (GWR), level transmitter 23b is a tape-and-

float, and level transmitter 23c is a differential pressure sensor. All three level transmitters sense liquid level using different technologies, each one with its own strengths and weaknesses. Better redundancy of measurement is obtained this way, since no single process condition or other random event is likely to fault more than one of the transmitters at any given time.

For instance, if the process liquid density happened to suddenly change, it would affect the measurement accuracy of the differential pressure transmitter (LT-23c), but not the radar transmitter nor the tape-and-float transmitter. If the process vapor density were to suddenly change, it might affect the radar transmitter (since vapor density generally affects dielectric constant, and dielectric constant affects the propagation velocity of electromagnetic waves, which in turn will affect the time taken for the radar pulse to strike the liquid surface and return), but this will not affect the float transmitter's accuracy nor will it affect the differential pressure transmitter's accuracy. Surface turbulence of the liquid inside the vessel may severely affect the float transmitter's ability to accurately sense liquid level, but it will have little effect on the differential pressure transmitter's reading nor the radar transmitter's measurement (assuming the radar transmitter is shrouded in a *stilling well*.

If the selector function takes either the median (middle) measurement or an average of the best 2-out-of-3 ("2oo3"), none of these random process occurrences will greatly affect the selected measurement of liquid level inside the vessel. True redundancy is achieved here, since the three level transmitters are not only less likely to (all) fail simultaneously than for any single transmitter to fail, but also because the level is being sensed in three completely different ways.

A crucial requirement for redundancy to be effective is that all redundant components must have precisely the same process function. In the case of redundant DCS components such as processors, I/O cards, and network cables, each of these redundant components must do nothing more than serve as "backup" spares for their primary counterparts. If a particular DCS node were equipped with two processors – one as the primary and another as a secondary (backup) – but yet the backup processor were tasked with some detail specific to it and not to the primary processor (or vice-versa), the two processors would *not* be truly redundant to each other. If one processor were to fail, the other would not perform *exactly* the same function, and so the system's operation would be affected (even if only in a small way) by the processor failure.

Likewise, redundant sensors must perform the exact same process measurement function in order to be truly redundant. A process equipped with triplicate measurement transmitters such as the previous example were a vessel's liquid level was being measured by a guided-wave radar, tape-and-float, and differential pressure based level transmitters, would enjoy the protection of redundancy if and only if all three transmitters sensed the exact same liquid level over the exact same calibrated range. This often represents a challenge, in finding suitable locations on the process vessel for three different instruments to sense the exact same process variable. Quite often, the pipe fittings penetrating the vessel (often called *nozzles*) are not conveniently located to accept multiple instruments at the points necessary to ensure consistency of measurement between them. This is often the case when an existing process vessel is retrofitted with redundant process transmitters. New construction is usually less of a problem, since the necessary nozzles and other accessories may be placed in their proper positions during the design stage[29].

If fluid flow conditions inside a process vessel are excessively turbulent, multiple sensors installed

[29] Of course, this assumes good communication and proper planning between all parties involved. It is not uncommon for piping engineers and instrument engineers to mis-communicate during the crucial stages of process vessel design, so that the vessel turns out not to be configured as needed for redundant instruments.

to measure the same variable will sometimes report significant differences. Multiple temperature transmitters located in close proximity to each other on a distillation column, for example, may report significant differences of temperature if their respective sensing elements (thermocouples, RTDs) contact the process liquid or vapor at points where the flow patterns vary. Multiple liquid level sensors, even of the same technology, may report differences in liquid level if the liquid inside the vessel swirls or "funnels" as it enters and exits the vessel.

Not only will substantial measurement differences between redundant transmitters compromise their ability to function as "backup" devices in the event of a failure, such differences may actually "fool" a redundant system into thinking one or more of the transmitters has already failed, thereby causing the deviating measurement to be ignored. To use the triplicate level-sensing array as an example again, suppose the radar-based level transmitter happened to register two inches greater level than the other two transmitters due to the effects[30] of liquid swirl inside the vessel. If the selector function is programmed to ignore such deviating measurements, the system degrades to a duplicate-redundant instead of triplicate-redundant array. In the event of a dangerously low liquid level, for example, only the radar-based and float-based level transmitters will be ready to signal this dangerous process condition to the control system, because the pressure-based level transmitter is registering too high.

[30] If a swirling fluid inside the vessel encounters a stationary baffle, it will tend to "pile up" on one side of that baffle, causing the liquid level to actually be greater in that region of the vessel than anywhere else inside the vessel. Any transmitter placed within this region will register a greater level, regardless of the measurement technology used.

32.4.5 Proof tests and self-diagnostics

A reliability enhancing technique related to preventive maintenance of critical instruments and functions, but generally not as expensive as component replacement, is periodic *testing* of component and system function. Regular "proof testing" of critical components enhances the MTBF of a system for two different reasons:

- Early detection of developing problems
- Regular "exercise" of components

First, proof testing may reveal weaknesses developing in components, indicating the need for replacement in the near future. An analogy to this is visiting a doctor to get a comprehensive exam – if this is done regularly, potentially fatal conditions may be detected early and crises averted.

The second way proof testing increases system reliability is by realizing the beneficial effects of regular function. The performance of many component and system types tends to degrade after prolonged periods of inactivity[31]. This tendency is most prevalent in mechanical systems, but holds true for some electrical components and systems as well. Solenoid valves, for instance, may become "stuck" in place if not cycled for long periods of time. Bearings may corrode and seize in place if left immobile. Both primary- and secondary-cell batteries are well known for their tendency to fail after prolonged periods of non-use. Regular cycling of such components actually *enhances* their reliability, decreasing the probability of a "stagnation" related failure well before the rated useful life has elapsed.

An important part of any proof-testing program is to ensure a ready stock of spare components is kept on hand in the event proof-testing reveals a failed component. Proof testing is of little value if the failed component cannot be immediately repaired or replaced, and so these warehoused components should be configured (or be easily configurable) with the exact parameters necessary for immediate installation. A common tendency in business is to focus attention on the engineering and installation of process and control systems, but neglect to invest in the support materials and infrastructure to keep those systems in excellent condition. High-reliability systems have special needs, and this is one of them.

[31] The father of a certain friend of mine has operated a used automobile business for many years. One of the tasks given to this friend when he was a young man, growing up helping his father in his business, was to regularly drive some of the cars on the lot which had not been driven for some time. If an automobile is left un-operated for many weeks, there is a marked tendency for batteries to fail and tires to lose their air pressure, among other things. The salespeople at this used car business jokingly referred to this as *lot rot*, and the only preventive measure was to routinely drive the cars so they would not "rot" in stagnation. Machines, like people, suffer if subjected to a lack of physical activity.

Methods of proof testing

The most direct method of testing a critical system is to stimulate it to its range limits and observe its reaction. For a process transmitter, this sort of test usually takes the form of a full-range calibration check. For a controller, proof testing would consist of driving all input signals through their respective ranges in all combinations to check for the appropriate output response(s). For a final control element (such as a control valve), this requires full stroking of the element, coupled with physical leakage tests (or other assessments) to ensure the element is having the intended effect on the process.

An obvious challenge to proof testing is how to perform such comprehensive tests without disrupting the process in which it functions. Proof-testing an out-of-service instrument is a simple matter, but proof-testing an instrument installed in a working system is something else entirely. How can transmitters, controllers, and final control elements be manipulated through their entire operating ranges without actually disturbing (best case) or halting (worst case) the process? Even if all tests may be performed at the required intervals during shut-down periods, the tests are not as realistic as they could be with the process operating at typical pressures and temperatures. Proof-testing components during actual "run" conditions is the most realistic way to assess their readiness.

One way to proof-test critical instruments with minimal impact to the continued operation of a process is to perform the tests on only some components, not all. For instance, it is a relatively simple matter to take a transmitter out of service in an operating process to check its response to stimuli: simply place the controller in manual mode and let a human operator control the process manually while an instrument technician tests the transmitter. While this strategy admittedly is not comprehensive, at least proof-testing some of the instruments is better than proof-testing none of them.

Another method of proof-testing is to "test to shutdown:" choose a time when operations personnel plan on shutting the process down anyway, then use that time as an opportunity to proof-test one or more critical component(s) necessary for the system to run. This method enjoys the greatest degree of realism, while avoiding the inconvenience and expense of an unnecessary process interruption.

Yet another method to perform proof tests on critical instrumentation is to accelerate the speed of the testing stimuli so that the final control elements will not react fully enough to actually disrupt the process, but yet will adequately assess the responsiveness of all (or most) of the components in question. The nuclear power industry sometimes uses this proof-test technique, by applying high-speed pulse signals to safety shutdown sensors in order to test the proper operation of shutdown logic, without actually shutting the reactor down. The test consists of injecting short-duration pulse signals at the sensor level, then monitoring the output of the shutdown logic to ensure consequent pulse signals are sent to the shutdown device(s). Various chemical and petroleum industries apply a similar proof-testing technique to safety valves called *partial stroke testing*, whereby the valve is stroked only part of its travel: enough to ensure the valve is capable of adequate motion without closing (or opening, depending on the valve function) enough to actually disrupt the process.

Redundant systems offer unique benefits and challenges to component proof-testing. The benefit of a redundant system in this regard is that any one redundant component may be removed from service for testing without any special action by operations personnel. Unlike a "simplex" system where removal of an instrument requires a human operator to manually take over control during the

duration of the test, the "backup" components of a redundant system should do this automatically, theoretically making the test much easier to conduct. However, the challenge of doing this is the fact that the portion of the system responsible for ensuring seamless transition in the event of a failure is in fact a component liable to failure itself. The only way to test this component is to actually disable one (or more, in highly redundant configurations) of the redundant components to see whether or not the remaining component(s) perform their redundant roles. So, proof-testing a redundant system harbors no danger if all components of the system are good, but risks process disruption if there happens to be an undetected fault.

Let us return to our triplicate level transmitter system once again to explore these concepts. Suppose we wished to perform a proof-test of the pressure-based level transmitter. Being one of three transmitters measuring liquid level in this vessel, we should be able to remove it from service with no preparation (other than notifying operations personnel of the test, and of the potential consequences) since the selector function should automatically de-select the disabled transmitter and continue measuring the process via the remaining two transmitters. If the proof-testing is successful, it proves not only that the transmitter works, but also that the selector function adequately performed its task in "backing up" the tested transmitter while it was removed. However, if the selector function happened to be failed when we disable the one level transmitter for proof-testing, the selected process level signal could register a faulty value instead of switching to the two remaining transmitters' signals. This might disrupt the process, especially if the selected level signal went to a control loop or to an automatic shutdown system. We could, of course, proceed with the utmost caution by having operations personnel place the control system in "manual" mode while we remove that one transmitter from service, just in case the redundancy does not function as designed. Doing so, however, fails to fully test the system's redundancy, since by placing the system in manual mode before the test we do not allow the redundant logic to fully function as it would be expected to in the event of an actual instrument failure.

Regular proof-testing is an essential activity to realize optimum reliability for any critical system. However, in all proof-testing we are faced with a choice: either test the components to their fullest degree, in their normal operating modes, and risk (or perhaps guarantee) a process disruption; or perform a test that is less than comprehensive, but with less (or no) risk of process disruption. In the vast majority of cases, the latter option is chosen simply due to the costs associated with process disruption. Our challenge as instrumentation professionals is to formulate proof tests that are as comprehensive as possible while being the least disruptive to the process we are trying to regulate.

Instrument self-diagnostics

One of the great advantages of digital electronic technology in industrial instrumentation is the inclusion of *self-diagnostic* ability in field instruments. A "smart" instrument containing its own microprocessor may be programmed to detect certain conditions known to indicate sensor failure or other problems, then signal the control system that something is wrong. Though self-diagnostics can never be perfectly effective in that there will inevitably be cases of undetected faults and even false positives (declarations of a fault where none exists), the current state of affairs is considerably better than the days of purely analog technology where instruments possessed little or no self-diagnostic capability.

Digital field instruments have the ability to communicate self-diagnostic error messages to their host systems over the same "fieldbus" networks they use to communicate regular process data. FOUNDATION Fieldbus instruments in particular have extensive error-reporting capability, including a "status" variable associated with every process signal that propagates down through all function blocks responsible for control of the process. Detected faults are efficiently communicated throughout the information chain in the system when instruments have full digital communication ability.

"Smart" instruments with self-diagnostic ability but limited to analog (e.g. 4-20 mA DC) signaling may also convey error information, just not as readily or as comprehensively as a fully digital instrument. The NAMUR recommendations for 4-20 mA signaling (NE-43) provide a means to do this:

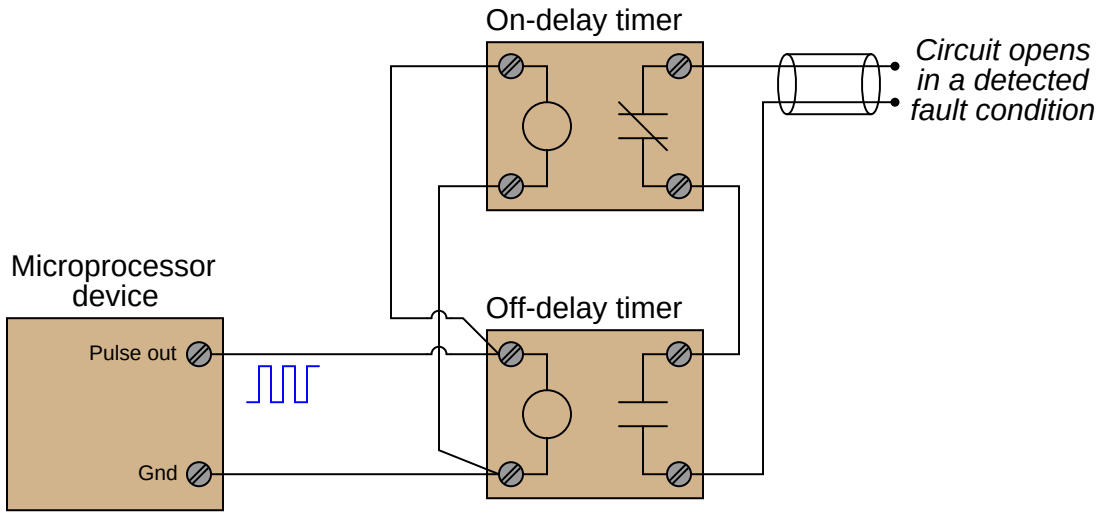
Signal level	**Fault condition**
Output $\leq$ 3.6 mA	Sensing transducer failed low
3.6 mA $<$ Output $<$ 3.8 mA	Sensing transducer failed (detected) low
3.8 mA $\leq$ Output $<$ 4.0 mA	Measurement under-range
21.0 $>$ Output $\geq$ 20.5 mA	Measurement over-range
Output $\geq$ 21.0 mA	Sensing transducer failed high

Proper interpretation of these special current ranges, of course, demands a receiver capable of accurate current measurement outside the standard 4-20 mA range. Many control systems with analog input capability are programmed to recognize the NAMUR error-indicating current levels.

A challenge for any self-diagnostic system is how to check for faults in the "brain" of the unit itself: the microprocessor. If a failure occurs within the microprocessor of a "smart" instrument – the very component responsible for performing logic functions related to self-diagnostic testing – how would it be able to detect a fault in logic? The question is somewhat philosophical, equivalent to determining whether or not a neurologist is able to diagnose his or her own neurological problems.

One simple method of detecting gross faults in a microprocessor system is known as a *watchdog timer*. The principle works like this: the microprocessor is programmed to output continuous a low-frequency pulse signal, with an external circuit "watching" that pulse signal for any interruptions or freezing. If the microprocessor fails in any significant way, the pulse signal will either skip pulses or "freeze" in either the high or low state, thus indicating a microprocessor failure to the "watchdog" circuit.

One may construct a watchdog timer circuit using a pair of solid-state timing relays connected to the pulse output channel of the microprocessor device:



Both the on-delay and off-delay timers receive the same pulse signal from the microprocessor, their inputs connected directly in parallel with the microprocessor's pulse output. The off-delay timer immediately actuates upon receiving a "high" signal, and begins to time when the pulse signal goes "low." The on-delay timer begins to time during a "high" signal, but immediately de-actuates whenever the pulse signal goes "low." So long as the time settings for the on-delay and off-delay timer relays are greater than the "high" and "low" durations of the watchdog pulse signal, respectively, neither relay contact will open as long as the pulse signal continues in its regular pattern.

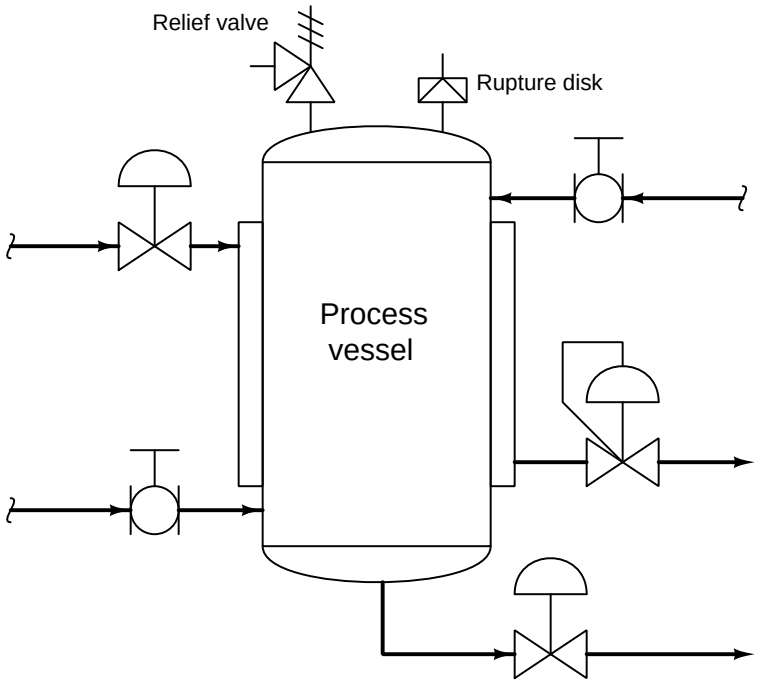
When the microprocessor is behaving normally, outputting a regular watchdog pulse signal, the off-delay timer's contact will hold in a closed state because it keeps getting energized with each "high" signal and never has enough time to drop out during each "low" signal. Likewise, the on-delay timer's contact will remain in its normally closed state because it never has enough time to pick up during each "high" signal before being de-actuated with each "low" signal. Both timing relay contacts will be in a closed state when all is well.

However, if the microprocessor's pulse output signal happens to freeze in the "low" state (or skip a "high" pulse), the off-delay timer will de-actuate, opening its contact and signaling a fault. Conversely, if the microprocessor's pulse signal happens to freeze in the "high" state (or skip a "low" pulse), the on-delay timer will actuate, opening its contact and signaling a fault. Either timing relay opening its contact signals an interruption or cessation of the watchdog pulse signal, indicating a serious microprocessor fault.

32.5 Overpressure protection devices

Fluid pressure exerts force on any surface area it contacts, as described by the formula $F = PA$. One practical consequence of this fact is that process vessels and pipelines may catastrophically burst if subjected to excessive fluid pressure. If subjected to excessive vacuum, some vessels and piping may implode (collapse in on themselves). Not only do these potential failures pose operational problems, but they may also pose severe safety and environmental hazards, especially if the process fluid in question is toxic, flammable, or both.

Special safety devices exist to help prevent such unfortunately events from occurring, among them being *rupture disks*, *relief valves*, and *safety valves*. The following subsections describe each of these protective devices and their intended operation. In a P&ID, rupture disks and relief valves are represented by the following symbols:



A rupture disk acts like an electrical fuse for overpressure protection: when the burst pressure is exceeded, the disk ruptures to let fluids escape through it. Safety and relief valves work like self-resetting circuit breakers: they open to relieve pressure, then re-close to seal the process system once more.

Two common causes of process overpressure are *piping blockages* and overheating caused by *fires*. Although it may sound ridiculous, a number of fatal industrial accidents have been caused by something as simple as shut block valves that should have been left open. When fluid cannot escape a process vessel, the pumping forces may exceed the burst rating of the vessel, causing catastrophic failure. Fires may also cause overpressure conditions, owing to the expansion of process fluids inside sealed vessels. Overpressure protection devices play a crucial role in such scenarios, venting process fluid so as to avoid bursting the vessel. It should be mentioned that these two causes of overpressure

may have vastly differing protection requirements: the required flow rate of exiting fluid to safely limit pressure may be far greater in a "fire case" than it is for a "blockage case," which means overpressure protection devices sized for the latter may be insufficient to protect against the former.

Overpressure protection device selection is a task restricted to the domain of process safety engineers. Instrument technicians may be involved in the installation and maintenance of overpressure protection devices, but only a qualified and licensed engineer should decide which specific device(s) to use for a particular process system.

32.5.1 Rupture disks

One of the simplest forms of overpressure protection for process lines and vessels is a device known as a *rupture disk*. This is nothing more than a thin sheet of material (usually alloy steel) designed to rupture in the event of an overpressure condition. The amount of force applied to this thin metal sheet is given by the formula $F = PA$ (force equals pressure times area). The thin metal sheet is designed to rupture at a certain threshold of force equivalent to the burst pressure ($P = \frac{F}{A}$). Once the disk ruptures, the fluid vents through new-formed path, thus relieving pressure. Like an electrical fuse, a rupture disk is a one-time device which must be replaced after it has "blown."

A photograph of a small rupture disk (prior to being placed in service) appears here:



The particular rupture disk shown here has a burst pressure of 30 PSI at a temperature of 130 ^{o}C. Temperature is an important factor in the rating of a rupture disk, as the physical strength of the thin metal rupture element changes with temperature. This metal disk is usually quite thin, usually in the order of 0.002 to 0.060 inches in thickness.

Some modern rupture disks use a *graphite* rupture element instead of metal. Not only does graphite exhibit better corrosion resistance to process fluids than metal, but it also does not fatigue in the same way that metal will over time. Burst pressure for a graphite rupture disk, therefore, may be more consistent than with a metal disk. A significant disadvantage of graphite rupture disks, however, is their tendency to shatter upon bursting. Metal rupture disks merely tear, but a graphite rupture disk tends to break into small pieces which are then carried away by the exiting

fluid. These graphite shards may exit as shrapnel, or even lodge inside the mechanism of a valve if one is installed downstream of the rupture disk.

32.5.2 Direct-actuated safety and relief valves

Pressure Relief Valves (PRVs) and *Pressure Safety Valves* (PSVs) are special types of valves designed to open up in order to relieve excess pressure from inside a process vessel or piping system. These valves are normally shut, opening only when sufficient fluid pressure develops across them to relieve that process fluid pressure and thereby protect the pipes and vessels upstream. Unlike regular control valves, PRVs and PSVs are actuated by the process fluid pressure itself rather than by some external pressure or force (e.g. pneumatic signal pressure, electrical motor or solenoid coil).

While the terms "Relief Valve" and "Safety Valve" are sometimes interchanged, there is a distinct difference in operation between them. A *relief valve* opens in direct proportion to the amount of overpressure it experiences in the process piping. That is, a PRV will open slightly for slight overpressures, and open more for greater overpressures. Pressure Relief Valves are commonly used in liquid services. By contrast, a *safety valve* opens fully with a "snap action" whenever it experiences a sufficient overpressure condition, not closing until the process fluid pressure falls significantly below that "lift" pressure value. In other words, a PSV's action is *hysteretic*[32]. Pressure Safety Valves are commonly used in gas and vapor services, such as compressed air systems and steam systems.

Safety valves typically have two pressure ratings: the pressure value required to initially open ("lift") the valve, and the pressure value required to reseat (close) the valve. The difference between these two pressure is called the *blowdown* pressure. A safety valve's lift pressure will always exceed its reseat pressure, giving the valve a hysteretic behavior.

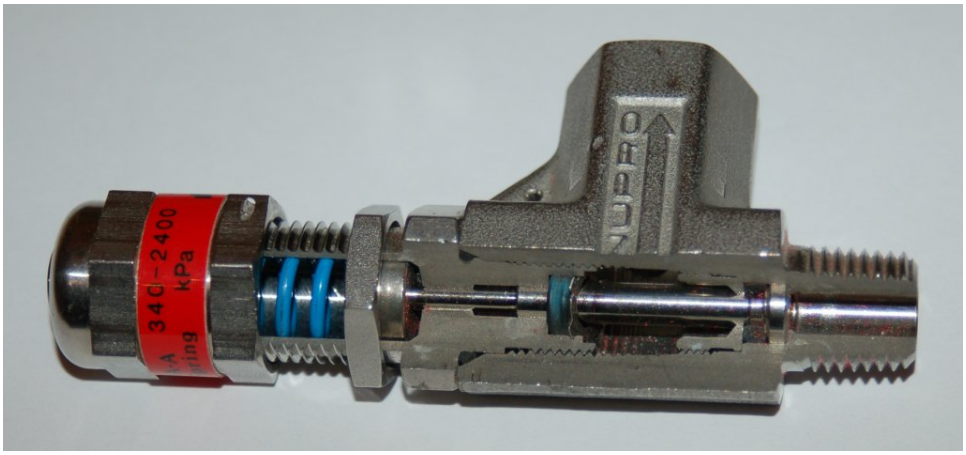
[32] A simple "memory trick" I use to correctly distinguish between relief and safety valves is to remember that a **s**afety valve has **s**nap action (both words beginning with the letter "s").

This photograph shows a Varec pressure relief valve on an industrial hot water system, designed to release pressure to atmosphere if necessary to prevent damage to process pipes and vessels in the system:

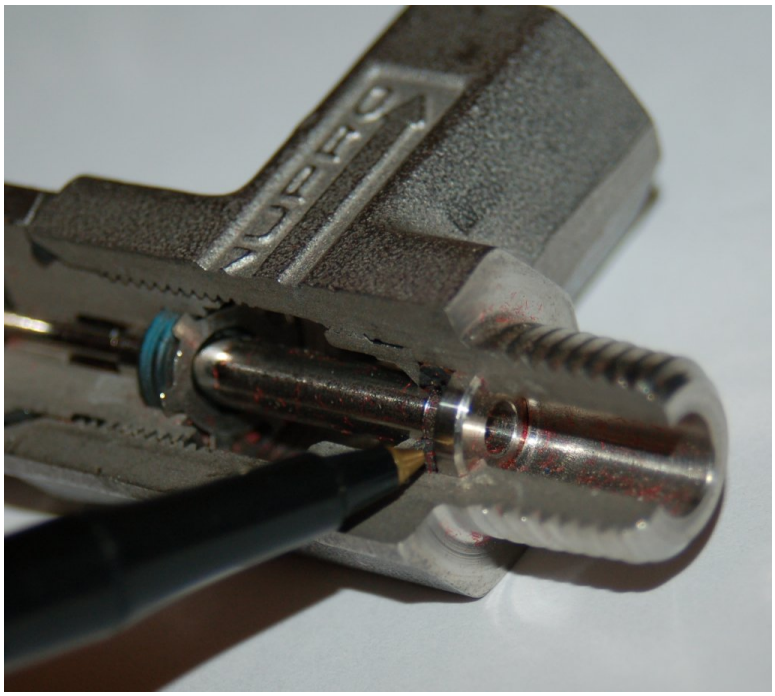


The vertical pipe is the atmospheric vent line, while the bottom flange of this PRV connects to the pressurized hot water line. A large spring inside the relief valve establishes the lift pressure.

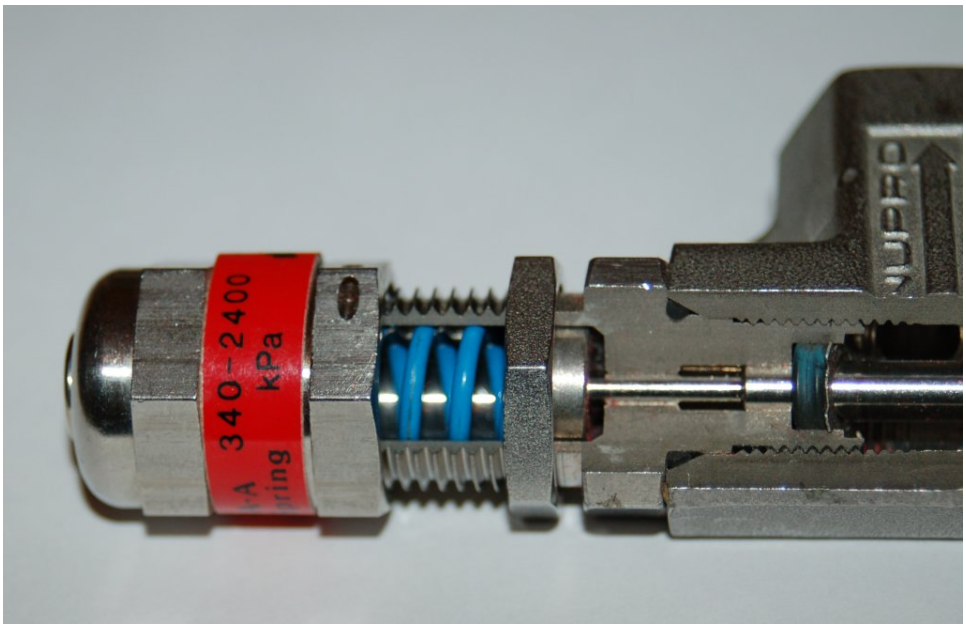
A miniature pressure relief valve manufactured by Nupro, cut away to show its internal components, appears in this next photograph. The pipe fittings on this valve are 1/4 inch NPT, to give a sense of scale:



A close-up photograph shows the plug and seat inside this PRV, pointed to by the tip of a ball-point pen:



A simple tension-adjusting mechanism on a spring establishes this valve's lift pressure. The spring exerts a force on the stem to the right, pressing the plug against the face of the seat. A knob allows manual adjustment of spring tension, relating directly to lift pressure:



The operation of this relief valve mechanism is quite simple: process fluid pressure entering the right-hand side fitting exerts force against the plug, which normally blocks passage of the fluid through to the side fitting. The area of the plug serves as a piston for the fluid pressure to push against, the amount of force predicted by the familiar force-pressure-area formula $F = PA$. If the fluid pressure exerts enough force on the plug's end to lift it off the seat against the restraining force of the spring (on the left-hand side of the valve mechanism), the plug lifts and vents fluid pressure through the side port.

It is worthy to note that most relief valve mechanisms work on the exact same principle of actuation: *the valve's plug serves as its own actuator.* The pressure difference across this plug provides all the motive force necessary to actuate the valve. This simplicity translates to a high degree of reliability, a desirable quality in any safety-related system component.

Another style of overpressure valve appears in this next photograph. Manufactured by the Groth corporation, this is a combination pressure/vacuum safety valve assembly for an underground tank, designed to vent excess pressure to atmosphere *or* introduce air to the tank in the event of excess vacuum forming inside:



Even when buried, the threat of damage to the tank from overpressure is quite real. The extremely large surface area of the tank's interior walls represents an incredible amount of force potential even with low gas pressures[33]. By limiting the amount of differential gas pressure which may exist between the inside and outside of the tank, the amount of stress applied to the tank walls by gas pressure or vacuum is correspondingly limited.

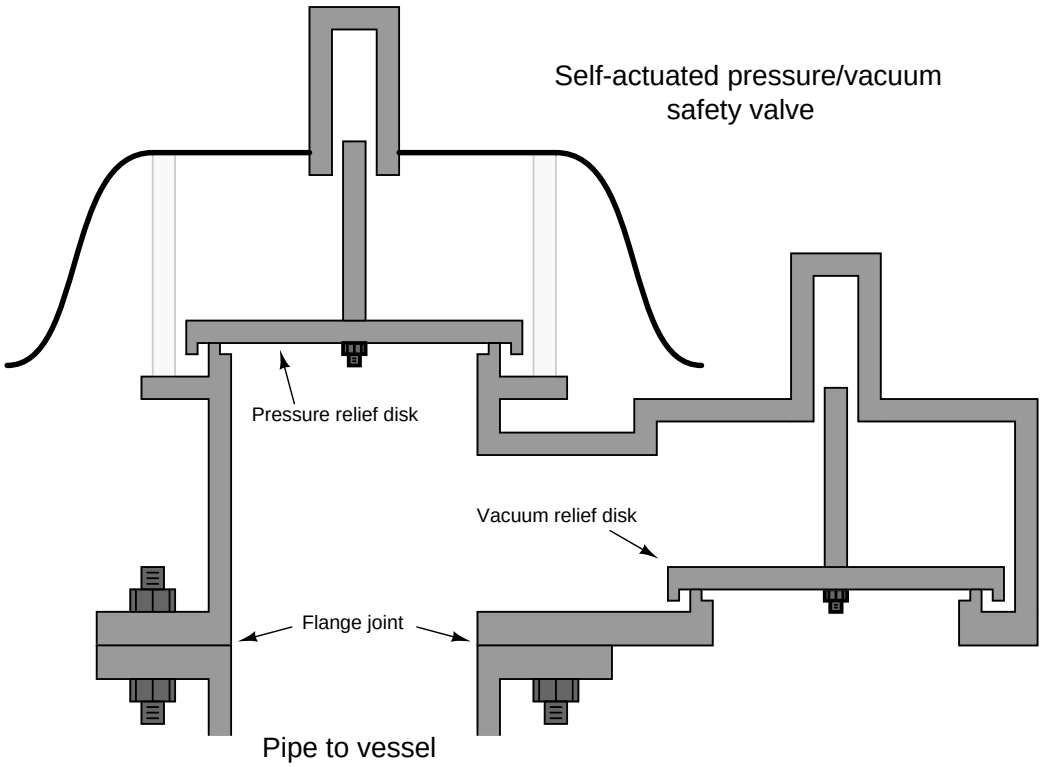
Large storage tanks – whether above-ground or subterranean – are typically thin-wall for reasons of economics, and cannot withstand significant pressures or vacuums. An improperly vented storage tank may burst with only slight pressure inside, or collapse inwardly with only a slight vacuum inside. Combination pressure/vacuum safety valves such as this Groth model 1208 unit reduce the chances of either failure from happening.

Of course, an alternative solution to this problem is to continuously vent the tank with an open vent pipe at the top. If the tank is always vented to atmosphere, it cannot build up either a pressure or a vacuum inside. However, continuous venting means vapors could escape from the tank if the liquid stored inside is volatile. Escaping vapors may constitute product loss and/or negative environmental impact, being a form of *fugitive emission*. In such cases it is prudent to vent the tank

[33] To illustrate, consider a (vertical) cylindrical storage tank 15 feet tall and 20 feet in diameter, with an internal gas pressure of 8 inches water column. The total force exerted radially on the walls of this tank from this very modest internal pressure would be in excess of *39000 pounds!* The force exerted by the same pressure on the tank's circular lid would exceed *13000 pounds* (6.5 tons)!

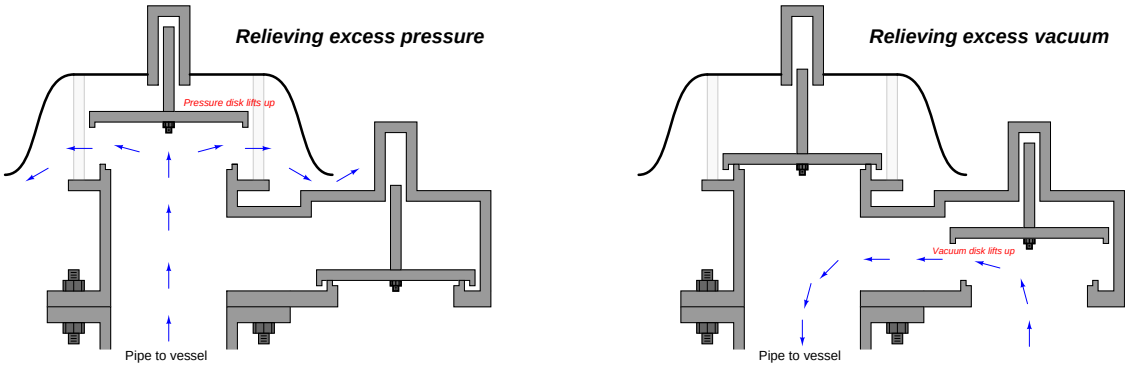
with an automatic valve such as this only when needed to prevent pressure-induced stress on the tank walls.

An illustration shows the interior construction of this safety valve:



Like the miniature Nupro relief valve previously shown, the trim of this Groth safety valve acts as its own actuator: process gas pressure directly forces the vent plug off its seat, while process gas vacuum forces the vacuum plug off its seat. The lift pressure and vacuum ratings of the Groth valve are quite low, and so no spring is used to provide restraining force to the plugs. Rather, the *weight* of the plugs themselves holds them down on their seats against the force of the process gas.

This set of illustrations shows a pressure/vacuum safety valve in both modes of operation:



In each mode, the respective disk lifts up against the force of its own weight to allow gases to flow through the valve. If a greater lift pressure (or lift *vacuum*) rating is desired, precise weights may be fixed to the top of either disk. Greater weights equate to greater pressures, following the familiar equation $P = \frac{F}{A}$, where F is the force of gravity acting on the disk and weight(s) and A is the area of the disk.

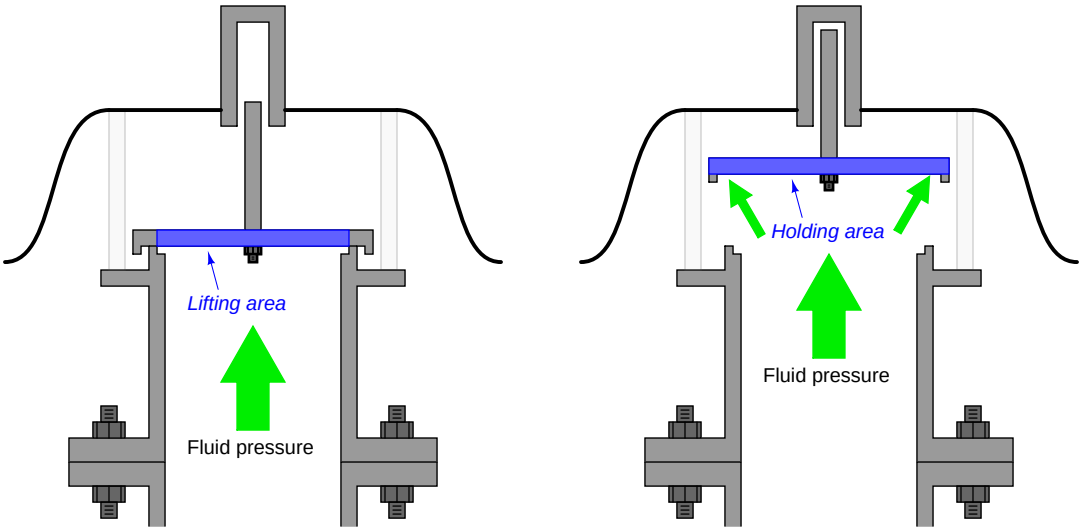
For example, suppose the disk in one of these safety valves weighs 8 pounds and has a diameter of 9 inches. The surface area for a circular disk nine inches in diameter is 63.62 square inches ($A = \pi r^2$), making the lift pressure equal to 0.126 PSI ($P = \frac{F}{A}$). Such low pressures are often expressed in units other than PSI in order to make the numbers more manageable. The lift pressure of 0.126 PSI for this safety valve might alternatively be described as 3.48 inches water column or 0.867 kPa.

A close inspection of this valve design also provides clues as to why it is technically a *safety* valve rather than a *relief* valve. Recall that the distinction between these two types of overpressure-protection valves was that a relief valve opens proportionally to the experienced overpressure, while a safety valve behaves in a "snap" action manner[34], opening at the lift pressure and not closing again until a (lower) re-seating pressure is achieved.

The "secret" to achieving this snap-action behavior characteristic of safety valves is to design the valve's plug in such a way that it presents a larger surface area for the escaping process fluid to act upon once open than it does when closed. This way, less pressure is needed to hold the valve open than to initially lift it from a closed condition.

[34] Think: a safety valve has snap action!

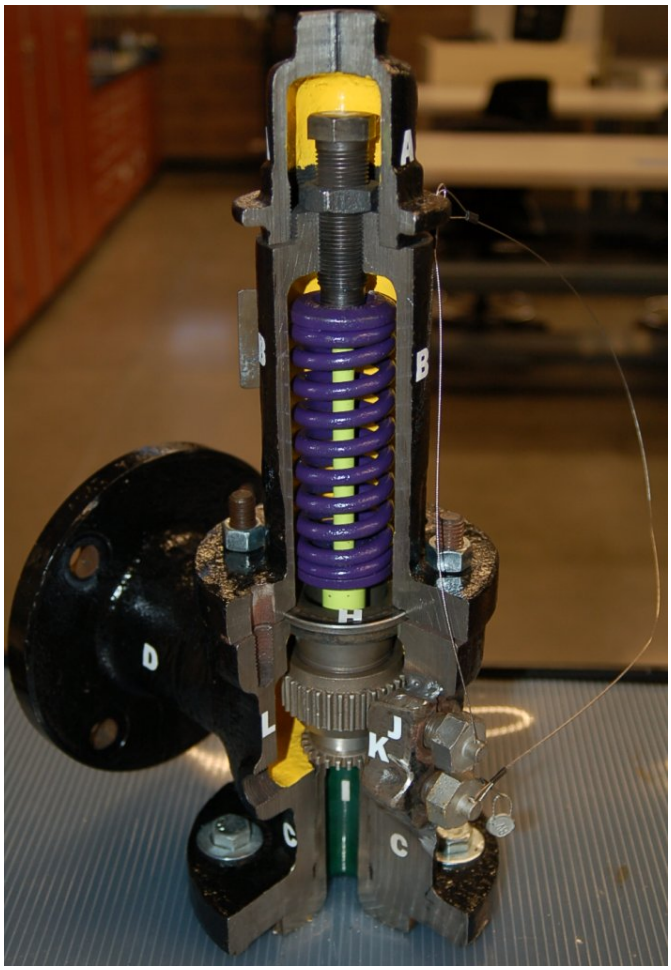
Examining the pressure-relief mechanism of the Groth valve design closer, we see how the plug's diameter exceeds that of the seating area, with a "lip" extending down. This wide plug, combined with the lip forms an effective surface area when the plug is lifted that is larger than that exposed to the process pressure when the plug is seated. Thus, the process fluid finds it "easier" to hold the plug open than to initially lift it off the seat. This translates into a reseating pressure that is less than the lift pressure, and a corresponding "snap action" when the valve initially lifts off the seat.



The extra area on the plug's lower surface enclosed by the lip (i.e. the holding area minus the lifting area) is sometimes referred to as a *huddling chamber*. The size of this "huddling chamber" and the length of the lip establishes the degree of hysteresis (blowdown) in the safety valve's behavior.

A certain class of overpressure valve called a *safety relief valve* is designed with an adjustable "blowdown ring" to allow variations in the huddling chamber's geometry. Essentially, the blowdown ring acts as an inner lip on the valve seat to complement the outer lip on the plug. Adjusting this inner lip farther away from the plug allows more process fluid to escape laterally without touching the plug, thereby minimizing the effect of the huddling chamber and making the valve behave as a simple relief valve with no snap-action. Adjusting the blowdown ring closer to the plug forces the escaping fluid to travel toward the plug's face before reversing direction past the outer lip, making the huddling chamber more effective and therefore providing snap-action behavior. This adjustability allows the safety relief valve to act as a simple relief valve (i.e. opening proportional to overpressure) or as a safety valve (snap action) with varying amounts of blowdown ($P_{blowdown} = P_{lift} - P_{reseat}$) as determined by the user. This blowdown ring's position is typically locked into place with a seal to discourage tampering once the valve is installed in the process.

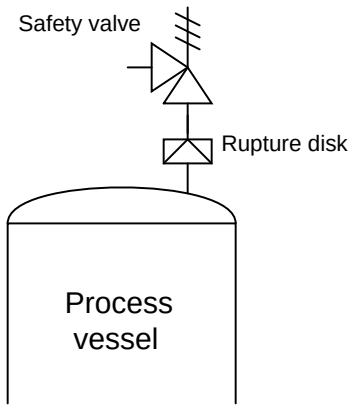
This next photograph shows a cutaway of a safety relief valve manufactured by Crosby, mounted on a cart for instructional use at Bellingham Technical College:



The adjusting bolt marked by the letter "A" at the top of the valve determines the lift pressure setting, by adjusting the amount of pre-load on the spring. Like the Nupro and Groth valves shown previously, the Crosby valve's plug serves as its own actuator, the actuating force being a function of differential pressure across the valve and plug/seat area ($F = PA$).

The toothed gear-like component directly left of the letter "J" is called a *guide ring*, and it functions as a blowdown adjustment. This ring forms a "lip" around the valve seat's edge much like the lip shown in the Groth valve diagrams. If the guide ring is turned to set it at a lower position (extending further past the seat), the volume of the huddling chamber increases, thereby increasing the blowdown value (i.e. keeping the valve open longer than it would be otherwise as the pressure falls).

An interesting combination of overpressure-protection technologies sometimes seen in industry are rupture disks combined with safety valves. Placing a rupture disk before a safety valve provides the benefits of ensuring zero leakage during normal operation as well as isolating the safety valve from potentially corrosive effects of the process fluid:

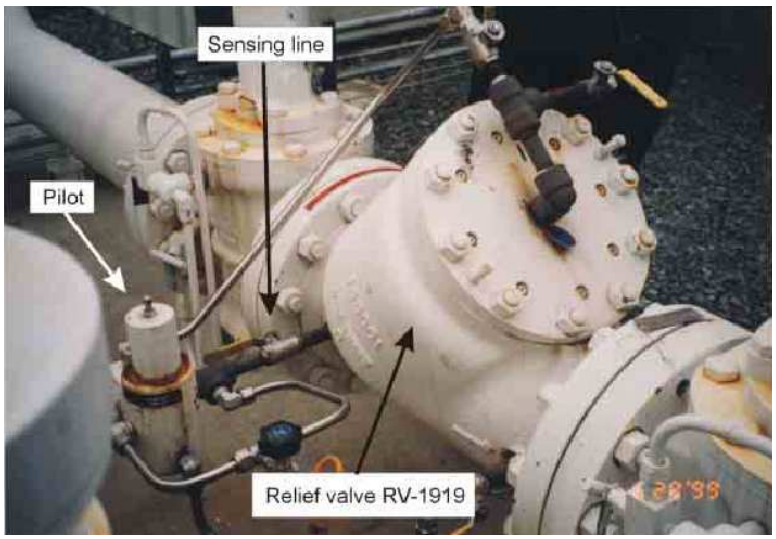


Potential problems with this strategy include the possibility of accumulating vapor pressure between the rupture disk and the safety valve (thereby increasing the effective burst pressure of the disk), and also the possibility of rupture disk shards becoming lodged in the safety valve mechanism, restricting flow and/or preventing re-closure.

32.5.3 Pilot-operated safety and relief valves

While many safety and relief valves actuate by the direct action of the process fluid forcing against the valve plug mechanism, others are more sophisticated in design, relying on a secondary pressure-sensing mechanism to trigger and direct fluid pressure to the main valve assembly to actuate it. This pressure-sensing mechanism is called a *pilot*, and usually features a widely-adjustable range to give the overall valve assembly a larger variety of applications.

In a pilot-operated overpressure-protection valve, the "lift" pressure value is established by a spring adjustment in the pilot mechanism rather than by an adjustment made to the main valve mechanism. A photograph[35] of a pilot-operated pressure relief valve used on a liquid petroleum pipeline appears here:



The relief valve mechanism itself is the white-painted flanged valve found in the center-right region of the photograph (RV-1919). This particular relief valve happens to be a Fisher model 760 with 8-inch, ANSI 300# flanges. The actuating pilot mechanism is the small unit connected to the relief valve body via stainless-steel tubing. When this pilot senses fluid pressure in the pipeline exceeding the lift pressure, it switches fluid pressure to the piston actuating mechanism of the main relief valve, opening it to relieve fluid pressure from the pipeline. Thus, the lift pressure value for the relief valve is set within the pilot rather than within the main valve mechanism. Altering this lift pressure setting is a matter of adjusting spring tension within the pilot mechanism, and/or replacing components within the pilot mechanism.

[35] This photograph courtesy of the National Transportation Safety Board's report of the 1999 petroleum pipeline rupture in Bellingham, Washington. Improper setting of this relief valve pilot played a role in the pipeline rupture, the result of which was nearly a quarter-million gallons of gasoline spilling into a creek and subsequently igniting. One of the lessons to take from this event is the importance of proper instrument maintenance and configuration, and how such technical details concerning industrial components may have consequences reaching far beyond the industrial facility where those components are located.

32.6 Safety Instrumented Functions and Systems

A *Safety Instrumented Function*, or *SIF*, is one or more components designed to execute a specific safety-related task in the event of a specific dangerous condition. The over-temperature shutdown switch inside a clothes dryer or an electric water heater is a simple, domestic example of an SIF, shutting off the source of energy to the appliance in the event of a detected over-temperature condition. Safety Instrumented Functions are alternatively referred to as *Instrument Protective Functions*, or *IPF*s.

A *Safety Instrumented System*, or *SIS*, is a collection of SIFs designed to bring an industrial process to a safe condition in the event of any dangerous detected conditions. Also known as *Emergency Shutdown* (ESD) or *Protective Instrument Systems* (PIS), these systems serve as an additional "layer" of protection against process equipment damage, adverse environmental impact, and/or human injury beyond the protection normally offered by a properly operating regulatory control system. Like all automatic control systems, an SIS consists of three basic sections: (1) Sensor(s) to detect a dangerous condition, (2) Controller to decide when to shut down the process, and (3) Final control element(s) to actually perform the shutdown action necessary to bring the process to a safe condition. Sensors may consist of process switches and/or transmitters separate from the regulatory control system. The controller for an SIS is usually called a *logic solver*, and is also separate from the regular control system. The final control elements for an SIS may be special on/off valves (often called "chopper" valves) or override solenoids used to force the normal control valve into a shutdown state.

Some industries, such as chemical processing and nuclear power, have extensively employed safety instrumented systems for many decades. Likewise, automatic shutdown controls have been standard on steam boilers and combustion furnaces for years. The increasing capability of modern instrumentation, coupled with the realization of enormous costs (both social and fiscal) resulting from industrial disasters has pushed safety instrumentation to new levels of sophistication and new breadths of application. It is the purpose of this section to explore some common safety instrumented system concepts as well as some specific industrial applications.

One of the challenges inherent to safety instrumented system design is to balance the goal of maximum safety against the goal of maximum economy. If an industrial manufacturing facility is equipped with enough sensors and layered safety shutdown systems to virtually ensure no unsafe condition will ever prevail, that same facility will be plagued by "false alarm" and "spurious trip" events[36] where the safety systems malfunction in a manner detrimental to the profitable operation of the facility. In other words, a process system designed with an emphasis on automatic shut-down will probably shut down more frequently than it actually needs to. While the avoidance of unsafe process conditions is obviously a noble goal, it cannot come at the expense of economically practical operation or else there will be no reason for the facility to exist at all[37]. A safety system must fulfill

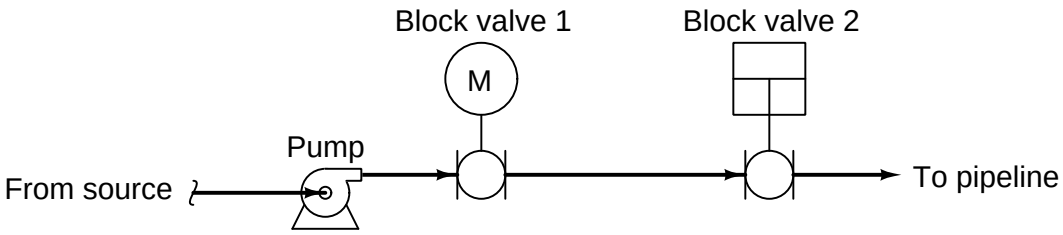
[36] Many synonyms exist to describe the action of a safety system needlessly shutting down a process. The term "nuisance trip" is often (aptly) used to describe such events. Another (more charitable) label is "fail-to-safe," meaning the failure brings the process to a safe condition, as opposed to a dangerous condition.

[37] Of course, there do exist industrial facilities operating at a financial loss for the greater public benefit (e.g. certain waste processing operations), but these are the exception rather than the rule. It is obviously the point of a *business* to turn a profit, and so the vast majority of industries simply cannot sustain a philosophy of safety at *any* cost. One could argue that a "paranoid" safety system even at a waste processing plant is unsustainable, because too many "false trips" result in inefficient processing of the waste, posing a greater public health threat the longer it remains unprocessed.

its intended protective function, but not at the expense of compromising the intended purpose of the facility.

This tension is understood well within the electric power generation and distribution industries. Faults in high-voltage electrical lines can be very dangerous, as well as destructive to electrical equipment. For this reason, special protective devices are placed within power systems to monitor conditions and halt the flow of electricity if those conditions become threatening. However, the very presence of these devices means it is possible for power to accidently shut off, causing unnecessary power outages for customers. In the electrical industry, the word "dependability" refers to the probability that the protective systems will cut power when required. By contrast, the word "security" is used in the electrical industry to refer to the avoidance of unnecessary outages. We will apply these terms to general process systems.

To illustrate the tension between dependability and security in a fluid process system, we may analyze a double-block shutoff valve[38] system for a petroleum pipeline:



The safety function of these block valves is, of course, to shut off flow from the petroleum source to the distribution pipeline in the event that the pipeline suffers a leak or rupture. Having two block valves in "series" adds an additional layer of safety, in that only one of the block valves need shut to fulfill the safety (dependability) function. Note the use of two different valve actuator technologies: one electric (motor) and the other a piston (either pneumatic or hydraulically actuated). This diversity of actuator technologies helps avoid common-cause failures, helping to ensure both valves will not simultaneously fail due to a single cause.

However, the typical operation of the pipeline demands both block valves be open in order for petroleum to flow through it. The presence of redundant (dual) block valves, while increasing safety, decreases security for the pipeline. If *either* of the two block valves happened to fail shut when there was no need to shut off the pipeline, flow through the pipeline would needlessly halt. Having two series-plumbed block valves instead of one block valve increases the probability of unnecessary pipeline shutdowns.

[38] As drawn, these valves happen to be ball-design, the first actuated by an electric motor and the second actuated by a pneumatic piston. As is often the case with redundant instruments, an effort is made to diversify the technology applied to the redundant elements in order to minimize the probability of common-cause failures. If both block valves were electrically actuated, a failure of the electric power supply would disable both valves. If both block valves were pneumatically actuated, a failure of the compressed air supply would disable both valves. The use of one electric valve and one pneumatic valve grants greater independence of operation to the double-block valve system.

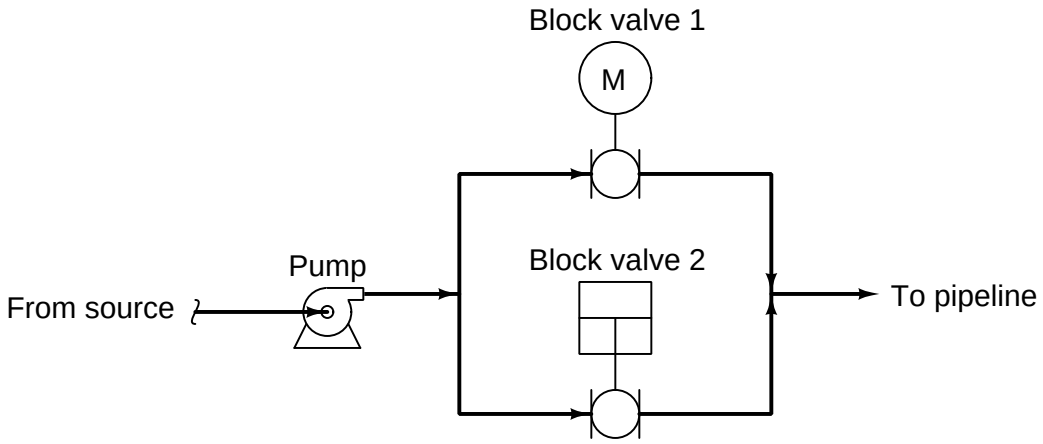
A precise notation useful for specifying dependability and security in redundant systems compares the number of redundant elements necessary to achieve the desired result compared to the total number of redundant elements. If the desired result for our double-block valve array is to shut down the pipeline in the event of a detected leak or rupture, we would say the system is *one out of two* (1oo2) redundant for dependability. In other words, only one out of the two redundant valves needs to function properly (shut off) in order to bring the pipeline to a safe condition. If the desired result is to allow flow through the pipeline when the pipeline is leak-free, we would say the system is *two out of two* (2oo2) redundant for security. This means *both* of the two block valves need to function properly (open up) in order to allow petroleum to flow through the pipeline.

This numerical notation showing the number of essential elements versus number of total elements is often referred to as *MooN* ("M out of N") notation, or sometimes as *NooM* ("N out of M") notation[39]. When discussing safety instrumented systems, the ISA standard 84 defines redundancy in terms of the number of agreeing channels necessary to perform the safety (shutdown) function – in other words, the ISA's usage of "MooN" notation implies dependability, rather than security.

A complementary method of quantifying dependability and security for redundant systems is to label in terms of how many element failures the system may sustain while still achieving the desired result. For this series set of double block valves, the safety (shutdown) function has a *fault tolerance* of one (1), since one of the valves may fail to shut when called upon but the other valve remains sufficient in itself to shut off the flow of petroleum to the pipeline. The normal operation of the system, however, has a fault tolerance of zero (0). Both block valves must open up when called upon in order to establish flow through the pipeline.

[39] For what it's worth, the ISA safety standard 84 defines this notation as "MooN," but I have seen sufficient examples of the contrary ("NooM") to question the authority of either label.

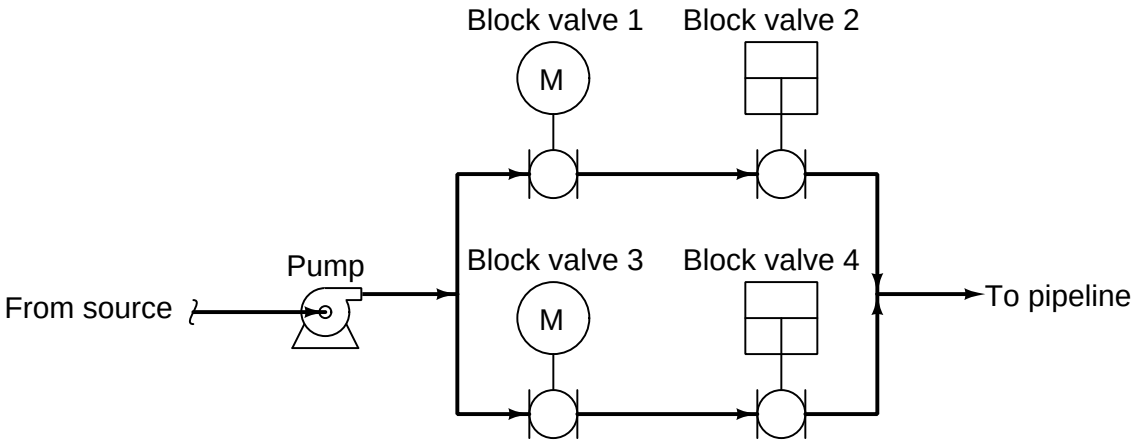
It should be clearly evident that a series set of block valves emphasizes dependability (the ability to shut off flow through the pipeline when needed) at the expense of security (the ability to allow normal flow through the pipeline when there is no leak). We may now analyze a parallel block valve scheme to compare its redundant characteristics:



In this system, the safety (dependability) redundancy function is 2oo2, since *both* block valves would have to shut off in order to bring the pipeline to a safe condition in the event of a detected pipeline leak. However, security would be 1oo2, since only one of the two valves would have to open up in order to establish flow through the pipeline. Thus, a parallel block valve array emphasizes production (the ability to allow flow through the pipeline) over safety (the ability to shut off flow through the pipeline).

Another way to express the redundant behavior of the parallel block valve array is to say that the safety function has a fault tolerance of zero (0), while the production function has a fault tolerance of one (1).

One way to avoid compromises between dependability and security is to increase the number of redundant components, forming arrays of greater complexity. Consider this quadruple block valve array, designed to serve the same function on a petroleum pipeline:



In order to fulfill its safety function of shutting off the flow of petroleum to the pipeline, both parallel pipe "branches" must be shut off. At first, this might seem to indicate a two-out-of-four (2oo4) dependability, because all we would need is for one valve in each branch (two valves total) out of the four valves to shut off in order to shut off flow to the pipeline. We must remember, however, that we do not have the luxury of assuming idealized faults. If only two of the four valves function properly in shutting off, they just might happen to be two valves *in the same branch*, in which case two valves properly functioning is not enough to guarantee a safe pipeline condition. Thus, this redundant system actually exhibits *three*-out-of-four (3oo4) dependability (i.e. it has a safety fault tolerance of one), because we need three out of the four block valves to properly shut off in order to *guarantee* a safe pipeline condition.

Analyzing this quadruple block valve array for security, we see that three out of the four valves need to function properly (open up) in order to guarantee flow to the pipeline. Once again, it may appear at first as though all we need are two of the four valves to open up in order to establish flow to the pipeline, but this will not be enough if those two valves happen to be in different parallel branches. So, this system exhibits three-out-of-four (3oo4) security (i.e. it has an production fault tolerance of one).

32.6.1 SIS sensors

Perhaps the simplest form of sensor providing process information for a safety instrumented function is a *process switch*. Examples of process switches include temperature switches, pressure switches, level switches, and flow switches[40]. SIS sensors must be properly calibrated and configured to indicate the presence of a dangerous condition. They must be separate and distinct from the sensors used for regulatory control, in order to ensure a level of safety protection beyond that of the basic process control system.

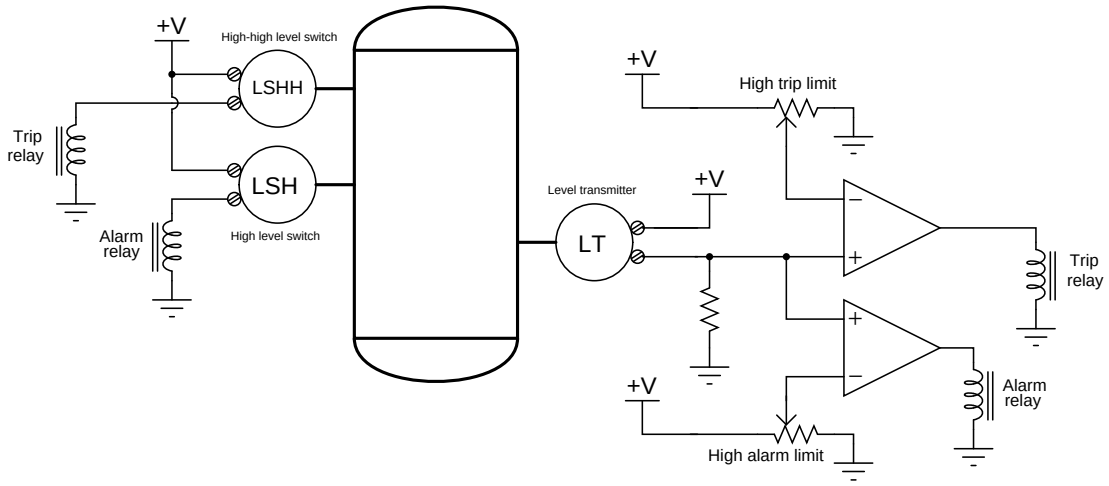
Referring to the clothes dryer and domestic water heater over-temperature shutdown switches, these high-temperature shutdown sensors are distinctly separate from the regulatory (temperature-controlling) sensors used to maintain the appliance's temperature at setpoint. As such, they should only ever spring into action in the event of a high-temperature *failure* of the basic control system. That is, the over-temperature safety switch on a clothes dryer or a water heater should only ever reach its high-temperature limit if the normal temperature control system of the appliance fails to do its job of regulating temperature to normal levels.

Industrial Safety Instrumented Systems (SIS) always use dedicated transmitters and/or process switches to detect abnormal process conditions. As a rule, one should always use independent sensors for safety shutdown, and never rely on the regulatory control sensor(s) for safety functions. In the electric power industry we see this same segregation of functions: separate instrument transformers (PTs and CTs) are used to sense line voltage and line current for metering and control (regulatory) versus for protective relay (safety shutdown) equipment. It would be foolish to depend on one sensor for both functions. We see this general rule applied even in home appliances such as electric water heaters: the safety shutdown temperature switch is a separate component from the thermostat switch used to regulate water temperature. This way, a failure in the regulatory sensor does not compromise the integrity of the safety function.

A modern trend in safety instrumented systems is to use continuous process transmitters rather than discrete process switches to detect dangerous process conditions. Any process transmitter – analog or digital – may be used as a safety shutdown sensor if its signal is compared against a "trip" limit value by a comparator relay or function block. This comparator function provides an on-or-off (discrete) output based on the transmitter's signal value relative to the trip point.

[40] For a general introduction to process switches, refer to chapter 9 beginning on page 655.

A simplified example of a continuous transmitter used as a discrete alarm and trip device is shown here, where analog comparators generate discrete "trip" and "alarm" signals based on the measured value of liquid in a vessel. Note the necessity of *two* level switches on the other side of the vessel to perform the same dual alarm and trip functions:



Benefits to using a continuous transmitter instead of discrete switches include the ability to easily change the alarm or trip value, and better diagnostic capability. The latter point is not as obvious as the former, and deserves more explanation. A transmitter continuously measuring liquid level will produce an output signal that varies over time with the measured process variable. A "healthy" transmitter should therefore exhibit a continuously changing output signal, proportional to the degree of change in the process. Discrete process switches, in contrast to transmitters, provide no indication of "healthy" operation. The only time a process switch should ever change states is when its trip limit is reached, which in the case of a safety shutdown sensor indicates a dangerous (rare) condition. A process switch showing a "normal" process variable may indeed be functional and indicating properly, but it might also be failed and incapable of registering a dangerous condition should one arise – there is no way to tell by monitoring its un-changing status. The continuously varying output of a process transmitter therefore serves as an indicator[41] of proper function.

[41] Of course, the presence of some variation in a transmitter's output over time is no guarantee of proper operation. Some failures may cause a transmitter to output a randomly "walking" signal when in fact it is not registering the process at all. However, being able to measure the continuous output of a process transmitter provides the instrument technician with far more data than is available with a discrete process switch. A safety transmitter's output signal may be correlated against the output signal of another transmitter measuring the same process variable, perhaps even the transmitter used in the regulatory control loop. If two transmitters measuring the same process variable agree closely with one another over time, chances are extremely good are both functioning properly.

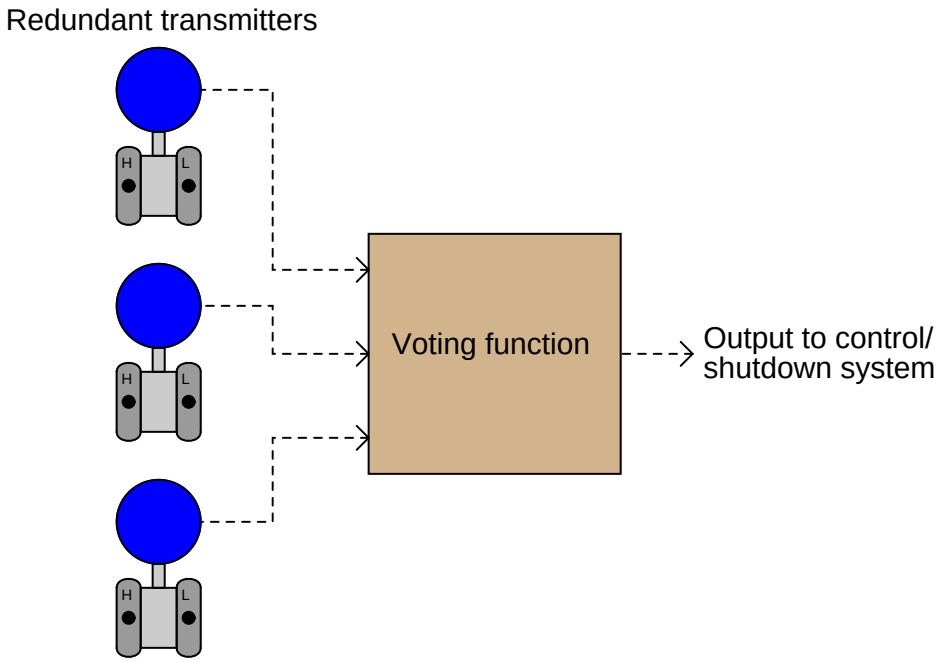
In applications where Safety Instrumented Function (SIF) reliability is paramount, *redundant* transmitters may be installed to yield additional reliability. The following photograph shows triple-redundant transmitters measuring liquid flow by sensing differential pressure dropped across an orifice plate:



A single orifice plate develops the pressure drop, with the three differential pressure transmitters "tubed" in parallel with each other, all the "high" side ports connected together through common[42] impulse tubing and all the "low" side ports connected together through common impulse tubing. These particular transmitters happen to be FOUNDATION Fieldbus rather than 4-20 mA analog electronic. The yellow instrument tray cable (ITC) used to connect each transmitter to a segment coupling device may be clearly seen in this photograph.

[42] It should be noted that the use of a single orifice plate and of common (parallel-connected) impulse lines represents a point of common-cause failure. A blockage at one or more of the orifice plate ports, or a closure of a manual block valve, would disable all three transmitters. As such, this might not be the best method of achieving high flow-measurement reliability.

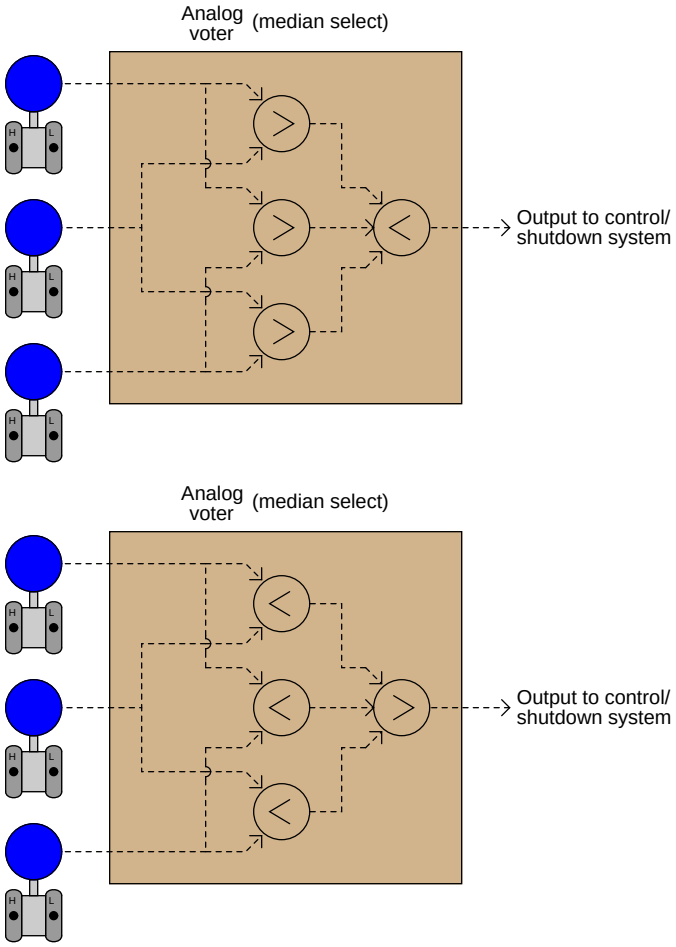
The "trick" to using redundant transmitters is to have the system self-determine what the actual process value is in the event one or more of the redundant transmitters disagree with each other. *Voting* is the name given to this important function, and it often takes the form of signal selector functions:



Multiple selection criteria are typically offered by "voting" modules, including *high*, *low*, *average*, and *median*. A "high" select voter would be suitable for applications where the dangerous condition is a large measured value, the voting module selecting the highest-valued transmitter signal in an effort to err on the side of safety. This would represent a 1oo3 safety redundancy (since only one transmitter out of the three would have to register beyond the high trip level in order to initiate the shutdown). A "low" select voter would, of course, be suitable for any application where the dangerous condition is a small measured value (once again providing a 1oo3 safety redundancy).

The "average" selection function merely calculates and outputs the mathematical average of all transmitter signals – a strategy prone to problems if one of the redundant transmitters happens to fail in the "safe" direction (thus skewing the average value away from the "dangerous" direction and thereby possibly causing the system to respond to an actual dangerous condition later than it should).

The *median select* criterion is very useful in safety systems because it effectively ignores any measurements deviating substantially from the others. Median selector functions may be constructed of high- and low-select function blocks in either of the following[43] manners:



Three transmitters filtered through a median select function effectively provide a 2oo3 safety redundancy, since just a single transmitter registering a value beyond the safety trip point would be ignored by the voting function. *Two* or more transmitters would have to register values past the trip point in order to initiate a shutdown.

It should be stressed that redundant transmitter strategies are only effective if the transmitters all sense the exact same process variable, and if their failure modes are independent (i.e. no common-cause failure modes exist). If, for example, a set of redundant transmitters are attached to the

[43] The best way to prove to yourself the median-selecting abilities of both function block networks is to perform a series of "thought experiments" where you declare three arbitrary transmitter signal values, then follow through the selection functions until you reach the output. For any three signal values you might choose, the result should always be the same: the *median* signal value is the one chosen by the voter.

process at different points such that they may legitimately sense different measurement values, the effectiveness of their redundancy will be compromised. Similarly, if a set of redundant transmitters are susceptible to failure from a shared condition (e.g. multiple liquid level transmitters that may be fooled by changes in process fluid density), then reliability will suffer.

32.6.2 SIS controllers (logic solvers)

Control hardware for safety instrumented functions should be separate from the control hardware used to regulate the process, if only for the simple reason that the SIF exists to bring the process to a safe state in the event of any unsafe condition arising, including dangerous failure of the basic regulatory controls. If a single piece of control hardware served the dual purposes of regulation *and* shutdown, a failure within that hardware resulting in loss of regulation (normal control) would not be protected because the safety function would be disabled by the same fault.

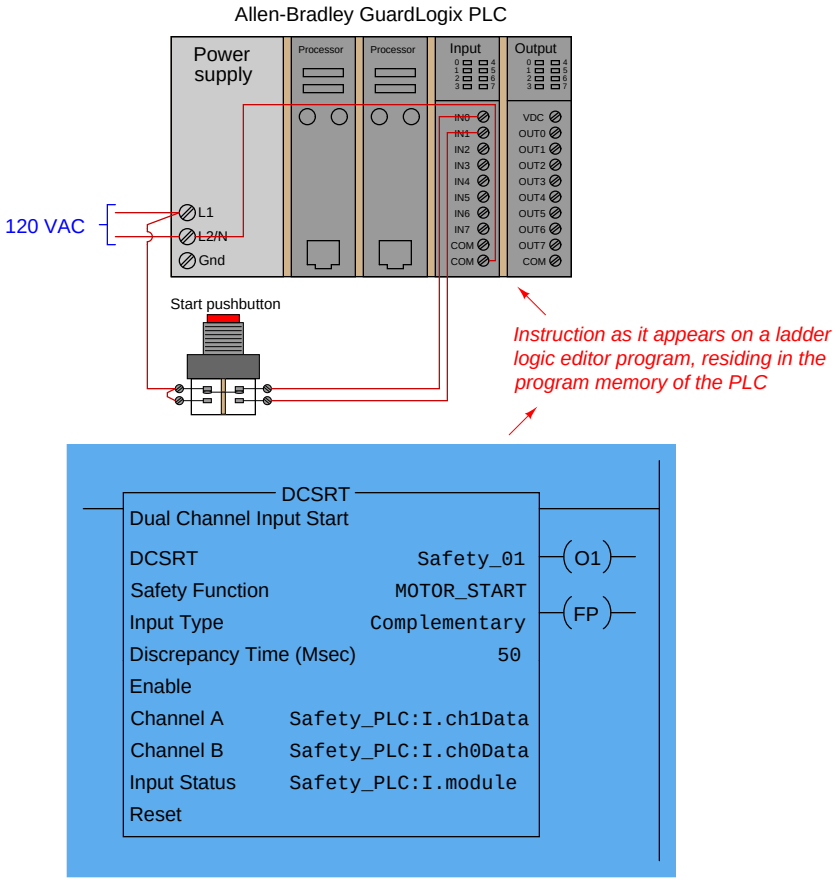
Safety controls are usually discrete with regard to their output signals. When a process needs to be shut down for safety reasons, the steps to implement the shutdown often take the form of opening and closing certain valves fully rather than partially. This sort of all-or-nothing control action is most easily implemented in the form of discrete signals triggering solenoid valves or electric motor actuators. A digital controller specially designed for and tasked with the execution of safety instrumented functions is usually called a *logic solver*, or sometimes a *safety PLC*, in recognition of this discrete-output nature.

A photograph of a "safety PLC" used as an SIS in an oil refinery processing unit is shown here, the controller being a Siemens "Quadlog" model:



Some logic solvers such as the Siemens Quadlog are adaptations of standard control systems (in the case of the Quadlog, its standard counterpart is called APACS). In the United States, where Rockwell's Allen-Bradley line of programmable logic controllers holds the dominant share of the PLC market, a version of the ControlLogix 5000 series called *GuardLogix* is manufactured specifically for safety system applications. Not only are there differences in hardware between standard and safety controllers (e.g. redundant processors), but some of the programming instructions are unique to these safety-oriented controllers as well.

An example of a safety-specific programming instruction is the GuardLogix `DCSRT` instruction, which compares two redundant input channels for agreement before activating a "start" bit which may be used to start some equipment function such as an electric motor:



In this case, the `DCSRT` instruction looks for two discrete inputs to be in the correct complementary states (Channel A = 1 and Channel B = 0) before allowing a motor to start. These states must not conflict for a time-span longer than 50 milliseconds, or else the `DCSRT` instruction will set a "Fault Present" (FP) bit. As you can see, the form-C pushbutton contacts are wired to two discrete inputs on the GuardLogix PLC, giving the PLC dual (complementary) indication of the switch status.

For specialized and highly critical applications, dedicated safety controllers exist which share no

legacy with standard control platforms. Triconex and ICS-Triplex are two such manufacturers, producing *triple-modular redundant* (TMR) control systems implementing 2oo3 voting at the hardware level, with redundant signal conditioning I/O circuits, redundant processors, and redundant communication channels between all components. The nuclear power industry boasts a wide array of application-specific digital control systems, with triple (or greater!) component redundancy for extreme reliability. An example of this is Toshiba's TOSMAP system for boiling-water nuclear power reactors, the digital controller and electro-hydraulic steam turbine valve actuator subsystem having a stated MTBF[44] of over 1000 years!

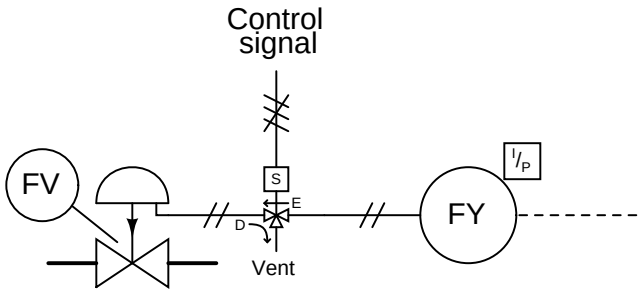
32.6.3 SIS final control elements

When a dangerous condition in a volatile process is sensed by process transmitters (or process switches), triggering a shutdown response from the logic solver, the final control elements must move with decisive and swift action. Such positive response may be obtained from a standard regulatory control valve (such as a globe-type throttling valve), but for more critical applications a rotary ball or plug valve may be more suitable. If the valve in question is used for safety shutdown purposes only and not regulation, it is often referred to as a *chopper* valve for its ability to "chop" (shut off quickly and securely) the process fluid flow. A more formal term for this is an *Emergency Isolation Valve*, or *EIV*.

Some process applications may tolerate the over-loading of both control and safety functions in a single valve, using the valve to regulate fluid flow during normal operation and fully stroke (either open or closed depending on the application) during a shutdown condition. A common method of achieving this dual functionality is to install a solenoid valve in-line with the actuating air pressure line, such that the valve's normal pneumatic signal may be interrupted at any moment, immediately driving the valve to a fail-safe position at the command of a discrete "trip" signal.

[44] *MTBF* stands for *Mean Time Between Failure*, and represents the reliability of a large collection of components or systems. For any large batch of identical components or systems constantly subjected to ordinary stresses, MTBF is the theoretical length of time it will take for 63.2% of them to fail based on ordinary failure rates within the lifetime of those components or systems. Thus, MTBF may be thought of as the "time constant" (τ) for failure within a batch of identical components or systems.

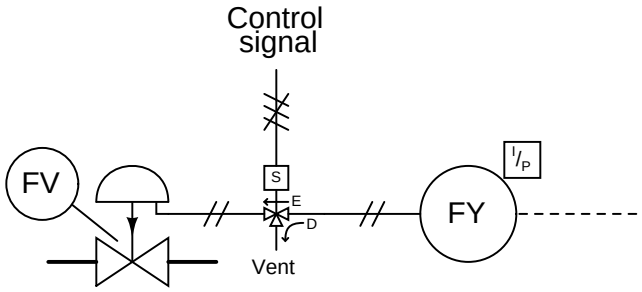
Such a "trip" solenoid (sometimes referred to as a *dump* solenoid, because it "dumps" all air pressure stored in the actuating mechanism) is shown here, connected to a fail-closed (air-to-open) control valve:



Compressed air passes through the solenoid valve from the I/P transducer to the valve's pneumatic diaphragm actuator when energized, the letter "E" and arrow showing this path in the diagram. When de-energized, the solenoid valve blocks air pressure coming from the I/P and vents all air pressure from the valve's actuating diaphragm as shown by the letter "D" and arrow. Venting all actuating air pressure from a fail-closed valve will cause the valve to fail closed, obviously.

If we wished to have the valve fail open on demand, we could use the exact same solenoid and instrument air plumbing, but swap the fail-closed control valve for a fail-open control valve. When energized (regular operation), the solenoid would pass variable air pressure from the I/P transducer to the valve actuator so it could serve its regulating purpose. When de-energized, the solenoid would force the valve to the fully-open position by "dumping" all air pressure from the actuator.

For applications where it is safer to lock the control valve in its last position than to have it fail either fully closed or fully open, we might elect to use a solenoid valve in a different manner:



Here, de-energization of the solenoid valve causes the I/P transducer's air pressure output to vent, while trapping and holding all air pressure inside the actuator at the trip time. Regardless of the valve's "natural" fail-safe state, this system forces the valve to lock position[45] until the solenoid is re-energized.

[45] This is assuming, of course, that there are no air leaks anywhere in the actuator, tubing, or solenoid which would cause the trapped pressure to decrease over time.

An example of a trip solenoid installed on a control valve appears in the following photograph. This valve also happens to have a *hand jack* wheel installed in the actuating mechanism, allowing a human operator to manually override the valve position by forcing it closed (or open) when the hand wheel is turned sufficiently:



Of all the components of a Safety Instrumented System (SIS), the final control elements (valves) are generally the least reliable, contributing most towards the system's probability of failure on demand (PFD). Sensors generally come in at second place in their contribution toward unreliability, and logic solvers a distant third place. Redundancy may be applied to control elements by creating valve networks where the failure of a single valve does not cause the system as a whole to fail. Unfortunately, this approach is extremely expensive, as valves have both high capital and high maintenance costs compared to SIS sensors and logic solvers.

A less expensive approach than redundancy to increasing safety valve reliability is to perform regular proof tests of their operation. This is commonly referred to in the industry as *partial stroke testing*. Rather than proof-test each safety valve to its full travel, which would interrupt normal process operations, the valve is commanded to move only part of its full travel. If the valve responds well to this "partial stroke" test, there is a high probability that it is able to move all the way, thus fulfilling the basic requirements of a proof test without actually shutting the process down[46].

[46] Of course, if there is opportunity to fully stroke the safety valve to the point of process shutdown without undue interruption to production, this is the superior way of performing valve proof tests. Such "test-to-shutdown" proof testing may be scheduled at a time convenient to operations personnel, such as at the beginning of a planned process shutdown.

32.6.4 Safety Integrity Levels

A common way of ranking the dependability of a Safety Instrumented Function (SIF) is to use a simple numerical scale from one to four, with four being extremely dependable and one being only moderately dependable:

SIL number	**Required Safety Availability** (RSA)	**Probability of Failure on Demand** (PFD)
1	90% to 99%	0.1 to 0.01
2	99% to 99.9%	0.01 to 0.001
3	99.9% to 99.99%	0.001 to 0.0001
4	99.99% to 99.999%	0.0001 to 0.00001

The Required Safety Availability (RSA) value is synonymous with *dependability*: the probability[47] that a Safety Instrumented Function will perform its duty when faced with a dangerous process condition. Conversely, the Probability of Failure on Demand (PFD) is synonymous with *undependability*: the mathematical complement of RSA (PFD = 1 − RSA), expressing the probability that the SIF will fail to perform as needed, when needed.

Conveniently, the SIL number matches the minimum number of "nines" in the Required Safety Availability (RSA) value. For instance, a safety instrumented function with a Probability of Failure on Demand (PFD) of 0.00073, will have an RSA value of 99.927%, which equates to a SIL 3 rating.

It is important to understand what SIL is, and what SIL is not. The SIL rating refers to the reliability of a safety *function*, not to individual components of a system nor to the entire process itself. An overpressure protection system on a chemical reactor process with a SIL rating of 2, for example, has a Probability of Failure on Demand between 0.01 and 0.001 *for the specific shutdown function as a whole*. This PFD value incorporates failure probabilities of the sensor(s), logic solver, final control element(s), and the process piping including the reactor vessel itself plus any relief valves and other auxiliary equipment. If there arises a need to improve the PFD of this reactor's overpressure protection, safety engineers have a variety of options at their disposal for doing so. The safety instruments themselves might be upgraded, a different redundancy strategy implemented, preventive maintenance schedules increased in frequency, or even process equipment changed to make an overpressure event less likely.

SIL ratings do not apply to an entire process. It is quite possible that the chemical reactor mentioned in the previous paragraph with an overpressure protection system SIL rating of 3 might have an over*temperature* protection system SIL rating of only 2, due to differences in how the two different safety systems function.

Adding to this confusion is the fact that many instrument manufacturers rate their products as approved for use in certain SIL-rated applications. It is easy to misunderstand these claims, thinking that a safety instrumented function will be rated at some SIL value simply because instruments rated for that SIL value are used to implement it. In reality, the SIL value of any safety function is a much more complex determination. It is possible, for instance, to purchase and install a pressure transmitter rated for use in SIL 2 applications, and have the safety function as a whole be less than

[47] *Probability* is a quantitative measure of a particular outcome's likelihood. A probability value of 1, or 100%, means the outcome in question is certain to happen. A probability value of 0 (0%) means the outcome is impossible. A probability value of 0.3 (30%) means it will happen an average of three times out of ten.

99% reliable (PFD greater than 0.01, or a SIL level no greater than 1) due to the effect of *Lusser's Law*[48].

As with so many other complex calculations in instrumentation engineering, there exist software packages with all the necessary formulae pre-programmed for engineers and technicians alike to use for calculating SIL ratings of safety instrumented functions. These software tools not only factor in the inherent reliability ratings of different system components, but also correct for preventive maintenance schedules and proof testing intervals so the user may determine the proper maintenance attention required to achieve a given SIL rating.

[48]Lusser's Law of Reliability states that the total reliability of a system dependent on the function of several independent components is the mathematical product of those components' individual reliabilities. For example, a system with three essential components, each of those components having an individual reliability value of 70%, will exhibit a reliability of only 34.3% because $0.7 \times 0.7 \times 0.7 = 0.343$. This is why a safety function may utilize a pressure transmitter rated for use in SIL-3 applications, but exhibit a much lower total SIL rating due to the use of an ordinary final control element.

32.6.5 SIS example: burner management systems

One "classic" example of an industrial automatic shutdown system is a *Burner Management System* (or *BMS*) designed to monitor the operation of a combustion burner and shut off the fuel supply in the event of a dangerous condition. Sometimes referred to as *flame safety systems*, these systems watch for such potentially dangerous conditions as *low fuel pressure*, *high fuel pressure*, and *loss of flame*. Other dangerous conditions related to the process being heated (such as *low water level* for a steam boiler) may be included as additional trip conditions.

The safety shutdown action of a burner management system is to halt the flow of fuel to the burner in the event of any hazardous detected condition. The final control element is therefore one or more shutoff valves (and sometimes a vent valve in addition) to positively stop fuel flow to the burner.

A typical ultraviolet flame sensor appears in this photograph:



This flame sensor is sensitive to ultraviolet light only, not to visible or infrared light. The reason for this specific sensitivity is to ensure the sensor will not be "fooled" by the visible or infrared glow of hot surfaces inside the firebox if ever the flame goes out unexpectedly. Since ultraviolet light is emitted *only* by an active gas-fueled flame, the sensor acts as a true flame detector, and not a heat detector.

One of the more popular models of fuel gas safety shutoff valve used in the United States for burner management systems is shown here, manufactured by Maxon:



This particular model of shutoff valve has a viewing window on it where a metal tag linked to the valve mechanism marked "Open" (in red) or "Shut" (in black) positively indicates the valve's mechanical status. Like most safety shutoff valves on burner systems, this valve is electrically actuated, and will automatically close by spring tension in the event of a power loss.

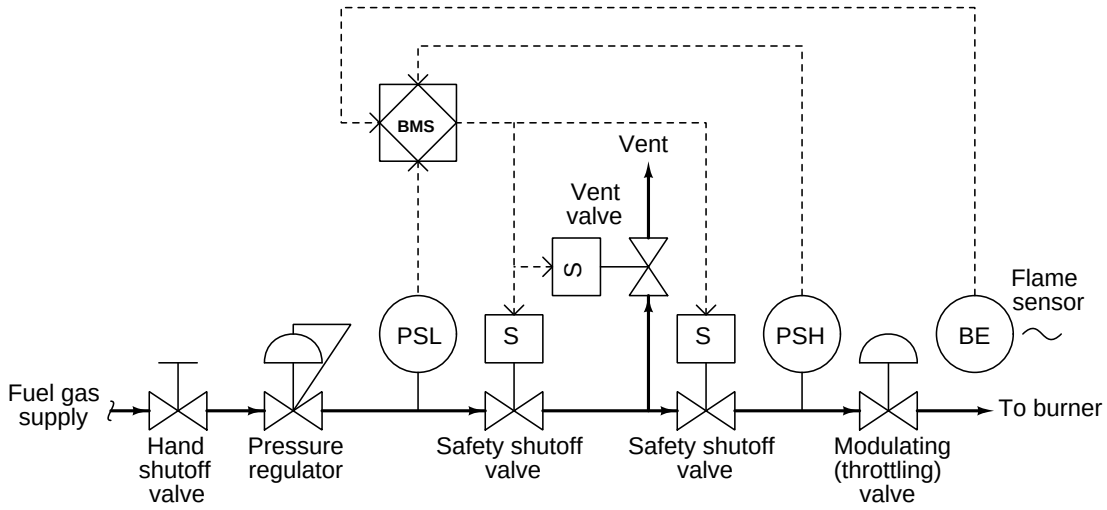
Another safety shutoff valve, this one manufactured by ITT, is shown here:



Close inspection of the nameplate on this ITT safety valve reveals several important details. Like the Maxon safety valve, it is electrically actuated, with a "holding" current indicated as 0.14 amps at 120 volts AC. Inside the valve is an "auxiliary" switch designed to actuate when the valve has mechanically reached the full "open" position. An additional switch, labeled *valve seal overtravel interlock*, indicates when the valve has securely reached the full "shut" position. This "valve seal" switch generates a *proof of closure* signal used in burner management systems to verify a safe shutdown condition of the fuel line. Both switches are rated to carry 15 amps of current at 120

VAC, which is important when designing the electrical details of the system to ensure the switch will not be tasked with too much current.

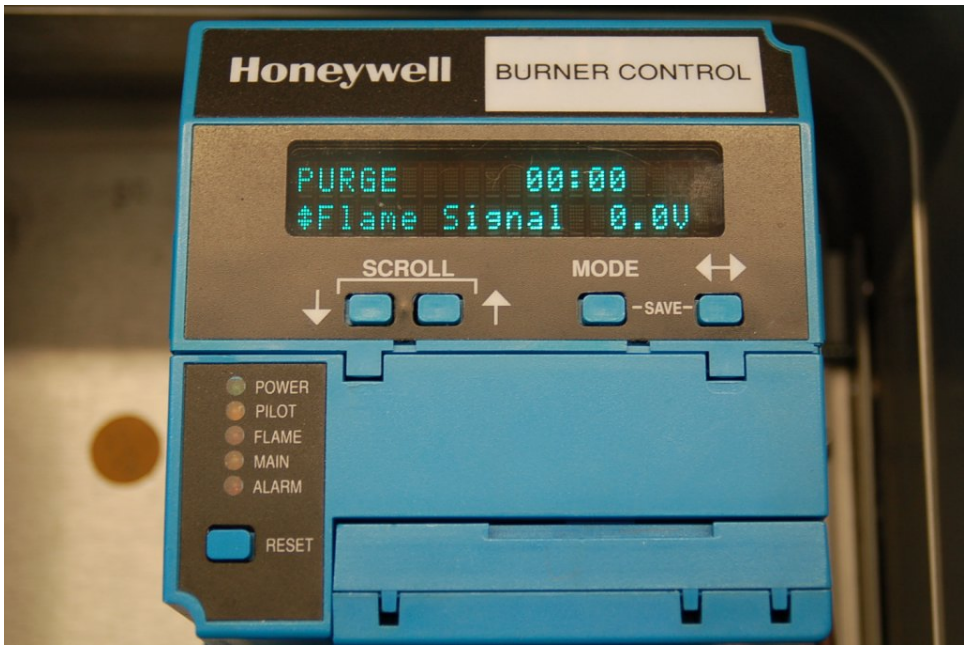
A simple P&ID for a gas-fired combustion burner system is shown here. The piping and valving shown is typical for a single burner. Multiple-burner systems are often equipped with individual shutoff valve manifolds and individual fuel pressure limit switches. Each burner, if multiple exist in the same furnace, *must* be equipped with its own flame sensor:



Note the use of double-block and bleed shutdown valves to positively isolate the fuel gas supply from the burner in the event of an emergency shutdown. The two block valves are specially designed for the purpose (such as the Maxon and ITT safety valves previously shown), while the bleed valve is often nothing more than an ordinary electric solenoid valve.

Most burner management systems are charged with a dual role: both to manage the safe shutdown of a burner in the event of a hazardous condition, *and* the safe start-up of a burner in normal conditions. Start-up of a large industrial burner system usually includes a lengthy *purge time* prior to ignition where the combustion air damper is left wide-open and the blower running for several minutes to positively purge the firebox of any residual fuel vapors. After the purge time, the burner management system will ignite the burner (or sometimes ignite a smaller burner called the *pilot*, which in turn will light the main burner). A burner management system executes all these pre-ignition and timing functions to ensure the burners will ignite safely and without incident.

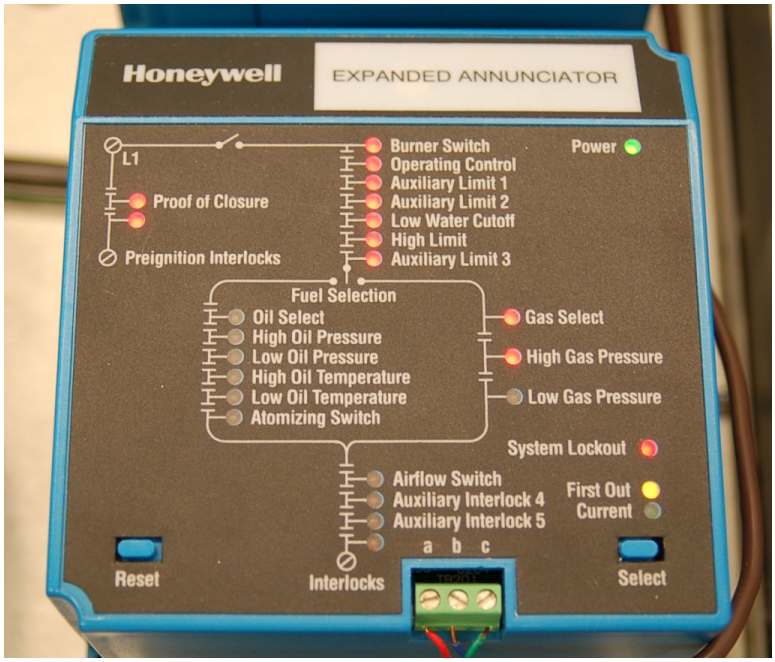
While many industrial burners are managed by electromechanical relay or analog electronic control systems, the modern trend is toward microprocessor-based digital electronic controls. One popular system is the Honeywell 7800 series burner control system, an example of which is shown in this photograph:



Microprocessor controls provide numerous advantages over relay-based and analog electronic burner management systems. Timing of purge cycles is far more accurate with microprocessor control, and the requisite purge time is more difficult to override[49]. Microprocessor-based burner controls usually have digital networking capability as well, allowing the connection of multiple controls to a single computer for remote monitoring.

[49] Yes, maintenance and operations personnel alike are often tempted to bypass the purge time of a burner management system out of impatience and a desire to resume production. I have personally witnessed this in action, performed by an electrician with a screwdriver and a "jumper" wire, overriding the timing function of a flame safety system during a troubleshooting exercise simply to get the job done faster. The electrician's rationale was that since the burner system was having problems lighting, and had been repeatedly purged in prior attempts, the purge cycle did not have to be full-length in subsequent attempts. I asked him if he would feel comfortable repeating those same words in court as part of the investigation of why the furnace exploded. He didn't think this was funny.

The Honeywell 7800 series additionally offers local "annunciator" modules to visually indicate the status of permissive (interlock) contacts, showing maintenance personnel which switches are closed and what state the burner control system is in:



The entire "gas train" piping system for a dual-fuel boiler at a wastewater treatment facility appears in the following photograph. Note the use of double-block and bleed valves on both "trains" (one for utility-supplied natural gas and the other for "sludge gas" produced by the facility's anaerobic digesters), the block valves for each train happening to be of different manufacture. A Honeywell 7800 flame safety control system is located in the blue enclosure:

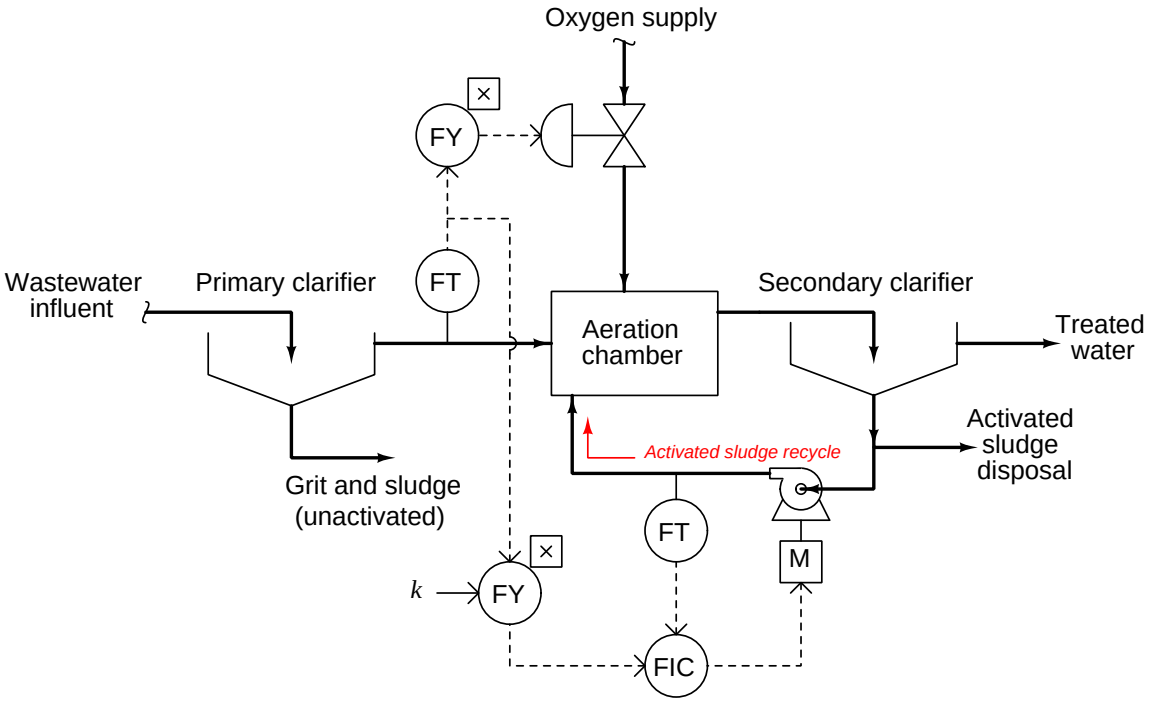


32.6.6 SIS example: water treatment oxygen purge system

One of the processes of municipal wastewater treatment is the aerobic digestion of organic matter by bacteria. This process emulates one of many waste-decomposition processes in nature, performed on an accelerated time frame for the needs of large wastewater volumes in cities. The process consists of supplying naturally occurring bacteria within the wastewater with enough oxygen to metabolize the organic waste matter, which to the bacteria is food. In some treatment facilities, this aeration is performed with ambient air. In other facilities, it is performed with nearly pure oxygen.

Aerobic decomposition is usually part of a larger process called *activated sludge*, whereby the effluent from the decomposition process is separated into solids (sludge) and liquid (supernatant), with a large fraction of the sludge recycled back to the aerobic chamber to sustain a healthy culture of bacteria and also ensure adequate retention time for decomposition to occur. Separating liquids from solids and recycling the solids ensures a short retention time for the liquid (allowing high processing rates) and a long retention time for the solids (ensuring thorough digestion of organic matter by the bacteria).

A simplified P&ID of an activated sludge water treatment system is shown here, showing how both the oxygen flow into the aeration chamber and the sludge recycle flow back to the aeration chamber are controlled as a function of influent wastewater flow:

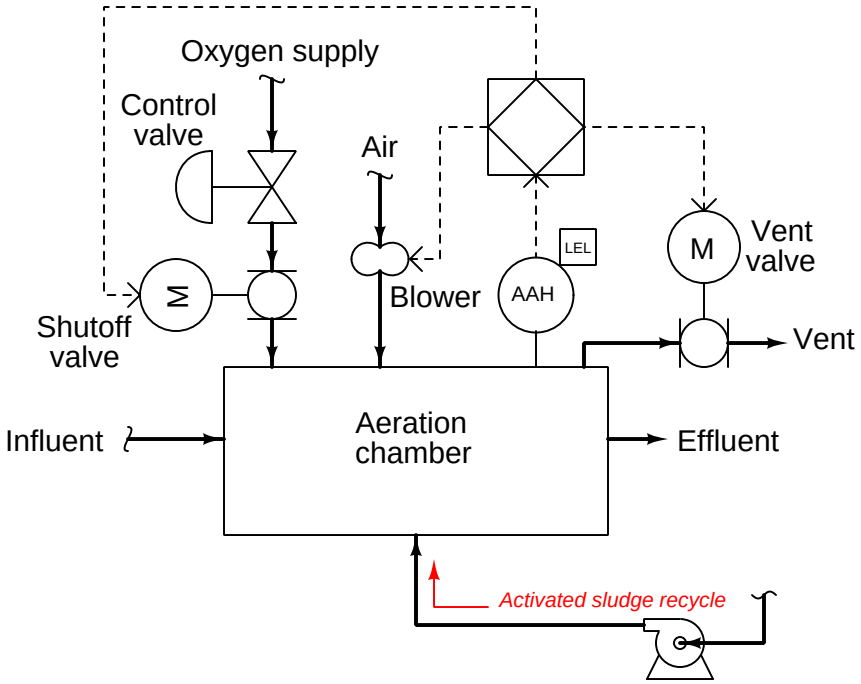


Aerobic decomposition performed with ambient air as the oxidizer is a very simple and safe process. Pure oxygen may be chosen instead of ambient air because it accelerates the metabolism of the bacteria, allowing more processing flow capacity in less physical space. For the same reason that pure oxygen accelerates bacterial metabolism, it also accelerates combustion of any flammable substances. This means if ever a flammable vapor or liquid were to enter the aeration chamber, there would be a risk of explosion.

Although flammable liquids are not a normal component of municipal wastewater, it is possible for flammable liquids to find their way to the wastewater treatment plant. One possibility is the event of a fuel carrier vehicle spilling its cargo, with gasoline or some other volatile fuel draining into a sewer system tunnel through holes in a grate. Such an occurrence is not normal, but certainly possible. Furthermore, it may occur without warning for the operations personnel to take preemptive action at the wastewater treatment plant.

To decrease this safety hazard, *Low Explosive Limit* (LEL) sensors installed on the aeration chamber detect and signal the presence of flammable gases or vapors inside the chamber. If any of the sensors register the presence of flammable substances, a safety shutdown system purges the chamber of pure oxygen by taking the following steps:

- Stop the flow of pure oxygen into the aeration chamber
- Open large vent valves to atmosphere
- Start air blowers to purge the chamber of residual pure oxygen



As with the P&ID, this diagram is a simplified representation of the real safety shutdown system. In a real system, multiple analytical high-alarm (LEL) sensors work to detect the presence of flammable gases or vapors, and the oxygen block valve arrangement would most likely be a double block and bleed rather than a single block valve.

The following photograph shows an LEL sensor mounted inside an insulated enclosure for protection from cold weather conditions at a wastewater treatment facility:



In this photograph, we see a purge air blower used to sweep the aeration chamber of pure oxygen (replacing it with ambient air) during an emergency shutdown condition:

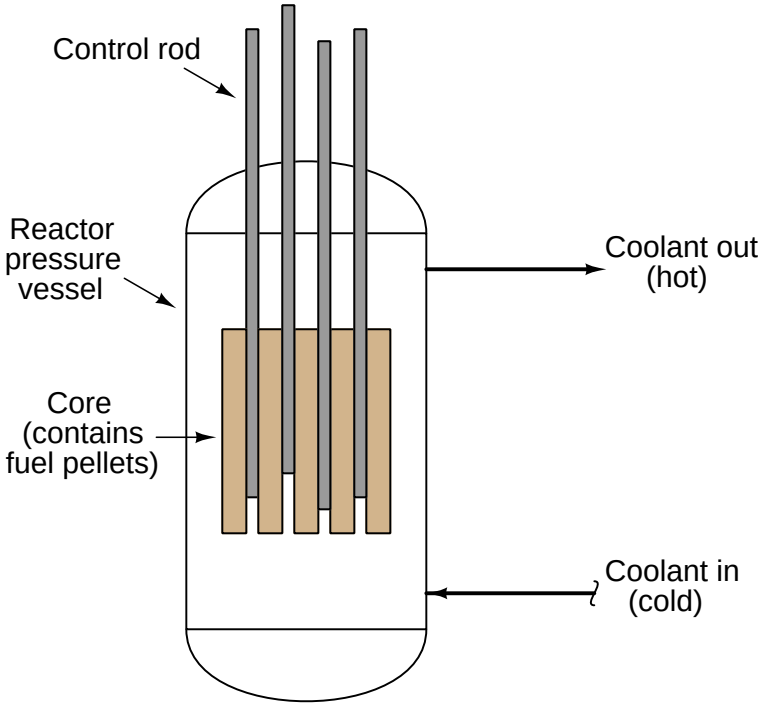


Since this is a centrifugal blower, providing no seal against air flow through it when stopped, an automatic purge valve located downstream (not to be confused with the manually-actuated vent valve seen in this photograph) is installed to block off the blower from the oxygen-filled chamber. This purge valve remains shut during normal operation, and opens only after the blower has started to initiate a purge.

32.6.7 SIS example: nuclear reactor scram controls

Nuclear fission is a process by which the nuclei of specific types of atoms (most notably uranium-235 and plutonium-239) undergo spontaneous disintegration upon the absorption of an extra neutron, with the release of significant thermal energy and additional neutrons. A quantity of fissile material subjected to a source of neutron particle radiation will begin to fission, releasing massive quantities of heat which may then be used to boil water into steam and drive steam turbine engines to generate electricity. The "chain reaction" of neutrons splitting fissile atoms, which then eject more neutrons to split more fissile atoms, is inherently exponential in nature, but may be regulated by natural and artificial feedback loops.

A simplified diagram of a pressurized[50] water reactor (PWR) appears here:



[50] Boiling-water reactors (BWR), the other major design type in the United States, output saturated steam at the top rather than heated water. Control rods enter a BWR from the bottom of the pressure vessel, rather than from the top as is standard for PWRs.

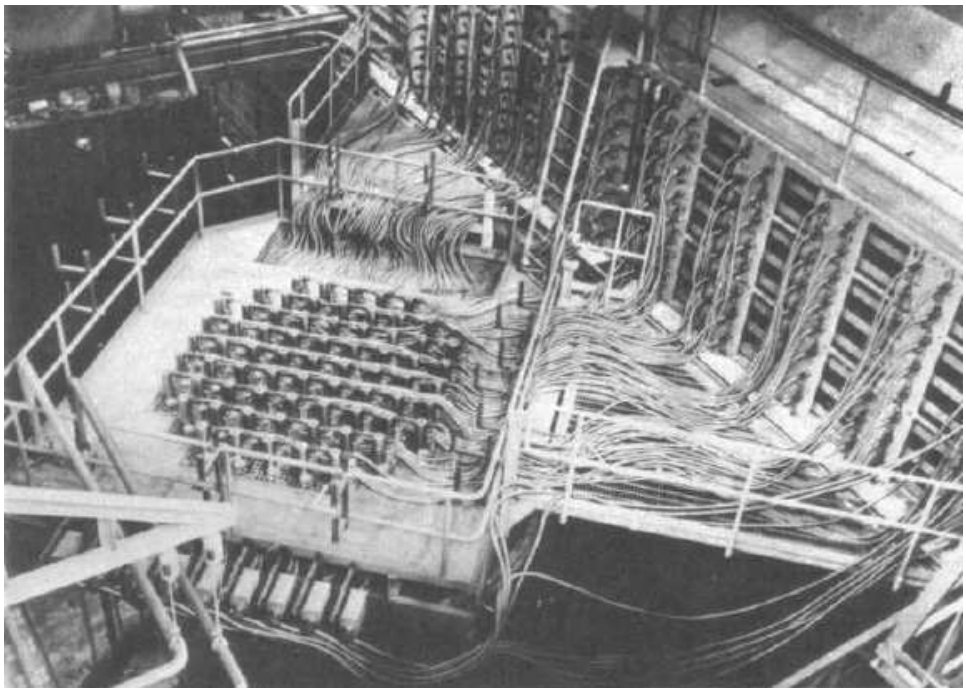
In the United States of America, nuclear reactors are designed to exhibit what is called a *negative temperature coefficient*, which means the chain reaction naturally slows as the temperature of the coolant increases. This physical tendency, engineered by the configuration of the reactor core and the design of the coolant system, adds a measure of self-stabilization to what would otherwise be an inherently unstable ("runaway") process. This is an example of a "natural" negative-feedback loop in action: a process by which the very laws of physics conspire to regulate the activity of the fission reaction.

Additional regulation ability comes from the insertion of special *control rods* into the reactor core, designed to absorb neutrons and prevent them from "splitting" more atoms. With enough control rods inserted into a reactor core, a chain reaction cannot self-sustain. With enough control rods withdrawn from a freshly-fueled reactor core, the chain reaction will grow to an intensity strong enough to damage the reactor. Control rod position thus constitutes the primary method of power control for a fission reactor, and also the first[51] means of emergency shutdown. These control rods are inserted and withdrawn in order to exert demand-control over the fission reaction. If the reaction rate is too low to meet demand, either a human operator or an automatic control system may withdraw the rods until the desired reactivity is reached. If the reaction rate becomes excessive, the rods may be inserted until the rate falls down to the desired level. Control rods are therefore the final control element (FCE) of an "artificial" negative-feedback loop designed to regulate reaction rate at a level matching power demand.

Due to the intense radiation flux near an operating power reactor, these control rods must be manipulated remotely rather than by direct human actuation. Nuclear reactor control rod actuators are typically special electric motors developed for this critical application.

[51] Other means of reactor shutdown exist, such as the purposeful injection of "neutron poisons" into the coolant system which act as neutron-absorbing control rods on a molecular level. The insertion of "scram" rods into the reactor, though, is by far the *fastest* method for quenching the chain-reaction.

A photograph[52] showing the control rod array at the top of the ill-fated reactor at Three-Mile Island nuclear power plant appears here, with a mass of control cables connecting the rod actuators to the reactor control system:



Rapid insertion of control rods into a reactor core for emergency shutdown purposes is called a *scram*. Accounts vary as to the origin of this term, whether it has meaning as a technical acronym or as a colloquial expression to evacuate an area. Regardless of its etymology, a "scram" is an event to be avoided if possible. Like all industrial processes, a nuclear reactor fulfills its intended purpose only when operating. Shutdowns represent not only loss of revenue for the operating company, but also loss of power to local utilities and possible disruption of critical public services (heating, cooling, water pumping, fire protection, traffic control, etc.). An emergency shutdown system at a nuclear power plant must fulfill the opposing roles of dependability and security, with an extremely high degree of instrument reliability.

The electric motor actuators intended for normal operation of control rods are generally too slow to use for scram purposes. Hydraulic actuators capable of overriding the electric motor actuation may be used for scram insertion. Some early pressurized-water reactor scram system designs used a simple mechanical latch, disengaging the control rods from their motor actuators and letting gravity draw the rods fully into the reactor core.

[52] This appears courtesy of the Nuclear Regulatory Commission's special inquiry group report following the accident at Three Mile Island, on page 159.

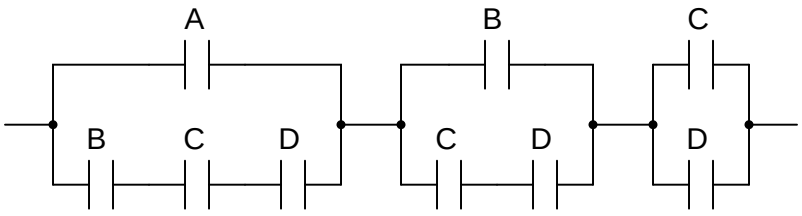
A partial list of criteria sufficient to initiate a reactor scram is shown here:

- Detected earthquake
- Reactor pressure high
- Reactor pressure low
- Reactor water level low (BWR only)
- Reactor differential temperature high
- Main steam isolation valve shut
- Detected high radioactivity in coolant loop
- Detected high radioactivity in containment building
- Manual shutdown switch(es)
- Control system power loss
- Core neutron flux high
- Core neutron flux rate-of-change (period) high

The last two criteria bear further explanation. Since each fission event (the "splitting" of one fuel atom's nucleus by an absorbed neutron) results in a definite amount of thermal energy release and also a definite number of additional neutrons released, the number of neutrons detected in the reactor core at any given moment is an approximate indication of the core's thermal power as well as its reactivity. Neutron radiation flux measurement is therefore a fundamental process variable for fission reactor control, and also for safety shutdown. If sensors detect an excessive neutron flux, the reactor should be "scrammed" to avoid damage due to overheating. Likewise, if sensors detect a neutron flux level that is *rising* at an excessive *rate*, it indicates the possibility of a runaway chain-reaction which should also initiate a reactor "scram."

In keeping with the high level of reliability and emphasis on safety for nuclear reactor shutdown controls, a common redundant strategy for sensors and logic is *two-out-of-four*, or *2oo4*. A contact logic diagram showing a 2oo4 configuration appears here:

2oo4 redundant logic for reactor scram systems



Any two contacts (A, B, C, or D) opening will interrupt power flow and "scram" the reactor

32.7 Review of fundamental principles

Shown here is a partial listing of principles applied in the subject matter of this chapter, given for the purpose of expanding the reader's view of this chapter's concepts and of their general interrelationships with concepts elsewhere in the book. Your abilities as a problem-solver and as a life-long learner will be greatly enhanced by mastering the applications of these principles to a wide variety of topics, the more varied the better.

- **Activation energy**: the amount of energy necessary to initiate a chemical reaction. Relevant to minimum ignition energy (MIE) for explosive mixtures of fuel and oxidizer, and also to intrinsic safety (preventing enough energy from reaching the process energy to possibly ignited a hazardous atmosphere).
- **Common-cause failures**: when multiple functions in a system depend on a single element, failure of that element will cause all dependent functions to fail. Relevant to design of safety shutdown systems and reliability calculations.
- **Defense-in-Depth**: a design philosophy relying on multiple layers of protection, the goal being to maintain some degree of protection in the event of one or more other layers failing.

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Chapter 34

Problem-solving and diagnostic strategies

The ability to solve complex problems is the most valuable technical skill an instrumentation professional can cultivate. A great many tasks associated with instrumentation work may be broken down into simple step-by-step instructions that any marginally qualified person may perform, but effective problem-solving is different. Problem-solving requires creativity, attention to detail, and the ability to approach a problem from multiple mental perspectives.

"Problem-solving" often refers to the solution of abstract problems, such as "word" problems in a mathematics class. However, in the field of industrial instrumentation it most often finds application in the form of "troubleshooting:" the diagnosis and correction of problems in instrumented systems. Troubleshooting is really just a form of problem-solving, applied to real physical systems rather than abstract scenarios. As such, many of the techniques developed to solve abstract problems work well in diagnosing real system problems. As we will see in this chapter, problem-solving in general and troubleshooting in particular are closely related to *scientific method*, where hypotheses are proposed, tested, and modified in the quest to discern cause and effect.

Like all skills, problem-solving may be improved with practice and persistence. The goal of this chapter is to outline several problem-solving tools and techniques.

34.1 Learn principles, not procedures

> "As to methods there may be a million and then some, but principles are few. The man who grasps principles can successfully select his own methods. The man who tries methods, ignoring principles, is sure to have trouble." – **Ralph Waldo Emerson**

Effective problem-solvers always reason from fundamental principles, rather than follow memorized procedures. Following this logic, the wise strategy for any student learning how to solve any type of problem is to internalize as many general principles as possible, and to connect common threads of principle to different applications and scenarios.

A good way for any student to self-check that their focus is indeed on principles rather than procedures is to continually ask themselves if they are able to explain *why* what they have learned is true. It is one thing to be able to explain what you have learned, but it is quite another to explain *why* your new knowledge is valid. "Why" may be the most important question you ever ask, and you need to ask it often.

34.2 Active reading

Learning from reading printed text is a kind of problem-solving activity it is own right. The problem is how to acquire new information from the pages of a book, and the solution requires active engagement of your mind as you read. This is often more difficult that it may seem at first, especially when the subject matter is complex and/or the source text is poorly written. Given the fact that much more technical information is available in text form than in any other format, and also the fact that continuous learning is absolutely essential in the field of instrumentation, active reading is an indispensable tool for student success as well as for continued professional growth in the field of instrumentation.

34.2.1 Don't limit yourself to one text!

A very common mistake made by struggling students is to limit themselves to the reading assigned by their instructor(s), when better books might be available. If homework assignments are given from a particular assigned text, it is understandable why a student might think this is the *only* appropriate text to use. However, textbook selection is an imperfect process, often influenced by factors other than optimal learning (e.g. instructor bias, publisher influence, etc.). Sadly, textbooks are sometimes chosen not on their merits as a learning resource but rather by whether or not the textbook comes complete with pre-made exercises and examinations to be used in class (thus relieving the instructors of much work creating these on their own). As a student you must take responsibility for your own learning and seek the best books available for learning, even if this means only using the assigned text as a source of homework problems!

Thanks to the internet, searching for high-quality books is very easy. Not only may you peruse titles and reviews, but many booksellers also allow limited access to these texts online so you can see firsthand how the books are written and decide whether or not each book would suit your needs.

If you are a student and you approach your instructor asking about other texts to read on the subject, you may find your instructor doesn't like the assigned text any better than you, and has alternatives ready to suggest. This is often the case in courses where the text has been selected by committee.

If you examine the personal library of any highly competent technician or engineer, you will likely find multiple books covering the same topics. Building your own collection of useful texts for learning is a sound strategy. Not only will you find yourself referencing these books throughout your career, but you will find their multiple explanations of common concepts an easier way to learn than by limiting yourself to just one point of view.

34.2.2 Marking versus outlining a text

A practical and common method to increase engagement while reading is to "mark" the paper pages of a book with notations, the idea being to note points of interest and thereby stimulate thinking as you read. For a brief primer on this subject I recommend Mortimer Adler's essay *How to Mark a Book*.

We will begin our exploration of active reading with an example taken from page 101 of the classic text *The Measurement of High Temperatures* written by George Burgess and Henry Louis Le Châtelier in 1912. One reason for choosing such an old text is that the style of writing adds another challenge to the task of reading. For those already familiar with the subject of temperature measurement, the archaic writing style will help give you the perspective of a new student, encountering something unfamiliar for the first time:

CHAPTER IV.

THERMOELECTRIC PYROMETER

Principle. -- The junction of two metals heated to a given temperature is the seat of an electromotive force which is a function of the temperature only, at least under certain conditions which we shall define further on. In a circuit including several different junctions at different temperatures, the total electromotive force is equal to their algebraic sum. In a closed circuit there is produced a current equal to the quotient of this resultant electromotive force and the total resistance.

Experiments of Becquerel, Pouillet, and Regnault. -- It was Becquerel who first had the idea to profit from the discovery of Seebeck to measure high temperature (1830). He used a platinum-palladium couple, and estimated the temperature of the flame of an alcohol lamp, finding it equal to 135°. In reality the temperature of a wire heated in a flame is not that of the gases in combustion; it is inferior to this.

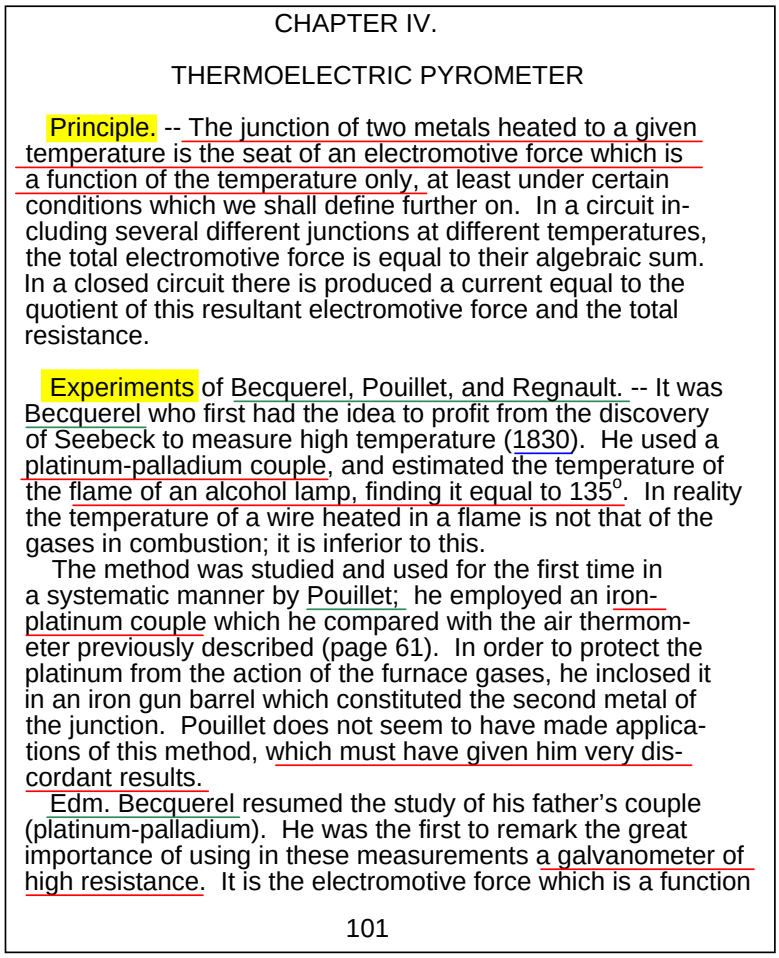
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Edm. Becquerel resumed the study of his father's couple (platinum-palladium). He was the first to remark the great importance of using in these measurements a galvanometer of high resistance. It is the electromotive force which is a function

101

This is the original text as it appears on page 101 of the book. Next, we will explore different ways of "marking" the text, as if we were a new reader to this subject.

The most rudimentary method of marking a book consists of underlining and/or highlighting with a felt-tipped pen:



CHAPTER IV.

THERMOELECTRIC PYROMETER

Principle. -- The junction of two metals heated to a given temperature is the seat of an electromotive force which is a function of the temperature only, at least under certain conditions which we shall define further on. In a circuit including several different junctions at different temperatures, the total electromotive force is equal to their algebraic sum. In a closed circuit there is produced a current equal to the quotient of this resultant electromotive force and the total resistance.

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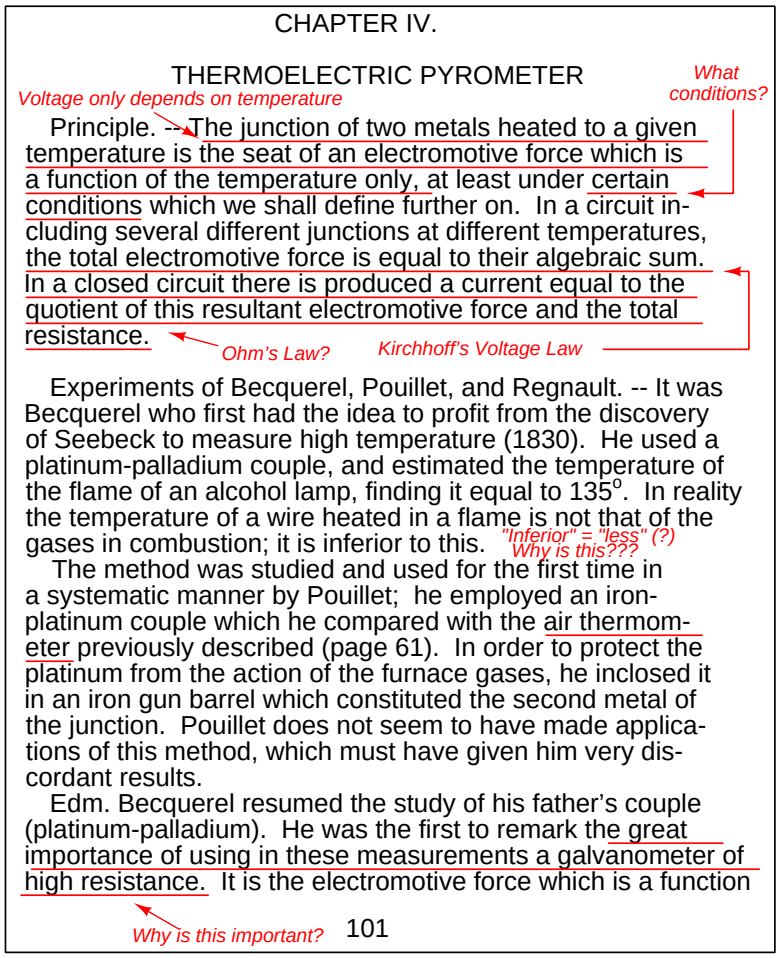
Here you see how the reader has highlighted the paragraph introductory words in yellow ink, underlined names and dates with green lines, and underlined concepts with red lines. While such exercises may have helped the reader remain awake as well as generate cues for later cramming, it is doubtful they assisted the reader in understanding the concepts. A large degree of blame for this rather shallow and unproductive practice may be laid on instructional curricula emphasizing memorization and execution of procedures over conceptual understanding. Simply put, *this is what you get when students expect to be quizzed on isolated bits of data:* you get poor study habits such as this. Unfortunately, not only will this approach fail to yield deep understanding of the concepts, but it also reduces the act of reading to drudgery. When such practices are so common, it's no wonder a great many students loathe academic reading.

The choice of highlighting versus underlining, or of one color over another, is relatively unimportant. Any process of "marking" a book merely by drawing attention to certain words within it suffers the same weakness: all you are doing is emphasizing specific words and phrases, not incorporating your own thoughts into the text. In order to be *actively engaged* in your reading, you must expose your own thoughts and reflections on what you read, not just emphasize specific statements made by the author. As Adler points out:

> "Understanding is a two-way operation; learning doesn't consist in being an empty receptacle. The learner has to question himself and question the teacher. He even has to argue with the teacher, once he understands what the teacher is saying. And marking a book is literally an expression of differences, or agreements of opinion, with the author."

The thrust of Adler's essay is that the reader gains the greatest understanding of a text by expressing their own thoughts about what they are reading: articulating questions, drawing conclusions, linking ideas, and otherwise being an active participant in the reading process rather than a passive observer.

With this in mind, let's re-approach the text. Instead of simply emphasizing words and phrases, we will now show how a reader may articulate some of their own thoughts on the same page:



Here you can see the focus has completely shifted away from facts and figures, and toward concepts. Note the questions raised by the reader, either doubting their own understanding of the book or the author's assertions. These questions need not be fully resolved at the conclusion of reading. Indeed, these may be excellent topics to raise in class once the student returns to school and has the instructor's attention. Questions should be taken as a positive sign for active reading and not as an indication of trouble: if you read a large body of prose and have no questions of your own, you probably weren't thinking deeply enough!

If this manner of marking the text seems messy and cluttered, one may opt to make all the notations on a separate piece of paper, or even typed into a computer for later printing and retrieval.

Even deeper engagement may be achieved if one takes the time to write an *outline* of ideas in the text. This is an exercise best done on a separate sheet of paper, or using a computer. An example is shown here, side-by-side with the original text. Note how an outline may include graphical sketches as well as words, and how it references the page number of the source text to make it easier to go back to the source and re-read if necessary:

CHAPTER IV.

THERMOELECTRIC PYROMETER

Principle. -- The junction of two metals heated to a given temperature is the seat of an electromotive force which is a function of the temperature only, at least under certain conditions which we shall define further on. In a circuit including several different junctions at different temperatures, the total electromotive force is equal to their algebraic sum. In a closed circuit there is produced a current equal to the quotient of this resultant electromotive force and the total resistance.

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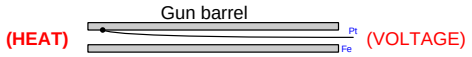
(From page 101)

Joining two different metals together and heating that junction produces a voltage dependent on temperature. What "certain conditions" affect voltage/temp relationship?

The voltages of multiple thermoelectric junctions adds in series, with current = V/R.

Becquerel used a platinum-palladium junction to try to measure a candle flame, but he got a result that was too low. Why is the wire's temp cooler than the flame?

Pouillet used a platinum-iron junction using a hollow iron tube as the second conductor. This tube protected the wire from -- maybe corrosion?



Becquerel's son discovered you need to use a meter with a high resistance. Why is this??

What you see here are notes for just one page of Burgess' and Le Châtelier's text. Although there is no absolute rule for how detailed one's notes should be, they should be expressed in *one's own words* rather than copying the author's words, and should at minimum express *at least one statement and/or question from the reader per paragraph of source text*, since each paragraph typically contains a distinct idea on the topic at hand. A complete outline of thermoelectric pyrometers would of course cover multiple pages of the source text, and not just the one page shown. Note how the outline includes cues for future learning – hints that there is more to this subject than what is immediately presented on this page of the text.

Writing your own outline of a text is especially useful when the text in question is densely packed with ideas and/or difficult to understand, because the act of outlining serves as a self-check for your own comprehension. With highlighting and underlining it is all too easy to lazily read the words and make these marks without really understanding what the text is saying. While outlining you know quite well when you haven't understood the text because you simply won't be able to express it in your own words. Instead of continuing to highlight in a state of blissful unawareness, you will find yourself stuck and unable to continue outlining. This prompts you to go back and scrutinize the text, going over it more carefully than before, until you find yourself able to continue outlining

again.

Perhaps the most common objection to outlining text as you read is that the process is slow. This raises a very important point, namely that *active reading should be slow*. Facts and figures may be skimmed, but complex ideas take time to penetrate into your mind. Not only will outlining force you to slow down when you need to, but it will also serve as a gauge for later study when you review your own notes to see if you still agree with them. In the course of studying some topic, you will often find that your understanding of that topic changes from your first impression. Seeing this change for yourself allows you to better understand how you learn, and thereby gives you practical insight into the workings of your own mind when it comes time to learn something new.

34.3 General problem-solving techniques

A variety of problem-solving techniques have been presented for students over the years which are all helpful in tackling problems both in the classroom and in the real world. Several of these techniques are presented here in this section.

34.3.1 Identifying and classifying all "known" conditions

An important step in solving certain types of problems, especially quantitative problems where calculations are necessary to obtain precise answers, it is often useful to list all the known quantities available to us relevant to the problem. Similarly, taking the time to list all relevant (and possibly relevant) mathematical formulae we might apply to the solution is a helpful step.

One way to save time applying the latter suggestion in a classroom setting is to keep a concise reference card or file filled with formulae you've been learning within that course. This reference may be referred to as often as necessary, without having to re-write the equations for each and every problem, thus eliminating unnecessary effort.

34.3.2 Re-cast the problem in a different format

Many people find it easier to grasp the nature of a problem – and by extension, that problem's solution – if they can look at an illustration of the problem. Therefore, a helpful step in solving problems described to you in words is to translate those words into a picture to look at.

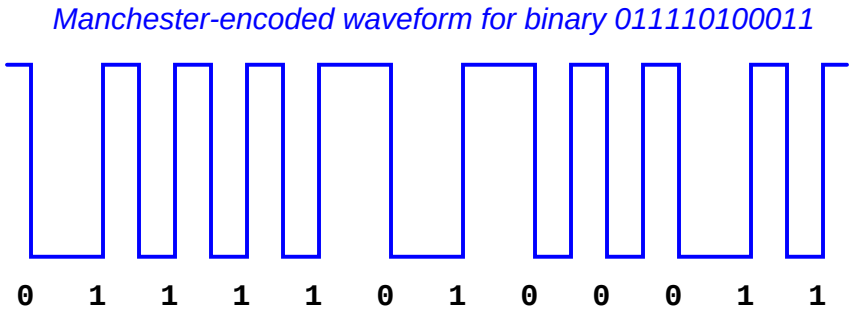
If you are one of those people for whom drawing is a challenge, take heart in the fact that this is a skill you can build. Practice is the key to honing this skill. With this in mind, make it a habit to sketch some kind of illustration for every problem you are asked to solve. If you are working in teams to solve a problem, a collaborative sketch goes a long way toward coordinating problem-solving efforts and ensuring everyone on the team has the same view of the problem.

For some people, describing a problem verbally is helpful in solving it. If your brain tends to work like this – understanding concepts and situations better when they are cast into clear prose – then you may find it helpful to first draft an explanatory paragraph of the problem in your own words. This is also an exercise lending itself well to team-based problem solving, as the entire team can help each other describe the nature of the problem.

34.3.3 Working backwards from a known solution

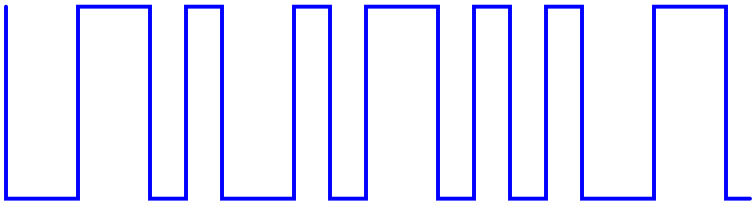
Sometimes we may gain insight into the solution of a problem by assuming we already know the answer to a similar problem, then working "backward" to find the problem from that assumed solution.

An application of this problem-solving strategy is found learning how to decode binary bits that have been encoded using the Manchester standard. With Manchester encoding, binary bits are represented by the rising and falling *edges* of square-shaped waveforms rather than high and low states themselves. For example:



Seeing this example, we note how each binary "0" bit is represented by a *falling* edge, while each "1" bit is represented by a *rising* edge.

Where most students encounter trouble is in situations where they have been given a Manchester encoded waveform and must decode it into its representative bit stream. Take this for example:



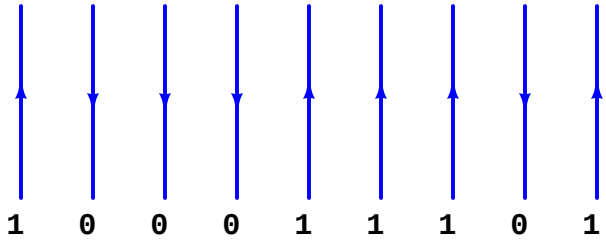
Most students' first inclination is to ask their instructor or their classmates for an algorithm to decode the waveform. "What steps should I take to figure out where the data bits are?" they will ask. This sort of "give me the answer" mind-set should always be discouraged, because it is the polar opposite of true problem-solving technique, where the student methodically searches for patterns and develops algorithms on their own.

A better approach is to encourage the strategy of working the problem backwards: begin with a known series of binary bits, and then develop a Manchester waveform from that. The act of *encoding* a binary string provides insight that will be useful in *decoding* the next Manchester waveform they encounter.

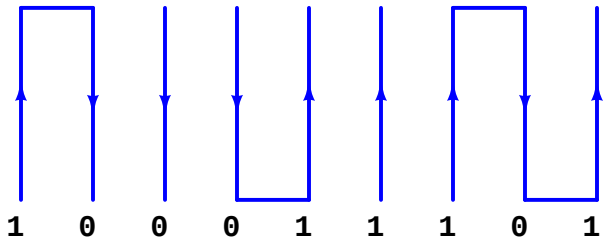
For example, let's begin with the binary string 100011101:



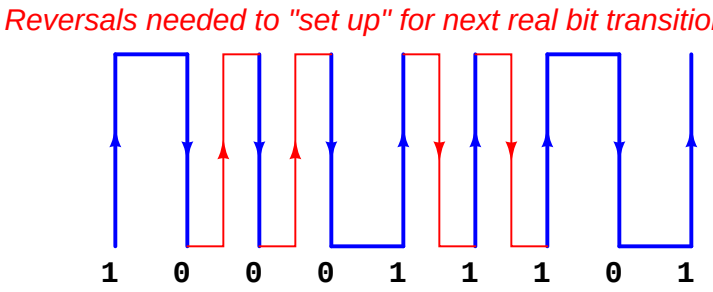
We may begin the process of encoding this into Manchester format by sketching the rising- and falling-edges we know we will need for each bit:



Next, we can try connecting the tops and bottoms of these pulse edges to form a complete waveform. Soon, however, we will find that this is only possible where opposite bit states are adjacent to each other. Where identical bits follow in sequence, we are faced with sequential rising edges or sequential falling edges, which we cannot simply bridge at the tops or bottoms to make a full pulse:



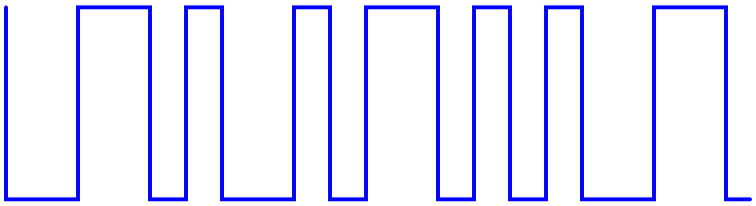
This observation leads to the realization of why we need *reversals* in a Manchester waveform. The only way to connect repeating bits' edges together is if the waveform goes through another rising or falling edge in order to be properly set up for the next edge we need to represent a bit:



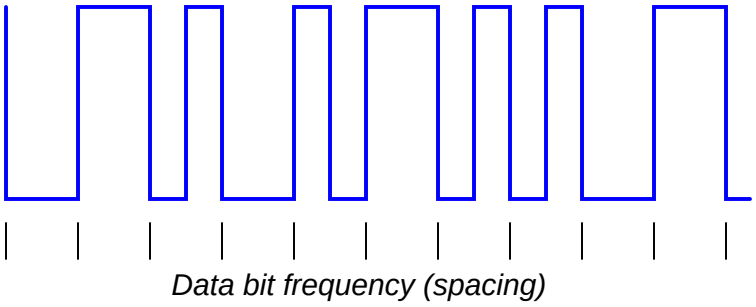
Here we see the power and utility of working a problem "backwards": it reveals to us the *reason why* things are the way they are. Without this understanding, problem-solving is nothing more than rote recall of algorithms, and limited in application. Any problem becomes simpler to solve once we fully understand its rationale.

Once we realize the purpose for reversals in a Manchester waveform, it becomes obvious to see that these reversals always fall *between* the bit transitions, and thus are always *out of step* with the frequency of the bits. Those edge transitions representing real data bits must always fall along a regular timed interval, with reversals being "half-steps" in between those intervals. We need only to look for the widest-spaced intervals in a Manchester waveform to distinguish those pulse edges representing real data bits, and then we know to ignore any pulse edges out of step with them.

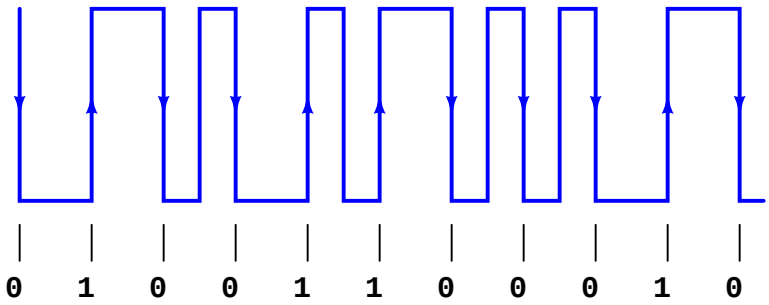
Returning to our sample problem, where we were given a Manchester waveform and asked to decode it:



First, we identify the real data bit edges by widest spacing:



Now that we know which pulse edges represent bits, we may ignore those that do not (the reversals), and decode the waveform:



34.3.4 Using thought experiments

One of the most powerful problem-solving techniques available for general use is something called a *thought experiment*. Scientists use experiments to confirm or refute hypotheses, testing their explanations by seeing whether or not they can successfully predict the outcome of a certain situation by comparing their predictions against real outcomes. While this technique is extremely useful, it might not always be practical or expedient. A useful alternative to real experiments is to mentally "model" the system and then imagine changing certain elements or variables within that model to deduce the effects.

Albert Einstein famously applied "thought experiments" to the formulation of his Theory of Relativity, for the very simple reason that he lacked the resources and technology to actually test his ideas in real life. Working as a patent clerk, he would imagine what might happen if an observer were to travel at or near the speed of light. One particular example of this is the anecdote of Einstein observing a clock tower as he rode a trolley traveling away from the tower. "What would an observer see," he wondered, "as he viewed the clock's face while traveling away from it at the speed of light?" Concluding that the clock's face would appear to be frozen in time was one of the surprising "experimental" results leading Einstein to a more rigorous examination of physical effects at extremely high velocities.

"Thought experiments" are useful in solving a wide variety of problems, because they allow us to test our understanding of a system's behavior. By imagining certain conditions or variables changing in a system and then asking ourselves what the effects will be, we probe our own understanding of that system, often times with the result being that we are able to predict its behavior under conditions that baffle us at first.

You will find "thought experiments" scattered throughout this book, used both as illustrations of problem-solving strategies and also as a tool to explain how certain technologies function. An example of this is the section explaining non-dispersive analyzers, which are instruments employing the absorption of light by certain species of chemicals in order to detect the presence and measure the quantities of those chemicals. Beginning in section 23.6 on page 1819, a series of "thought experiments" are used to explore the principles used to identify chemicals by light absorption. This series of virtual experiments becomes most valuable when this section explores the analyzer's ability to *selectively* measure the presence of one light-absorbing chemical to the exclusion of other light-absorbing chemicals within the same mixture.

34.3.5 Explicitly annotating your thoughts

Suppose you were asked to solve this multiplication problem, without the use of a calculating machine of any kind, but with access to paper and a writing tool:

$$\begin{array}{r} 3418 \\ \times\ 572 \\ \hline \end{array}$$

Your primary school education should have prepared you to solve elementary arithmetic problems of this kind, by a process of digit-by-digit multiplication and addition, to arrive at an answer of 1,955,096. The procedure, while tedious, is rather simple: manually multiply the top numeral three times over by successive digits of the bottom numeral, noting any "carried" quantities as you do so, then sum those three subtotals together (padded with zeros to represent the place of the bottom numeral's digit) to arrive at the final product.

Now suppose you were asked to solve the exact same multiplication problem, but this time doing the same digit-by-digit arithmetic all in your mind, without the use of a writing tool to annotate your work. Suddenly this elementary task becomes nearly impossible for anyone who isn't a mathematical savant. What made the difference between this problem as an elementary exercise and this same problem as a nearly impossible feat? The answer to this question is *short-term memory*: most people do not possess a good enough short-term memory to mentally manage all the intermediate calculations necessary to complete the calculation. This is why people learn to annotate their work when performing manual multiplication, so they don't have to rely on their limited short-term memories. The freedom to write your steps on paper converts what would otherwise be a Herculean feat of arithmetic into a rather trivial exercise.

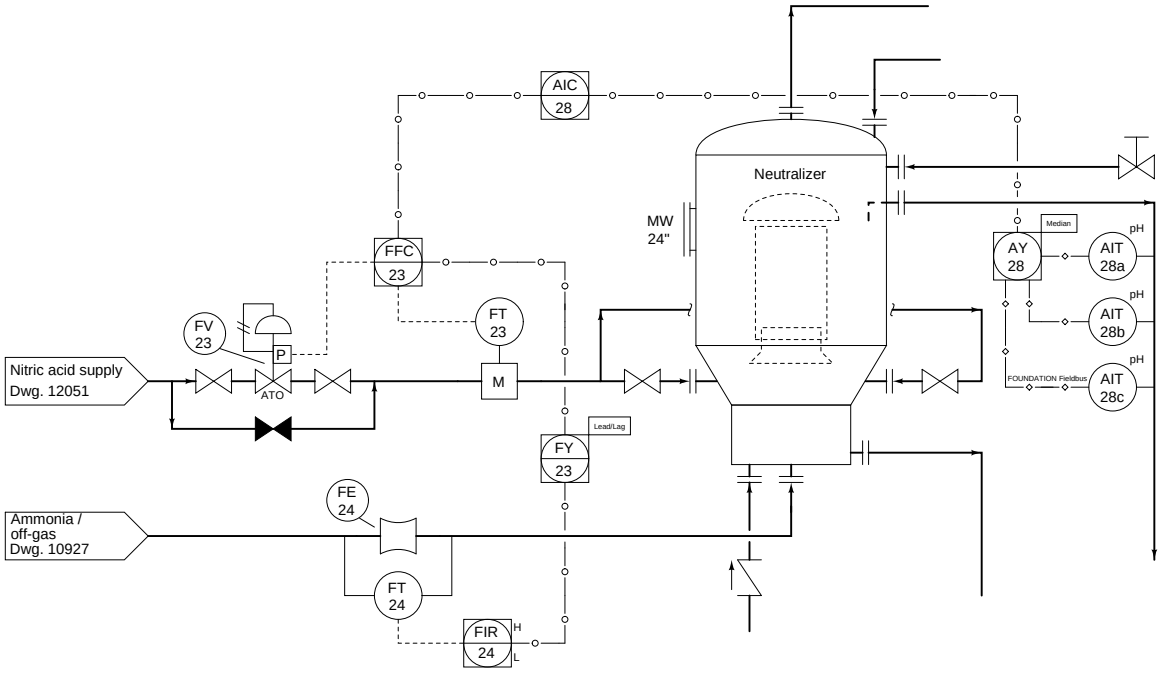
Annotating your intermediate steps as you solve a problem is actually an excellent general problem-solving strategy, applicable to far more than just arithmetic. Some examples of annotating intermediate steps are listed here:

- **Reading a complex document**: *annotating your thoughts, questions, and epiphanies as you read the text allows you to derive a better understanding of the text as a whole.*
- **Learning a new computer application**: *noting how features are accessed and identifying the necessary conditions for each feature to work helps you navigate the software more efficiently.*
- **Following a route on a map**: *marking where you started, where your destination is, and where you have traveled thus far helps you see how far you still need to go, and which alternative routes are open to you.*
- **Analyzing an electric circuit**: *annotating all calculated voltages, currents, and impedances on the diagram helps you keep track of what you know about the circuit and where to go next in your analysis.*
- **Troubleshooting a system fault**: *noting all your diagnostic steps and conclusions along the way helps you confirm or disprove hypotheses.*

Sadly, many students attempt to solve new types of problems analogously to performing multiplication without paper and pencil: they attempt to mentally manage all their intermediate steps, not writing anything down that would help them later. As a result, students tend to get "lost" when trying to solve new problems simply because they cannot readily reference of all their thoughts along the way. Most people simply give up when they begin to feel "lost" in solving a problem, thinking that if they cannot mentally picture the solution in its entirety then they have no hope of attaining it. Let's face it: how soon would you give up on multiplying 3418×572 without a calculator if you believed the only alternative was to manage all the arithmetic in your head?

One reason why students default to the "mental-only" approach when approaching new problems is that their educational experience has only presented annotation for specific types of problems. Thus, marking all the carry digits and subtotals is something they "only do" when performing multiplication by hand; marking calculated voltages and currents on a schematic diagram is something they "only do" when solving DC resistor circuits; taking notes when reading is something they "only do" when completing a book report. In other words, *students see annotation only in very specific contexts, and so they may fail to see just how widely applicable annotation is as a problem-solving strategy.* What teachers should do is model and encourage annotation as a problem-solving technique for *all* types of problems, not just for *some* types of problems.

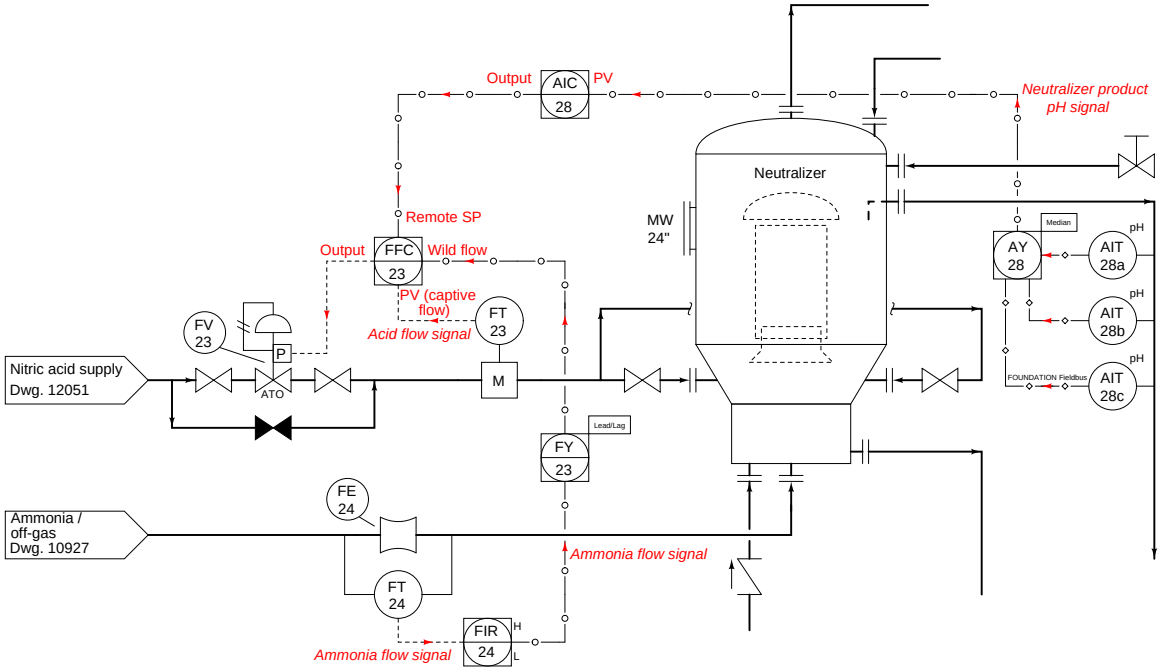
To illustrate how this might be done in the context of control system analysis, let us suppose we were asked to determine the effect of flow transmitter FT-24 failing with a low (no-flow) signal in this ratio control system, part of a process for manufacturing ammonium nitrate fertilizer:



Before it is possible to analyze the effects of a transmitter failure, we must first determine what the system ought to do in normal operation. Natural questions to ask might include the following:

- Where do the instrument signals come from and where do they go to?
- What does each instrument signal represent?
- What is the direction of action for each controller in the system?

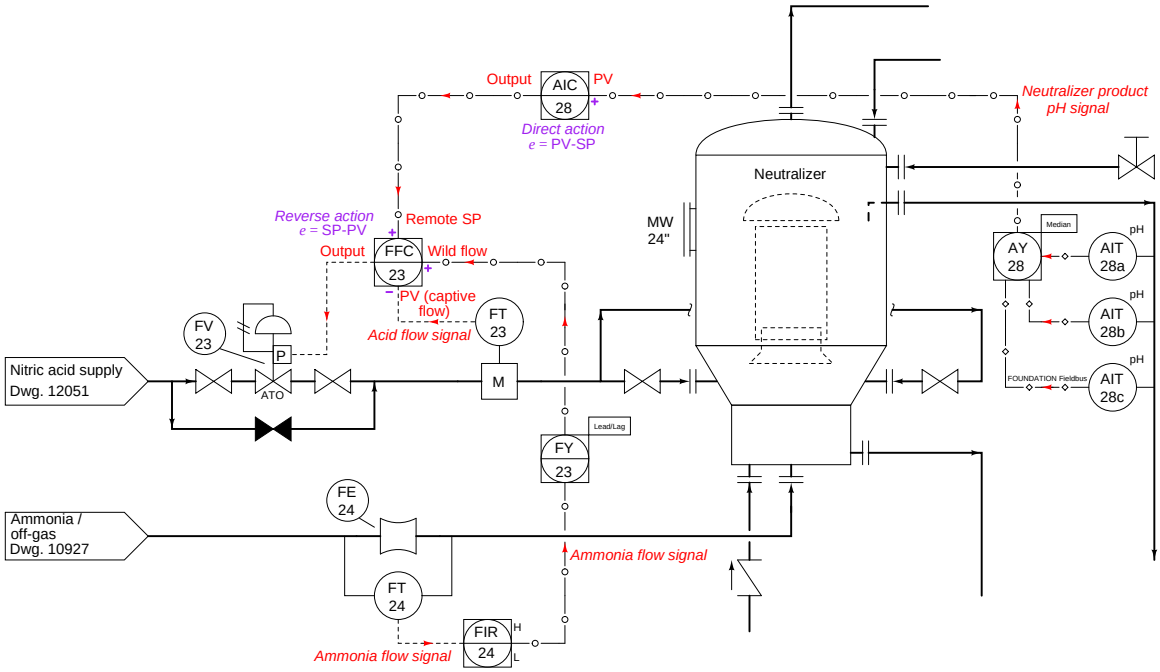
With just a basic understanding of ratio control systems, we may answer all of these questions by close examination of the P&ID segment, and also annotate those thoughts and conclusions on the diagram in order to help us analyze the system's response. Starting with the first two questions of where signals originate and terminate and what each signal represents, we may annotate this with arrows and text (shown in red):



We know that all transmitters *output* data, and so all signal arrows should point *away* from all transmitters and *toward* all controllers. We know that all valves *receive* data, which means arrows must point toward the control valve. The first tag letter of each transmitter (AIT, FT) tells us its measurement function: chemical pH and flow, respectively. The fact that FT-23 is mounted on the same pipe as FV-23 tells us FT-23 must send controller FFC-23 its process variable (captive flow), making the other flow signal (from FT-24) the "wild" flow in this ratio control scheme. AIC-28's task is to control pH exiting the neutralizer, so we know its output signal must call for a neutralizing reagent, in this case nitric acid. This tells us the signal between AIC-28 and FFC-23 must be a cascade output-setpoint link, with AIC-28 as the master controller and FFC-23 as the slave controller.

Now we turn to the question of controller action, since we know the direction of each controller's action (e.g. direct or reverse) is significant to how each controller will react to any given change in signal. Here, 'thought experiments" are useful as we imagine the process variable changing due to some load condition, and then determine how the controller must respond to bring that process variable back to setpoint.

When we annotate the action of each controller, it is best to use symbols more descriptive than the words "direct" and "reverse," especially due to the confusion this often causes when distinguishing the effects of a changing PV signal versus a changing SP signal. In this case, we will write a short formula next to each controller denoting its action according to how the error is calculated ($e = \text{PV} - \text{SP}$ for direct action and $e = \text{SP} - \text{PV}$ for reverse action). We may also write "+" and "−" symbols next to each input on each controller to further reinforce the direction of each signal's influence:

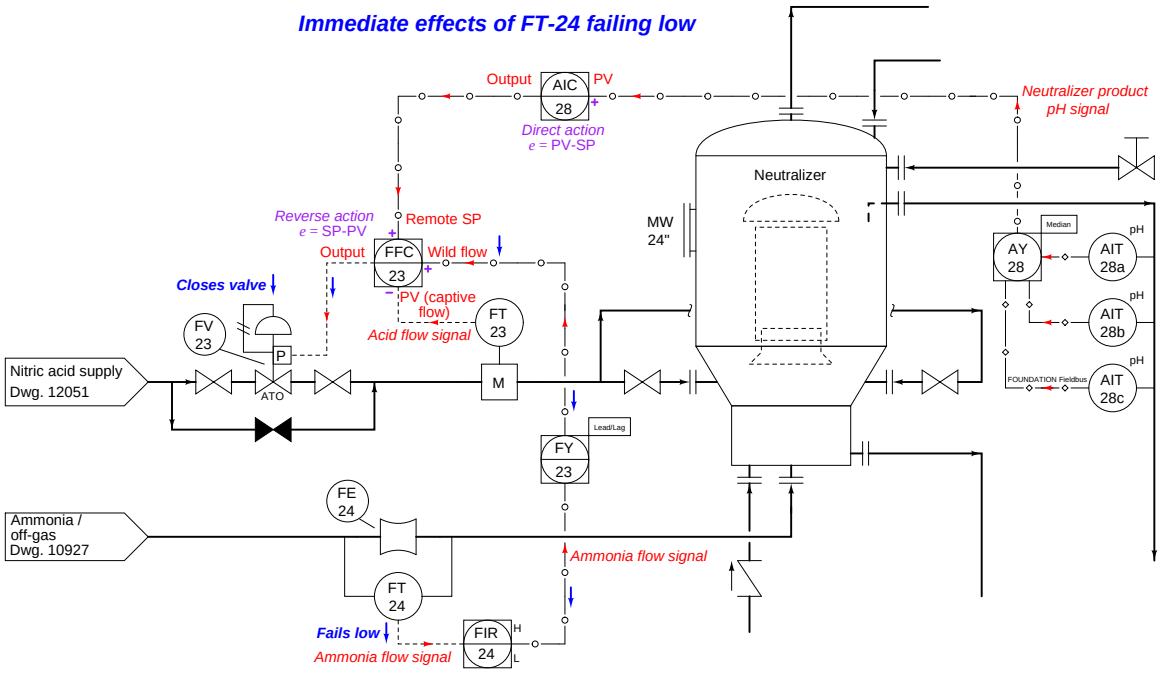


FFC-23 is the best controller to start with, since it is the slave controller (in the "inner-most" control loop of this cascade/ratio system). Here, we see that FFC-23 must be reverse-acting, for if FT-23 reports a higher flow we will want FV-23 to close down. This means the remote SP input must have a non-inverting effect on the output: a greater signal from AIC-28 will increase nitric acid flow into the neutralizer. Following this reasoning, we see that AIC-28 should be direct-acting, calling for more nitric acid flow into the neutralizer as product pH becomes more alkaline (pH increases).

The purpose of the ratio control strategy is to balance the "wild" flow of ammonia into the neutralizer with a proportional flow of nitric acid. This is in keeping with principles of chemical reactions (stoichiometry) and mass balance. Therefore, we would expect an increase in ammonia flow to call for a proportionate increase in nitric acid flow, giving the wild flow signal a non-inverting

effect on FFC-23.

Only at this point in time are we fully ready to analyze the effects of FT-24 failing with a low-flow signal. Once again, we may annotate the failure on the diagram as well, arbitrarily electing to use blue "up" and "down" arrows and bold text to indicate the directions of change for each signal immediately following the failure of FT-24:



As FT-24's signal fails low, the "wild" flow signal to FFC-23 goes low as well. Since we have already determined that input has a non-inverting effect on the ratio controller, we may conclude control valve FV-23 will close as a result, decreasing the flow of nitric acid into the neutralizer. This analysis becomes trivial after we've done the work of annotating the diagram with our own notes showing how the instruments are supposed to function. Without this set-up, the task of analyzing the effects of FT-24 failing low would be much more difficult.

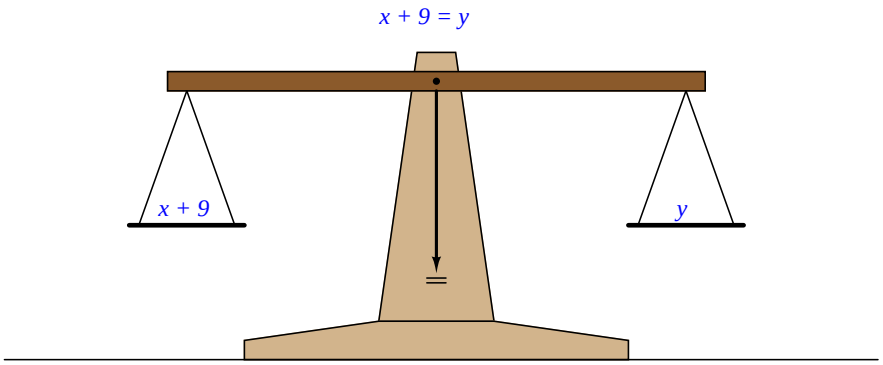
34.4 Mathematical problem-solving techniques

Some problem-solving techniques are unique to quantitative problems, involving mathematical calculations. In this section we will explore some useful tips to help you solve such problems.

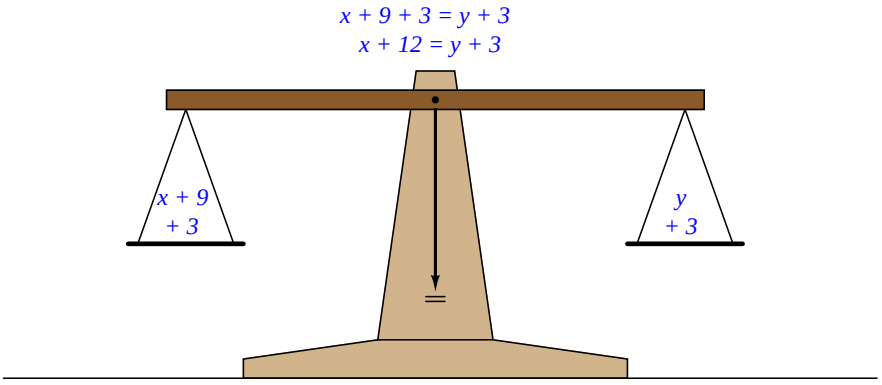
34.4.1 Manipulating algebraic equations

One of the most useful problem-solving techniques in all of algebra is the art of manipulating, or re-writing, equations to solve for a particular variable. The key to this technique is the fact that we may subject an equation to any mathematical operation we desire, so long as we apply that same operation to both sides of the equation.

Let us begin with an illustration showing why this is true. We will represent the equation $x + 9 = y$ as a balance-beam scale, showing the quantity $x + 9$ in one pan of the scale and y in the other pan:



The balance-beam analogy merely represents the fact that the two expressions $x + 9$ and y are equal; that is to say, they possess the same mathematical value as described by the equation $x + 9 = y$. It should be intuitively obvious that this equality will remain unaltered if we were to add some equal quantity to both sides of the equation, such as the number 3:



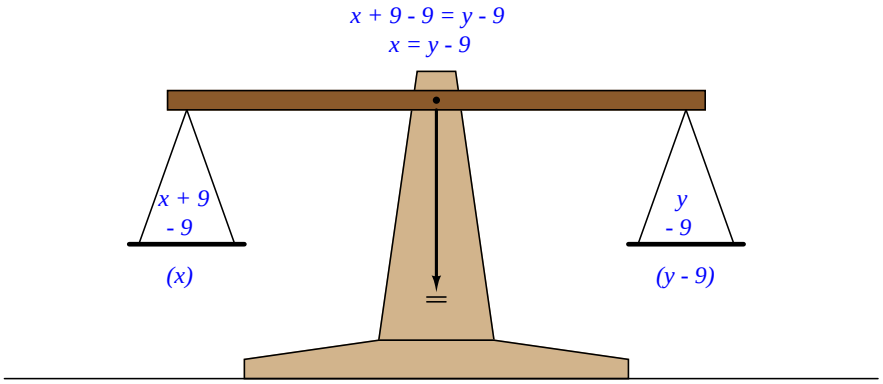
Adding 3 equally to both sides does not affect the balance

If the left-hand quantity is now larger by 3 units, and the right-hand quantity is also larger by 3 units, and those two quantities began as equal to each other, than those two quantities *must still remain equal to one another*. We have not altered the "truth" of the equation by adding 3 to both sides, any more than we would have upset the balance of a real scale by adding 3 units of mass to both pans.

We could similarly multiply each of these quantities by the same factor (say, 4) and still remain equal. The validity of the equation $x + 9 = y$ is not harmed by multiplying both sides by 4 to get $4(x + 9) = 4y$. Similarly, we could take the square-root of both sides to get $\sqrt{x + 9} = \sqrt{y}$ and still have an equality. We could reciprocate both sides to get $\frac{1}{x+9} = \frac{1}{y}$ and still have an equality. We could take the logarithm of both sides to get $\log(x + 9) = \log y$ and still have an equality. We could raise both sides to the power z to get $(x + 9)^z = y^z$ and still have an equality. The lesson here should be perfectly clear: if we start with an equation (two equal expressions), and apply the same mathematical operation to both sides of that equation, we'll still have an equation (two equal expressions).

This fact – that we may apply any mathematical operation equally to both sides of an equation without harming the validity of that equation – might seem at first to be pointless. However, it turns out to be a powerful tool for isolating variables within an equation. If we are creative in the operation(s) we choose to apply to an equation, we may do so in such a way that isolates one variable by itself, placing all other portions of the equation on the other side of the "equals" sign.

Returning to our expression $x + 9 = y$, suppose we were tasked with solving for x. In other words, we need to manipulate this equation to get x by itself on one side of the equals sign, with everything else on the other side of the equals sign. Clearly, the problem here is how to get rid of the "9" that's being added to x on the left-hand side of the equation. One way to do this is to subtract 9 from both sides of the equation, thus canceling the "9" on the left-hand side and shifting it over to the right-hand side. This results in x being left all by itself on the left-hand side of the equation, just like we want:



Subtracting 9 equally from both sides does not affect the balance

Although we have the mathematical freedom to do *anything* we wish to both sides of the equation so long as we do it equally, we can see here that only one operation will work to isolate x by itself. This, therefore, is the challenge of algebraic manipulation: *how to determine which operation(s) we should apply to both sides in order to end up with the equation re-written the way we want?*

A general principle to help us answer this challenge is that of *inverse functions*: that is, pairs of mathematical operations known to "un-do" each other. The inverse function to addition is subtraction, which is why we *subtracted* 9 from each side of the equation $x + 9 = y$ in order to "un-do" the *addition* of 9 to x on the left-hand side. In other words, we determine the mathematical operation being applied to the variable we wish to isolate, and then we intentionally apply the opposite mathematical operation to both sides of the equation to "strip away" that interfering operation.

Here is a table showing a few inverse mathematical functions:

Function $f(x)$	**Inverse function** $f^{-1}(x)$
Addition $(+)$	Subtraction $(-)$
Multiplication $(\times)$	Division $(\div)$
Power (x^n)	Root $(\sqrt[n]{x})$
Exponent (n^x)	Logarithm $(\log_n x)$

It should be noted that sometimes we can use the same function to "un-do" itself if we apply the quantity creatively. For example, instead of subtracting 9 from each side of the equation $x+9 = y$ to solve for x, we could have just as easily *added* -9 to both sides, which is mathematically equivalent. Likewise, instead of using division to "un-do" multiplication, we may simply multiply by the reciprocal, as in the following example where we wish to solve for x in the equation $\frac{5}{3}x = y$. Since we can see that the fraction $\frac{5}{3}$ is being *multiplied* by x, we may get rid of the fraction by multiplying both sides by its reciprocal $(\frac{3}{5})$, leaving x multiplied by a factor of 1, which is the same as x all by itself:

$$\frac{5}{3}x = y$$

$$\left(\frac{3}{5}\right)\frac{5}{3}x = \left(\frac{3}{5}\right)y$$

$$\frac{1}{1}x = \left(\frac{3}{5}\right)y$$

$$x = \frac{3}{5}y$$

Had we chosen to *divide* both sides of this equation by the fraction $\frac{5}{3}$, we still would have arrived at a mathematically correct result, but it would have taken the form of a *compound fraction*, which is not as "elegant" a presentation:

$$\frac{5}{3}x = y$$

$$\frac{\frac{5}{3}x}{\frac{5}{3}} = \frac{y}{\frac{5}{3}}$$

$$x = \frac{y}{\frac{5}{3}}$$

The task of solving for a variable becomes more complicated when there are multiple operations needing to be stripped away to isolate a particular variable. Take for example the equation $5x-4 = y$, where we wish to solve for x. Clearly, we must find a way to "un-do" the subtraction of 4, as well as the multiplication by 5, but which inverse operation should we apply first? The key here is to properly recognize the *order of operations* expressed in the original equation.

If we were to evaluate the equation $5x - 4 = y$ to arrive at a value for y given a value for x, our proper order of operations would be to first multiply x by 5, and then subtract 4. Multiplication/division precedes addition/subtraction, if there are no other influences such as parentheses. If our goal is to "un-do" each of these operations in order to arrive at x by itself, we must do so in *reverse order of operations*. This means first un-doing the subtraction of 4, and then un-doing the multiplication by 5. The following steps show how this is done:

Step	**Equation**	**Explanation**
1	$5x - 4 = y$	Original equation
2	$5x - 4 + 4 = y + 4$	Adding 4 to both sides
3	$5x = y + 4$	Simplifying
4	$\frac{5x}{5} = \frac{y+4}{5}$	Dividing both sides by 5
5	$x = \frac{y+4}{5}$	Simplifying

Note how these steps would have been different if the original equation were written with a different order of operations, such as $5(x - 4) = y$. With the parentheses forcing the order of operations such that the subtraction occurs before the multiplication, our steps for isolating x must reverse as well – we must first divide by 5, then add 4:

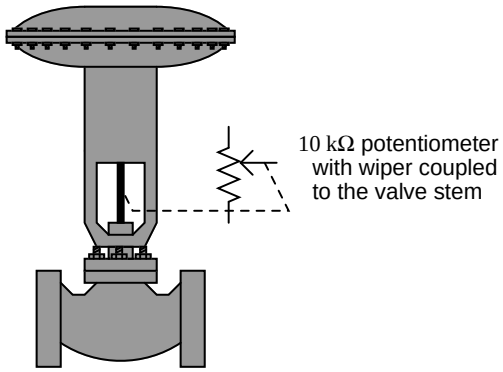
Step	**Equation**	**Explanation**
1	$5(x - 4) = y$	Original equation
2	$\frac{5(x-4)}{5} = \frac{y}{5}$	Dividing both sides by 5
3	$x - 4 = \frac{y}{5}$	Simplifying
4	$x - 4 + 4 = \frac{y}{5} + 4$	Adding 4 to both sides
5	$x = \frac{y}{5} + 4$	Simplifying

34.4.2 Linking formulae to solve mathematical problems

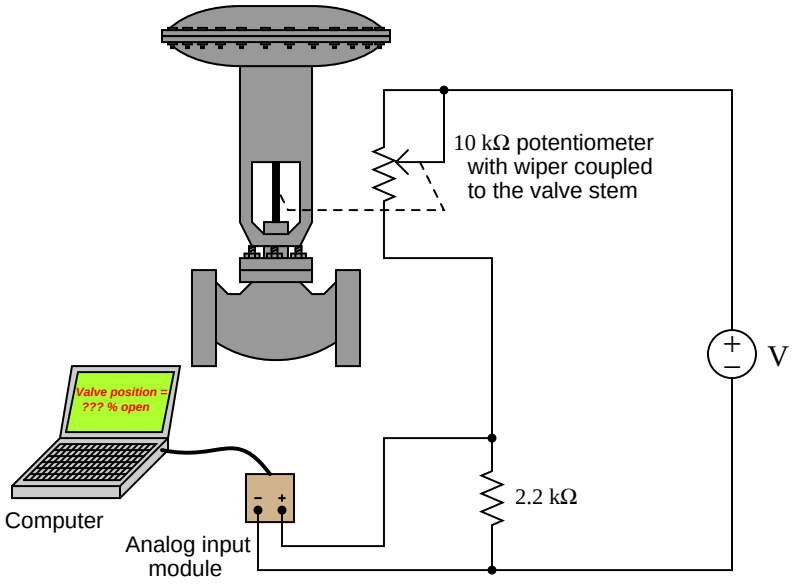
Most practical problems with mathematical solutions do not come to us with the proper formula pre-packaged for our use. Instead, we must identify relevant formulae relating the given variables together, and then use those formulae in combination to solve for the quantity we seek. The task of linking formulae to solve mathematical problems is one many students find quite challenging, and so it is worth our time to explore this in some detail.

Example: sensing valve position

For our first example we will consider a scenario where we wish to have a computer sense the stem position of a control valve. A linear potentiometer coupled to the valve stem will provide us with a suitable sensor to convert the valve stem's position into an electrical signal the computer can measure with appropriate analog I/O hardware:



This potentiometer may be used as one-half of a voltage divider network, with a constant-voltage power supply for the source, and a fixed resistor across which the computer's analog I/O can read a voltage drop. As valve stem position changes, the potentiometer's resistance in this circuit will change, thereby changing the voltage seen by the analog input module connected to the computer:



The problem we are now faced with is this: how do we program the computer to display the valve stem's position in *percent* (on a 0% to 100% range), when all it directly senses from the circuit is a varying DC voltage? It certainly would not be helpful to have the computer display the raw signal in units of volts. What we need is a mathematical formula to translate that sensed voltage drop across the 2200 ohm resistor into a percentage value representing valve stem position.

Our first step needs to be identifying all the relevant mathematical formulae in this problem. Since the computer is sensing the voltage drop across one of two resistors in a series circuit, it would appear the voltage divider formula is relevant here (relating source voltage and circuit resistances to voltage drops):

$$V = V_{source}\left(\frac{R}{R_{total}}\right)$$

We also need to somehow relate the valve's stem position (we'll use x to represent the per unit value of valve opening) to the resistance of the potentiometer. If 0% stem position (i.e. $x = 0$, valve fully closed) drives the potentiometer wiper fully down and 100% stem position (i.e. $x = 1$, valve fully open) drives the wiper fully up, the potentiometer's resistance in this circuit should be a simple proportion of its full 10000 ohm value:

$$R_{pot} = 10000x$$

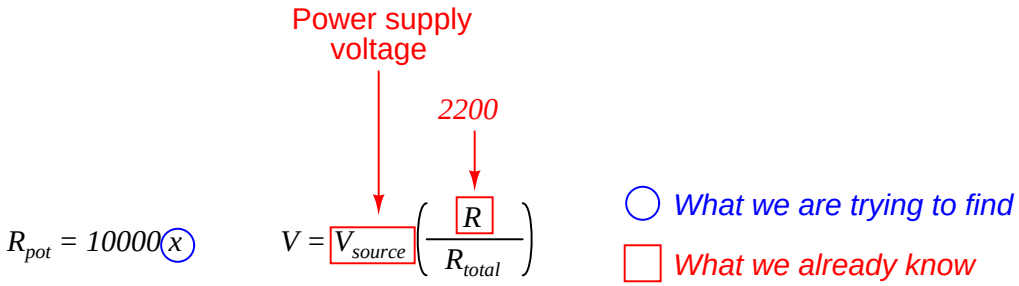
Now that we have a formula relating valve stem position to electrical resistance, and another formula relating electrical resistance to voltage drop, we are ready to link these two formulae together and derive a function expressing voltage drop as a function of valve stem position.

A useful strategy for identifying how multiple formulae link together is to "mark up" those formulae on paper to show how their variables relate to the given information as well as to each other. A standard I find easy to remember and apply is to draw a *circle* around the variable I'm need to solve the value of, and to draw a *square* or a *rectangle* around any variables whose values are given. To begin this process, I will write both the voltage divider and potentiometer proportion formulae near each other, then circle x (valve stem position, per unit) as the variable I need to solve for:

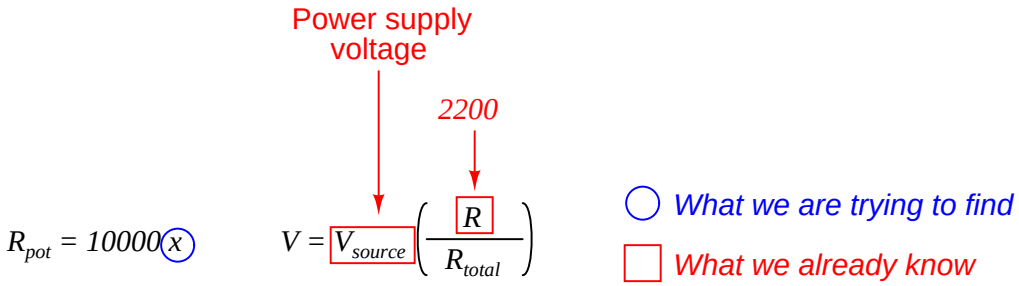
$$R_{pot} = 10000(x)$$

Of course, a basic rule of algebra is that for any *one* formula it is only possible to solve for the value of *one* variable. This means *all* other quantities in a formula must be known in order to solve for that one variable. In the R_{pot} formula we have x which we're trying to solve the value of, the full-scale resistance value of 10000 ohms which is given to us in the problem, and the potentiometer's manifested resistance value in the circuit (R_{pot}). Since 10000 is a constant, the only other piece of information we need in this formula to solve for x is the resistance value R_{pot}. The fact that the variable R_{pot} remains unmarked makes this point clear: *R_{pot} is the missing piece of the puzzle to solve for x.*

At this juncture we look to the other formula on hand to see if any of its variables will provide us this missing information. Here, it should be clear we need to find out where R_{pot} might fit in the voltage divider formula. The R variable in the numerator of the fraction in this formula refers to the resistance across which we are measuring the voltage drop V. In this case, R must be the 2200 ohm fixed resistor, so we will enclose R in a square to represent the fact we know its value. Since the power supply's voltage is constant, we may also enclose V_{source} in a rectangle to show that value will be available to us:

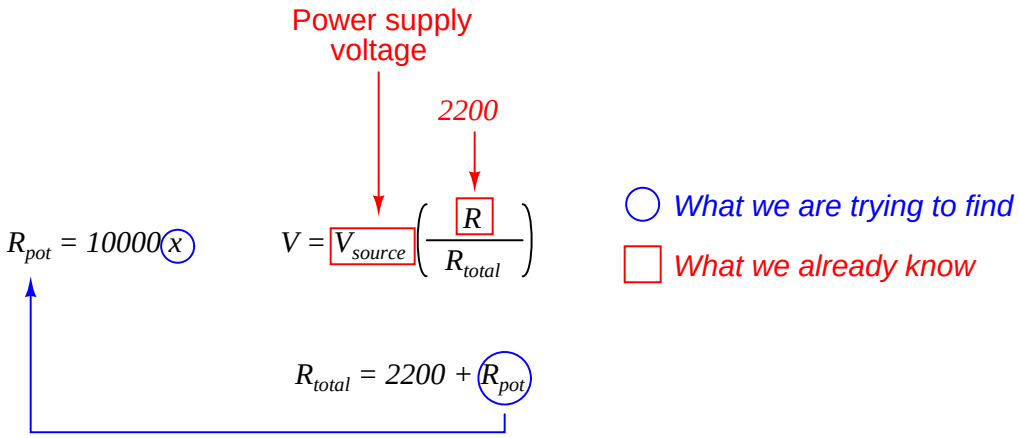


The only other resistance variable in the voltage divider formula is R_{total}, which refers to the total resistance of the series-connected resistors in a voltage divider circuit. This particular circuit has two resistors: the 2200 ohm fixed resistor and the potentiometer. We know that total resistance in a two-resistor series circuit is the sum of those two resistors' individual values ($R_{total} = R_1 + R_2$), we we will write this as a third formula in our collection, with R_1 being the 2200 ohm fixed resistor and R_2 being the potentiometer's resistance:



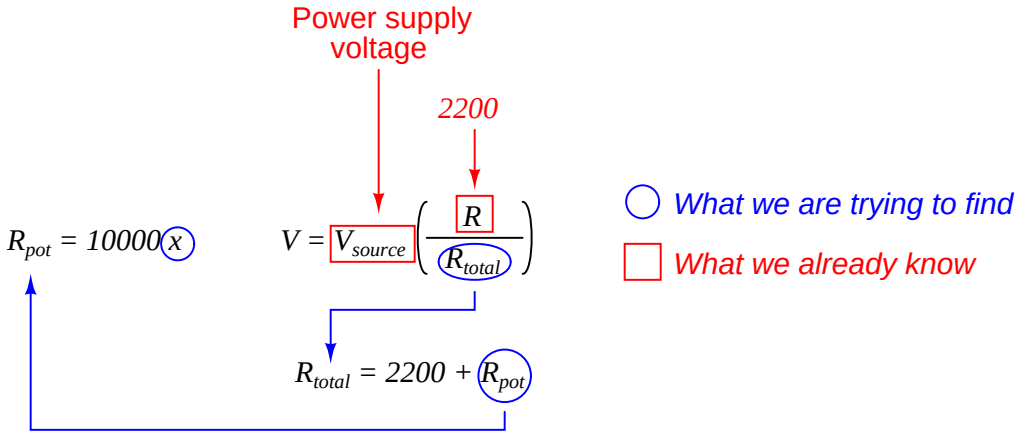
$$R_{total} = 2200 + R_{pot}$$

Now we are ready to link these formulae together. Recall how we needed to find the value of R_{pot} in the potentiometer's formula before we could calculate the value of x (valve stem position). Note how the R_{total} formula relates that potentiometer resistance value to the other resistances in the circuit. This means the total resistance formula can provide us the value of R_{pot} we need to solve for x. We will draw a circle around R_{pot} in the total resistance formula reminding us we need that value, then show a link between this and the potentiometer formula by drawing an arrow extending from one to the other:



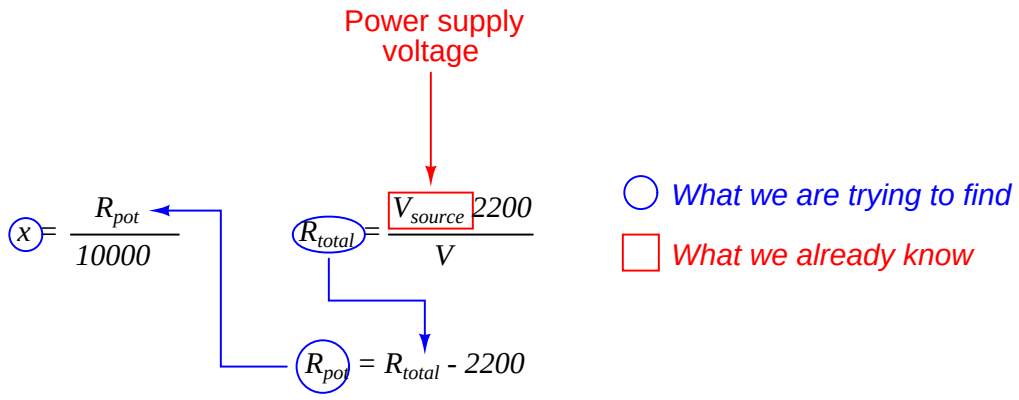
So far, this markup tells us R_{pot} is the missing puzzle piece to calculate x, and that R_{total} is the missing puzzle piece to calculate R_{pot}.

Looking at our three formulae, we see that the voltage divider formula is able to provide us with the value of R_{total} which we need to calculate R_{pot} which we need to calculate x. We will circle R_{total} in the voltage divider formula and draw another arrow showing the link:



The only variable unmarked and unlinked now is V, which is the voltage sensed by the computer's analog input module. This is now the one independent variable which will tell us the position of the valve stem (x).

Each arrow linking formulae together shows us where one formula will be *substituted* for a variable in another formula. The only thing we must do now prior to this substitution is algebraically manipulate each formula to solve for the one circled variable within it. I will skip these algebraic steps for brevity, and simply re-write the three formulae in manipulated form:



The logic chain of dependency linking these three formulae together is now crystal-clear: we begin with a measured voltage drop value of V to give us the value of R_{total}, which then plugs into the total resistance formula to give us the potentiometer's value R_{pot}, which then plugs into the potentiometer formula to give us valve stem position (x) as a per unit value. The algebraic substitutions are shown here:

Substituting $R_{total} - 2200$ for R_{pot} in the $x = \frac{R_{pot}}{10000}$ formula:

$$x = \frac{R_{total} - 2200}{10000}$$

Substituting $\frac{V_{source}2200}{V}$ for R_{total} in the $x = \frac{R_{total}-2200}{10000}$ formula:

$$x = \frac{\frac{V_{source}2200}{V} - 2200}{10000}$$

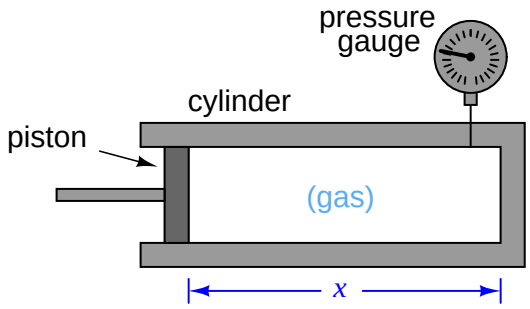
This final formula can now be programmed into the computer, telling the computer how to calculate x as a function of V.

To summarize this problem-solving strategy:

1. Begin by writing every formula you can think of relevant to the problem at hand.
2. Identify the final value you're trying to solve for, and circle it.
3. Identify all given values, and show them by drawing squares or rectangles around those variables.
4. Identify any and all "missing puzzle pieces" in the formula with the circled variable.
5. If there is another formula containing a "missing puzzle piece," circle that variable and draw an arrow linking it to the previous formula, then identify any and all "missing puzzle pieces" needed to solve for that variable. There should only be one circled variable per formula.
6. Repeat the previous step as often as needed until there are no missing pieces left.
7. Algebraically manipulate all formulae to solve for their circled variables.
8. Algebraically substitute all variables as shown by the arrows.

Example: gas pressure inside a cylinder

Let's apply this technique to a practical problem related to engine and compressor mechanisms – how to determine the amount of gas pressure generated inside a cylinder when a piston is moved to compress the gas, given the distance of the piston's motion (x):



Compression is nothing more than a forced reduction in volume, and so a good first step for us to take is to recall any mathematical formulae relating *pressure* and *volume* for a confined gas. In this case, the Ideal Gas Law is our clear choice, describing the relationship between gas pressure, volume, temperature, and the molar gas quantity inside any enclosed volume:

$$PV = nRT$$

Where,

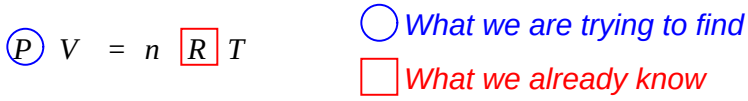
P = Absolute pressure (atmospheres)
V = Volume (liters)
n = Gas quantity (moles)
R = Universal gas constant (0.0821 L $\cdot$ atm / mol $\cdot$ K)
T = Absolute temperature (K)

However, the Ideal Gas Law formula contains no variable x to represent piston position. Somehow, we must find a way to relate x to the Ideal Gas Law formula in order to progress to a solution for this problem.

The first step in applying this formula is to identify which variables are given to us in the problem, and which we need to solve for. In order to solve for the value of any one variable in an equation, there cannot be any other unknown quantities – this is a basic law of algebra.

In this case, the variable we're ultimately trying to solve for is *pressure* (P). The only other quantity found in the Ideal Gas Law formula that we know the value of at this point is R, which is a natural constant. V, n, and T are all unknown to us at this point in time. Once we do know the values of these three variables, we may algebraically manipulate the Ideal Gas Law formula to solve for P.

Once again we will begin by "marking up" the formula by drawing a circle around the variable we wish to solve for, and drawing squares around any variables or constants for which we already know the values. Any variables left unmarked (i.e. not enclosed in a circle or a square) are unknown quantities which we *must* determine before we can solve for the circled variable:



This first step of identifying known and unknown quantities is critically important to the process of problem-solving, because it directs us to what we need to do next. An obstacle so many students and professionals alike experience is paralysis in problem-solving: they get part-way into solving a problem and encounter a point where they have no idea what to do or where to go next. This is why having a strategy to identify the next step is so vitally important.

After identifying these unknown variables of V, n, and T, we now know we need to either find other formulae to solve for those values, or investigate the real mechanism to see if we may directly measure their values.

It should be clear that temperature (T) is one of those variables we should be able to directly measure in the mechanism. We know in order to solve for pressure (P) given piston position (x), one of the real-world parameters we must first measure is temperature.

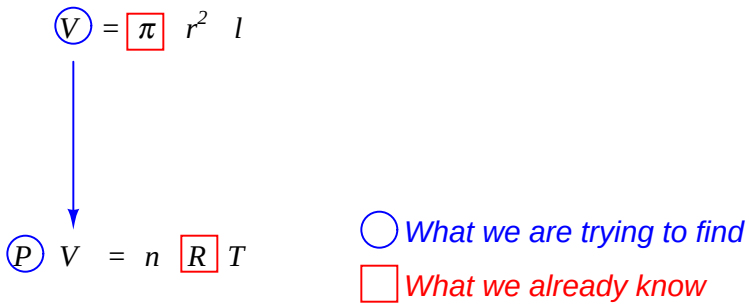
It should also be clear that volume (V) is a variable related to piston position (x), and as such we should be able to identify a formula relating the two, which we may then combine with the Ideal Gas Law formula to arrive at one step closer to our solution. Since the interior volume of the machine's cylinder is, well, *cylindrical*, we know that the formula for calculating the volume of a cylinder is what we need to incorporate next:

$$V = \pi r^2 l$$

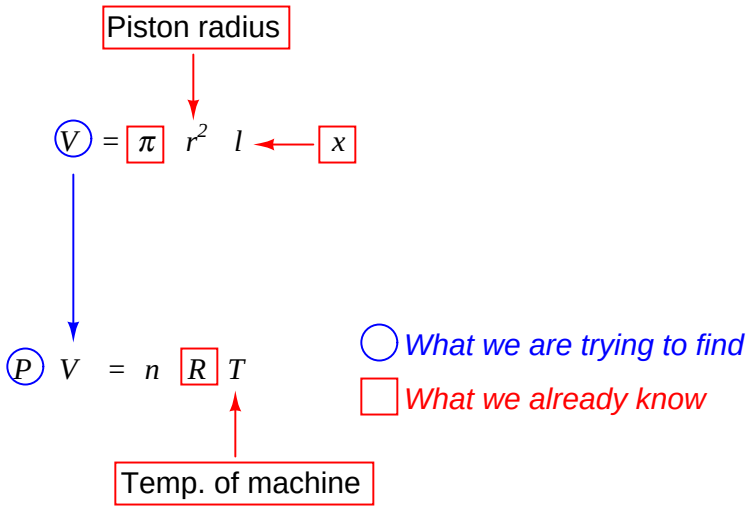
Where,

V = Volume of cylinder
$\pi \approx 3.1415927$
r = Radius of cylinder
l = Length of cylinder

As we did with the Ideal Gas Law formula, our next step is to identify in the cylinder volume formula those quantities we currently know versus those we don't (and therefore those we need to find). Marking up the formula with circles and squares is helpful, as is drawing an arrow between the variable we're trying to solve for with the cylinder formula and where it goes in the Ideal Gas Law formula. This arrow visually links the two formulae together, showing us where we will eventually need to perform algebraic substitution:



Pi (π) of course is a natural constant, just like R, so we enclose it in a square to show that we know its value. r is the radius of the cylinder, which in the mechanism's case is the same as the radius of the piston. As with temperature, this is a quantity we will need to determine in order to solve this problem. Length (l) is exactly the same as the piston position given to us in the problem: x, meaning we may re-write the cylinder volume formula as $V = \pi r^2 x$. Representing all of these relationships graphically:



The only unknown still existing is n, the number of moles of gas held inside the cylinder. This is not something one would be able to directly measure, like temperature, piston radius, or piston position. Either this is a quantity that would have to be provided to us as a given condition in the problem, or we must know something else about the problem in order to find the value of n. We

will set aside this issue for the time being and concentrate for now on how to combine the cylinder and Ideal Gas Law formulae.

As our graphical markup of the two formulae show us, volume (V) calculated by the cylinder formula is what gets "plugged in" to the Ideal Gas Law formula (V as well). In algebra, this operation is called *substitution*: replacing a single variable in an equation with another equation.

$$\text{Substituting} \quad V = \pi r^2 x \quad \text{for} \quad V \quad \text{in} \quad PV = nRT \quad \ldots$$

$$PV = nRT$$

$$P(\pi r^2 x) = nRT$$

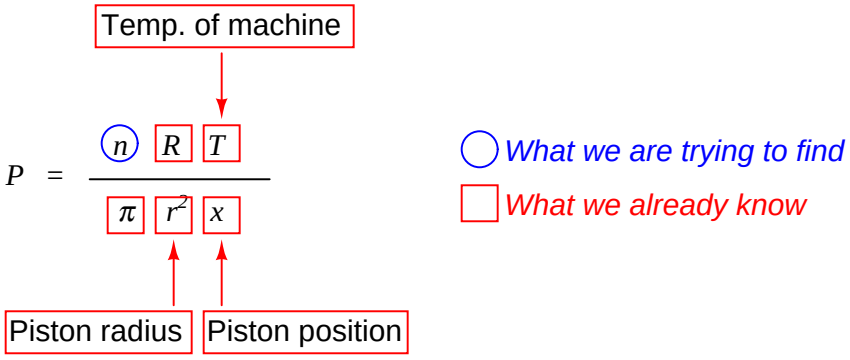
$$P\pi r^2 x = nRT$$

$$\text{Now, solving for} \quad P$$

$$P = \frac{nRT}{\pi r^2 x}$$

We now have a formula for P written as a function of x. Provided we know the temperature (T), the piston's dimensions, and the molecular quantity of gas inside the cylinder (n), we may plug any arbitrary value for piston position (x) and very easily calculate pressure (P).

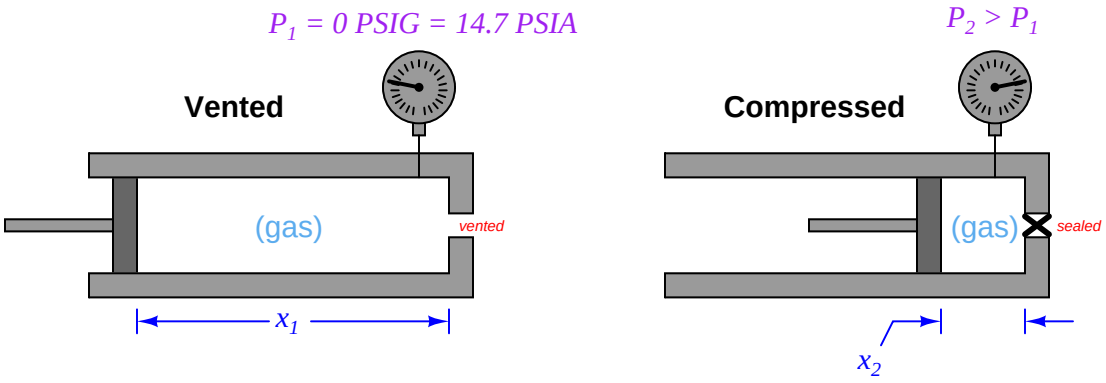
Returning to the question of knowing the value of molecular gas quantity (n), we might ask ourselves, "What if no one provided us with a value for n? How could we possibly solve for pressure then?" Recall once more the basic law of algebra telling us we can only solve for one unknown quantity in a single equation at one time. What would we need to know in our custom formula in order to solve for n? A graphic markup of the formula is helpful once again:



From this we can see it would be possible to calculate n, if only we knew P. Clearly, this is a Catch-22: our goal is to calculate the value of P from n, yet the only algebraic means we have of determining n is to first know the value of P.

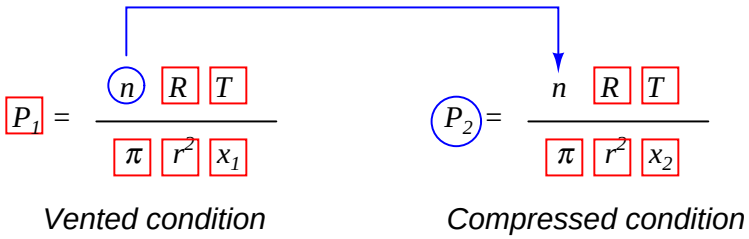
However, there is a solution to this conundrum. Most engine and compressor mechanisms work by moving the piston back and forth in coordination with valves to vent and direct the flow of gas into and out of the cylinder. This usually means there will be *other* piston positions where vent valve(s) are open to ensure atmospheric pressure inside the cylinder. In other words, there will be other values of x for which the pressure P will be known. In such a case where both x and P are known, we may solve for n, and then be confident that value of n will be the same for other piston positions because the vent valves close off before compression begins, trapping all gas molecules inside the cylinder.

Representing two such conditions, using subscripts to distinguish pressures and piston positions in each of the two conditions. Here, we will assume that the gas temperature is the same in both conditions, to simplify the problem:



$$P_1 = \frac{nRT}{\pi r^2 x_1} \qquad P_2 = \frac{nRT}{\pi r^2 x_2}$$

Now that we have a means to solve for n, we may use substitution again to replace n in the "compressed" formula (condition 2) with the n from the "vented" formula (condition 1). A key assumption in making this substitution is that we know n will actually be the same value for those two conditions, which is a safe assumption because compression machines seal off the cylinder (preventing gas entry or escape) prior to the compression stroke of the piston.



To do this substitution, we must first manipulate the first equation to solve for n, then replace n in the second equation with our manipulated first equation:

$$P_1 = \frac{nRT}{\pi r^2 x_1}$$

$$n = \frac{\pi r^2 x_1 P_1}{RT}$$

Substituting . . . n

$$P_2 = \frac{\left(\frac{\pi r^2 x_1 P_1}{RT}\right) RT}{\pi r^2 x_2}$$

Here we see something very interesting: following the substitution, many of the variables and constants cancel each other out, leaving us with a much simpler formula solving for P_2 in terms of x_1 and x_2. First, we cancel RT out of the numerator:

$$P_2 = \frac{\left(\frac{\pi r^2 x_1 P_1}{RT}\right) RT}{\pi r^2 x_2}$$

$$P_2 = \frac{\pi r^2 x_1 P_1}{\pi r^2 x_2}$$

Next, we cancel out πr^2 from numerator and denominator:

$$P_2 = \frac{x_1 P_1}{x_2}$$

$$P_2 = \frac{x_1}{x_2} P_1$$

Thus, the machine's gas pressure in the compressed condition is a simple ratio of its two piston positions multiplied by the absolute pressure in the vented condition.

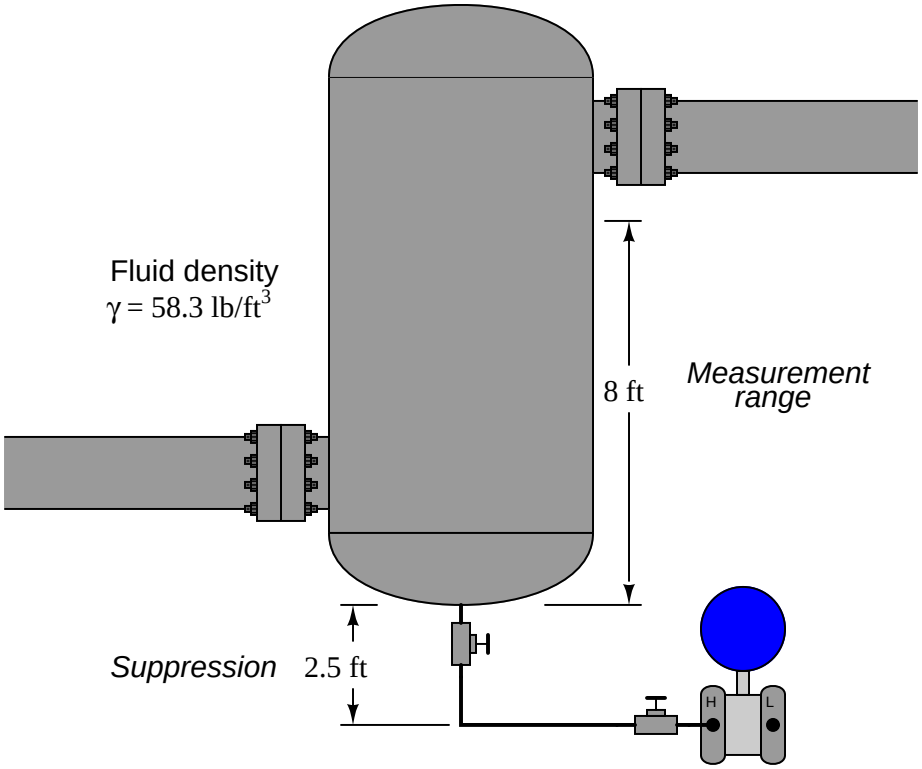
34.4.3 Double-checking calculations

When performing calculations to arrive at an answer to some problem, it is important to check your work. Even if all the algebraic work you've done is perfectly correct, it is still possible to commit simple "keystroke" errors while entering numbers or executing operations on your calculator, and/or to make simple mental-math calculation errors. For this reason, teachers from time immemorial have encouraged their students to *double-check* their work.

What many students tend to do, unfortunately, is check their work by following all the same steps one more time and seeing if they get the same answer as before. While this might make sense to do at first, it actually invites the exact same errors, because our short-term memory for calculator keystrokes and mental math operations tends to be quite good. If we made a keystroke error the first time, we are very likely to make the exact same keystroke error the second time, simply following muscle memory.

A good way to avoid this error is to check your mathematical work *backwards*, beginning with the answer you previously calculated, working backwards through the equations to see if you can arrive at one of the values given to you at the start of the problem. This technique forces you to approach the problem differently, using different keystrokes in different orders.

For example, suppose you were tasked with calculating the pressure generated by a vertical column of liquid inside a process vessel, in order to properly set the lower- and upper-range values (LRV and URV) for the hydrostatic level transmitter:



The pressure seen by the transmitter at a 0% level condition (empty vessel) should be only that produced by the 2.5 foot height of the impulse tubing filled with process fluid:

$$P = \gamma h$$

$$P = (58.3 \text{ lb/ft}^3)(2.5 \text{ ft})$$

$$P = 145.75 \text{ lb/ft}^2 = 1.012 \text{ PSI}$$

Likewise, the pressure seen by the transmitter at a 100% level condition should be that produced by the impulse tubing height *and* the vessel's internal fill of 8 feet:

$$P = \gamma h$$

$$P = (58.3 \text{ lb/ft}^3)(2.5 \text{ ft} + 8 \text{ ft})$$

$$P = (58.3 \text{ lb/ft}^3)(10.5 \text{ ft})$$

$$P = 612.15 \text{ lb/ft}^2 = 4.251 \text{ PSI}$$

If we wished to check our final answer of 4.25 PSI, we could work backwards from this result to try to calculate the total fluid height of 10.5 feet, or work backwards to calculate the fluid density of 58.3 lb/ft^3. Let's do the latter and see what we get:

$$\gamma = \frac{P}{h}$$

$$\gamma = \frac{612.15 \text{ lb/ft}^2}{10.5 \text{ ft}}$$

$$\gamma = 58.3 \text{lb/ft}^3$$

Although this technique may seem obvious, it nevertheless avoids the pitfalls of repeating keystroke errors, which is an error plaguing the work of many students!

A variation on this theme is to calculate a quantity not given or reached in any of the initial problems. Here, for example, we were tasked with determining the transmitter's lower- and upper-range values (pressures at 0% fill and 100% fill). One way to check our work is to see if a different fill condition such as 50% gives us a pressure value lying exactly half-way between the LRV of 1.012 PSI and the URV of 4.251 PSI.

If we imagine the vessel half-full, our total liquid height seen by the transmitter will be 4 feet plus the suppression value of 2.5 feet, or 6.5 feet total. Calculating hydrostatic pressure at this liquid height:

$$P = \gamma h$$

$$P = (58.3 \text{ lb/ft}^3)(2.5 \text{ ft} + 4 \text{ ft})$$

$$P = (58.3 \text{ lb/ft}^3)(6.5 \text{ ft})$$

$$P = 378.95 \text{ lb/ft}^2 = 2.632 \text{ PSI}$$

A simple way to check that 2.632 lies half-way in between 1.012 and 4.251 is to calculate the average value of 1.012 and 4.251:

$$\frac{1.012 + 4.251}{2} = 2.632$$

Sure enough, it checks out to be correct. This is good validation of our initial work in calculating the transmitter's LRV of 1.012 PSI and URV of 4.251 PSI.

Yet another variation on this same theme of checking your work different ways is to approach the problem differently altogether. We know we may calculate the LRV and URV pressure values for this process vessel and fluid if we convert the fluid's density into a specific gravity value (58.3 $\text{lb/ft}^3 = 0.9339$) and then calculate hydrostatic pressure as though we were dealing with inches water column, corrected for the specific gravity of this process fluid.

A height of 2.5 feet (0% level) is 30 inches, which when multiplied by the specific gravity value of 0.9339 yields 28.016 inches WC, which is equivalent to our first calculated pressure value of 1.012 PSI.

A height of 10.5 feet (100% level) is 126 inches, which when multiplied by the specific gravity value of 0.9339 yields 117.67 inches WC, which is equivalent to our first calculated pressure value of 4.251 PSI.

As you can see, solving for LRV and URV pressures using an entirely different technique yields the same answers as before, which is good validation of our original work.

It is important to note that double-checking your work in this manner merely *helps confirm* your original work, but it does not *conclusively prove* your original answers are correct. For example, one of the numbers we keep using in the URV calculations is the total liquid height of 10.5 feet. Suppose, though, that this height value was something we incorrectly calculated mentally by adding the suppression and the range heights. For example, suppose our actual given heights in the problem were a 2.5 feet suppression height and a 7 foot measurement range, and we mistakenly added 2.5 feet and 7 feet in our heads to get 10.5 feet. If that were the case, both the original URV pressure calculation and the subsequent double-checks might still agree with one another, but could still be wrong because they all depend on that same 10.5 foot height value.

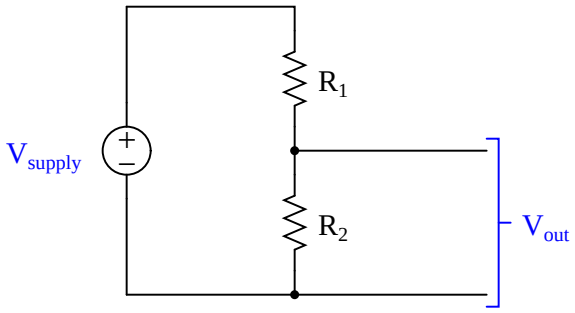
The most realistic attitude to maintain in all problem-solving is *scientific skepticism*: there is no such thing as *absolute proof* so long as the potential for error exists. Given that you are a human being – liable to all manner of fallacies and mistakes – absolute proof will forever lie outside your grasp. The best we can do is incrementally minimize the potential for errors and mistakes, and accept the inevitable uncertainty.

34.4.4 Maintaining relevance of intermediate calculations

A very common attitude I've noticed with students is something I like to call "any procedure, so long as it works." When learning to solve particular types of quantitative problems, the natural tendency is to identify procedural techniques for calculating the correct answer(s), and then practice those procedures until they can be performed without flaw. Unfortunately, it is all too easy to lose focus on principles when the learning emphasis is on procedure. The allure of a procedure guaranteed to yield correct answers overshadows the greater need to master and apply fundamental principles, leading to poor problem-solving ability masked by an illusion of competence.

This is a significant obstacle to deep and significant learning in the sciences, and there is no one solution to it. However, there is a way to identify and self-correct this behavior in some contexts, and it relies on a habit of identifying the real-world relevance of *all* intermediate calculations within a quantitative problem.

To illustrate, let us consider a simple voltage divider circuit comprised of two resistors, where some amount of supply voltage is divided into a smaller proportion to become the output voltage signal. The problem at hand is calculating the output voltage of this divider circuit, knowing the values of supply voltage and resistor resistances:



One of the basic formulae all beginning electronics students learn is the appropriately named *voltage divider formula*, shown here:

$$V_{out} = V_{supply}\left(\frac{R_2}{R_1 + R_2}\right)$$

Calculating V_{out} is a simple matter of substituting known values of voltage and resistance into this formula, then performing the necessary arithmetic. However, the procedure by which a student evaluates this formula – particularly in regard to the understanding of fundamental concepts – matters greatly to that student's mastery of circuit analysis.

Suppose a student evaluates the voltage divider formula according to the order of operations enforced by the parentheses. Note the sequence of steps in this procedure, and how each step yields a value relevant to the voltage divider circuit:

Step	Calculation	Unit of measurement	Meaning
1	$R_1 + R_2$	Ohms	Total circuit resistance (R_{total})
2	$R_2 \div R_{total}$	(unitless)	Voltage division ratio
3	$V_{supply} \times$ Ratio	Volts	Output voltage (V_{out})

Now let us suppose a student calculates the exact same output voltage for this divider circuit using a modified version of the voltage divider formula. This formula may be derived from the original version by applying the commutative property of multiplication, simply swapping the positions of R_2 and V_{supply}:

$$V_{out} = R_2 \left(\frac{V_{supply}}{R_1 + R_2} \right)$$

The proper order of operations for this modified formula will be different from the original version, but the final result (V_{out}) will be identical. Each step of the evaluation still yields a value relevant and applicable to the voltage divider circuit:

Step	Calculation	Unit of measurement	Meaning
1	$R_1 + R_2$	Ohms	Total circuit resistance (R_{total})
2	$V_{supply} \div R_{total}$	Amps	Circuit current (I)
3	$R_2 \times I$	Volts	Output voltage (V_{out})

Finally, let us suppose a student calculates the exact same output voltage for this divider circuit using another modified version of the voltage divider formula. This formula may be derived from the original version by applying the associative property of multiplication, grouping R_2 and V_{supply} together into the denominator of the fraction:

$$V_{out} = \frac{R_2 V_{supply}}{R_1 + R_2}$$

The proper order of operations for this modified formula will be different from the original version, but the final result (V_{out}) will be identical. Note, however, the irrelevance of the result in step #2:

Step	Calculation	Unit of measurement	Meaning
1	$R_1 + R_2$	Ohms	Total circuit resistance (R_{total})
2	$R_2 \times V_{supply}$	Volt-ohms	???
3	$R_{total} \times$ previous result	Volts	Output voltage (V_{out})

Despite the mathematical equivalence of this last formula to the prior two, step #2 makes no *conceptual* sense whatsoever. The product of resistance and voltage, while mathematically useful in solving for V_{out}, has no practical meaning within this or any other circuit.

Conceptual problem-solving is an important skill, because it is only by mastering fundamental concepts can one become proficient in solving any arbitrary problem involving those concepts. Procedural problem-solving is useful only when applied to the specific type of problem the procedure was developed for, and useless when faced with any other type of problem. A student who understands the meaning of each step as they evaluate a voltage divider circuit will have little problem solving for quantities in other electrical circuits. A student who has merely memorized a step-by-step procedure for evaluating a voltage divider will struggle trying to solve for quantities in other circuits, unless they happen to have memorized step-by-step procedures for all those other circuits as well.

Troubleshooting is also closely linked with conceptual understanding. In my years as an instructor of electronics and instrumentation, I have seen an almost perfect correlation between conceptual understanding and diagnostic proficiency: procedural thinkers are invariably poor troubleshooters, and poor troubleshooters are invariably procedural thinkers.

The practical lesson we may draw from this example of voltage divider circuit evaluation is the importance of identifying the meaning of every intermediate result. If you ever find yourself performing calculations, unable to explain the practical significance of every step you take, it means *you are thinking procedurally rather than conceptually.* Instructors may apply this standard to their students' work by asking students to explain the meaning of each and every calculation they perform.

34.5 Problem-solving by simplification

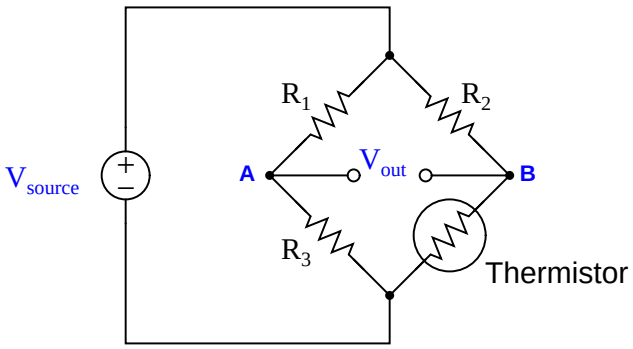
A whole class of problem-solving techniques focuses on *altering* the given problem into a simpler form that is easier to analyze. Once a solution is found to the simplified problem, fresh ideas for attacking the original problem often become clear. This section will highlight multiple techniques for problem-simplification, as well as other useful techniques for problem-solving.

The first step, however, to problem simplification is to "give yourself the right" to alter the problem into a different form! Many students tend to avoid this, for fear of "getting off track" and losing sight of the original problem. What is needed is a spirit of adventure: a willingness and a confidence to explore the possibilities. Do not think you *must* solve exactly the problem that is given to you at first. Modify the problem, solve the simpler version of that problem, then apply the lessons and patterns obtained from that solution to the original (more complex) problem!

34.5.1 Limiting cases

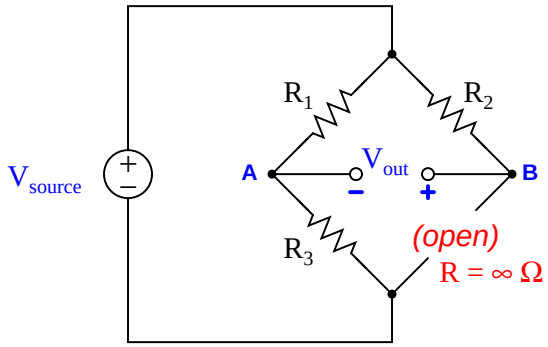
A powerful method for analyzing the effects of a change in some system is to consider the effects of "extreme" changes, which are often easier to visualize than subtle changes. Such "extreme" changes are examples of what is generally known in science as a *limiting case*: a special case of a more general rule or trend, possessing fewer possibilities. By virtue of the fact that limiting cases have fewer possibilities, applying a limiting case to a given problem generally simplifies that problem.

Consider, for example, this Wheatstone bridge circuit, where changes in the thermistor's resistance (with temperature) affect the output voltage of the bridge:



A realistic question to ask of this circuit is, "what will happen to V_{out} when the thermistor's resistance increases?" If our only goal is to arrive at a qualitative answer (e.g. increase/decrease, positive/negative), we may simplify the problem by considering the effects of the thermistor failing completely open, because an "open" fault is nothing more than an extreme example (a *limiting case*) of a resistance increase.

If we perform this "thought experiment" on the bridge circuit, the circuit becomes simpler because we have eliminated one resistor (the thermistor):



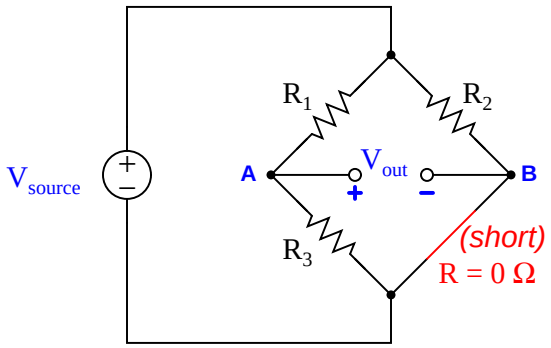
*Limiting case of **increased** thermistor resistance*

With the thermistor eliminated from the circuit, we see that test point **B** has lost its connection to the negative terminal of the voltage source. This can only mean one thing for the potential at test point **B**: it will become more positive (less negative). If the bridge circuit happened to be balanced prior to the thermistor fault, V_{out} will now be such that **B** is positive and **A** is negative by comparison.

Analyzing the results of this limiting case even further, we can see that resistor R_2 now carries zero current (thanks to the thermistor now being failed open), which means R_2 will now drop zero voltage. If R_2 drops no voltage at all, test point **B** must now be at the exact same potential as the positive terminal of the voltage source. This being the case, measuring V_{out} between test points **A** and **B** will be equivalent[1] to measuring voltage across R_1. Thus, the limiting case of V_{out} for an increase in thermistor resistance is V_{R1}, with **B** positive and **A** negative.

[1] With R_2 dropping zero voltage, test point **B** is now essentially common to the node at the top of the bridge circuit. With test point **A** already common with the lower terminal of R_1 and now test point **B** common to the upper terminal of R_1, V_{out} is exactly the same as V_{R1}.

Another realistic question to ask of this circuit is, "what will happen to V_{out} when the thermistor's resistance decreases?" Once again, the problem-solving technique of limiting cases helps us by transforming the four-resistor bridge circuit into a three-resistor bridge circuit. The limiting case of a resistance decrease would be a condition of no resistance: a *shorted* thermistor:

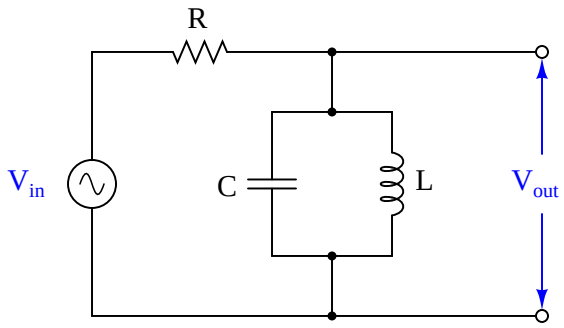


*Limiting case of **decreased** thermistor resistance*

With the thermistor shorted in this "thought experiment," we see that test point **B** now becomes electrically common with the negative terminal of the voltage source. This, of course, has the effect of making test point **B** as negative as it can possibly be. More specifically, by making test point **B** electrically common with the bottom node of the bridge, it makes V_{out} equal[2] to the voltage drop across R_3. Thus, the limiting case of V_{out} for a decrease in thermistor resistance is V_{R3}, with **A** positive and **B** negative.

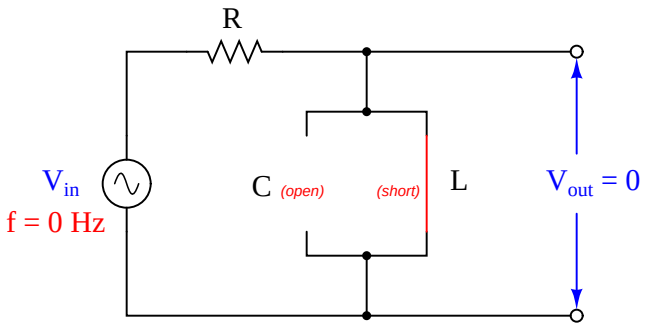
[2] As before, the limiting case of a thermistor fault causes test points **A** and **B** to become synonymous with the terminals of one of the remaining resistors, in this case R_3. Since point **A** is already common with the upper terminal of R_3 and the shorted fault has now made point **B** common with the lower terminal of R_3, V_{out} must be exactly the same as V_{R3}.

Let us consider another application of this problem-solving technique, this time to the analysis of a passive filter circuit:



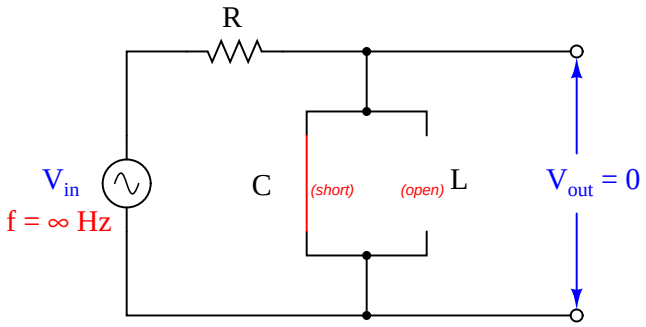
If the type of filter circuit shown here were unknown (i.e. the student could not identify it as a low-pass, high-pass, band-pass, or band-stop filter circuit at first sight), the technique of limiting cases could be applied to determine its behavior. In this case, the limit to apply is one of frequency: we may perform "thought experiments" whereby we imagine the input frequency being extremely low, versus being extremely high.

We know that the reactance of an inductor is directly proportional to frequency ($X_L = 2\pi f L$) and that the reactance of a capacitor is inversely proportional to frequency ($X_C = \frac{1}{2\pi f C}$). Therefore, at an extremely low frequency ($f = 0$ Hz), the inductor will act like a short while the capacitor acts like an open:



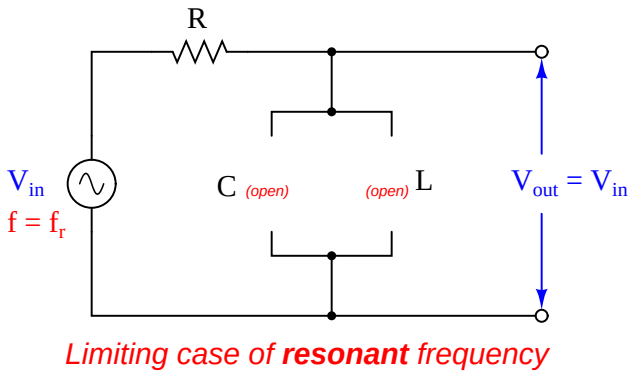
*Limiting case of **decreased** frequency*

Likewise, at extremely high frequencies ($f = \infty$), the capacitor will act like a short while the inductor acts like an open:



From these two limiting-case "thought experiments" we may conclude that the filter circuit is neither a low-pass nor a high-pass, because it neither passes low-frequency signals nor high-frequency signals. We may also conclude that it is not a band-stop filter, because that would pass both low-frequency and high-frequency signals. This means it must be a band-pass filter, by eliminating the other three alternatives.

If we would wish to confirm the band-pass nature of this filter by a positive experimental result rather than merely by eliminating what it is *not*, we could perform one more limiting-case "thought experiment:" a condition where the signal frequency exactly equals the resonant frequency of the LC network ($f = \frac{1}{2\pi\sqrt{LC}}$). Here, we must recall the principle that a parallel LC network has infinite impedance at its resonant frequency:



In this "thought experiment" we see that the LC network will be completely "open" and allow 100% of the input signal to appear at the output terminals. Thus, it becomes clear that this passive circuit functions as a *band-pass* filter.

As with the Wheatstone bridge circuit, the value of limiting-case analysis is that it acts to simplify the system by effectively eliminating components (replacing them with "shorts" or "opens"). Even in non-electrical problems, limiting cases works the same by simplifying a system's behavior so that it becomes easier to apprehend, and from these simplified cases we may usually determine behavioral trends of the system (e.g. which way it tends to respond as some variable increases or decreases).

34.6 Scientific system diagnosis

At the root of successful system diagnosis is a rigorous adherence to *scientific reasoning*. There exists no single algorithmic approach to solving problems, but rather a singular *mind-set* characterized by the following traits:

- Curiosity
- Persistence
- Attention to detail
- Diligence in checking conclusions
- Regular checking of assumptions
- A willingness to abandon ideas based on contrary evidence

Science is, at its heart, a methodology useful to identify causes and effects. Thus, it is well-suited to the problem of system diagnosis, where our goal is to quickly and accurately identify the cause(s) behind improper operation (effects).

34.6.1 Scientific method

Although no one technique seems to be universally recognized as "the scientific method," the following steps are commonly applied in science to determine causes and effects:

- Observe effects, and then create *hypotheses* (explanations accounting for those observations)
- Design a test for one or more of those hypotheses
- Perform the test (experiment), and collect data from it
- Validate or invalidate the hypotheses based on the data
- Repeat

Perhaps the most challenging step in this method is designing a good test for the hypotheses. By "test" I mean a trial that really challenges each hypothesis, and doesn't just collect more data to support it. A good way to help yourself devise a rigorous test of any hypothesis is to keep these two questions in mind:

"If this hypothesis is true, what other effects should we see if we look for them?"

. . . and . . .

"If this hypothesis is false, what other effects should we *not* see if we look for them?"

An ideal test (experiment) is one that answers both of these questions at once, providing both positive and negative evidence.

In contrast to scientific diagnosis is a technique a colleague of mine refers to as "Easter-egging," where the troubleshooter tries to find the problem by individually checking every component or possible fault they can think of, in serial fashion. The term "Easter-egging" invokes the image of children hunting for hidden eggs on Easter morning, randomly searching in every place they can think of where an egg might be hidden. There is no logical reasoning to "Easter-egging" and so it is a very inefficient method of solving system problems.

34.6.2 Occam's Razor

A very helpful principle in scientific testing is something called *Occam's Razor*, a rule stating that the simplest explanation for any observed effects is usually the most accurate. While not infallible, Occam's Razor is nevertheless a valid "gambling strategy" based on simple probability. In system troubleshooting, it means that a single fault is more likely to account for the symptoms than a set of coincidental faults. For this reason, it is generally good practice to enter a troubleshooting scenario with the assumption that only one thing is wrong, unless the data conclusively points otherwise.

34.6.3 Diagnosing intermittent problems

Intermittent faults are some of the most challenging to diagnose, for the simple reason that the relevant symptoms come and go. A persistent fault is easier to solve because the data is continuously there for inspection.

The key to troubleshooting intermittent faults is to set up test equipment to capture events that occur when you are not directly observing them. Some suggested methods include:

- Using the "Min/Max" capture mode on a digital multimeter (DMM)
- Using a data recorder or event logger to capture signal history
- Looking for evidence left by certain intermittent faults (e.g. if the suspected fault is high temperature at a certain location, looking for evidence such as charring or discoloration that would be caused by high temperature at some past time)
- Using videorecording equipment to capture events

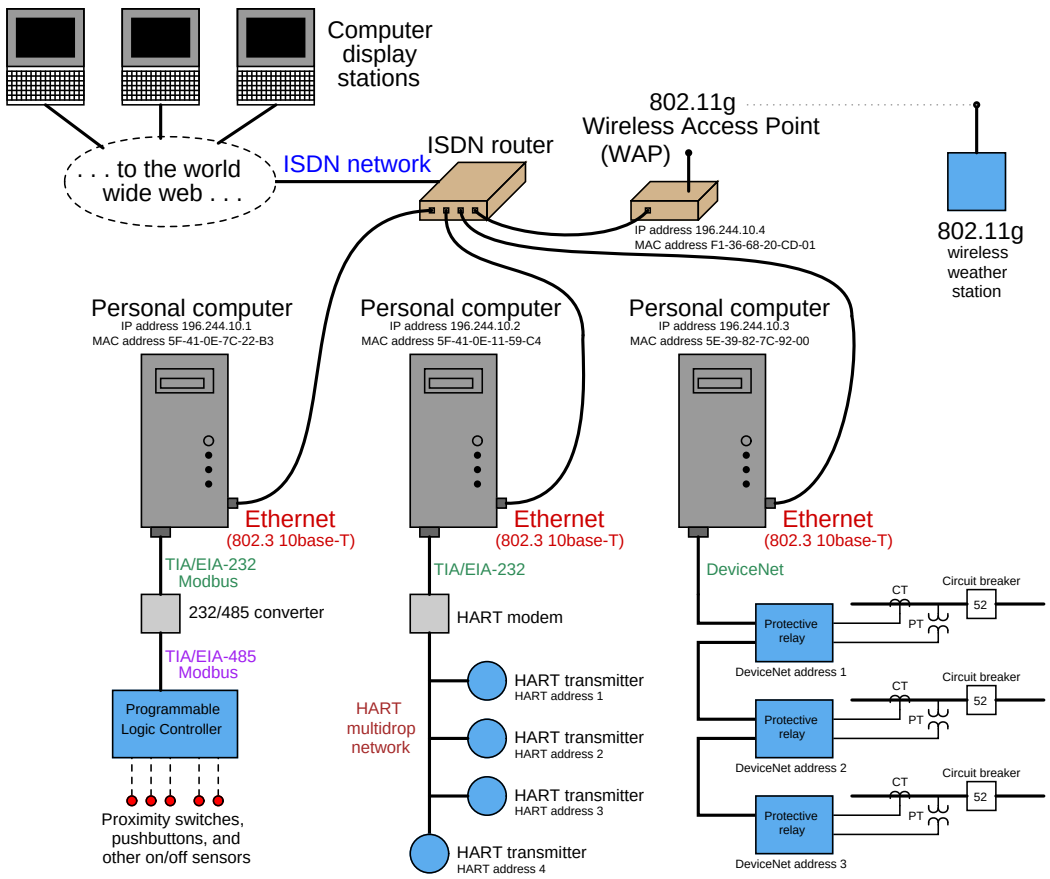
Perhaps one of the most useful features of modern digital multimeters is the ability to capture minimum and maximum signal levels. Many times I have used this feature on my own meter to monitor the highs and lows of some signal in order to capture evidence of an intermittent fault. This is also useful for monitoring signal changes that happen too fast to see on the display of a meter (e.g. detecting the peak pulse amplitude of a fast signal). While limited to the sample rate of the digital meter, it remains a powerful tool in the hands of a knowledgeable technician.

A colleague of mine once diagnosed a complex, intermittent problem on a natural gas compressor unit by setting up a video camera to film the control panel gauges on the compressor, then reviewing the video recording frame-by-frame after the camera had recorded a "trip" event. This kind of creativity is often key to diagnosing intermittent problems.

34.6.4 Strategy: tracing data paths

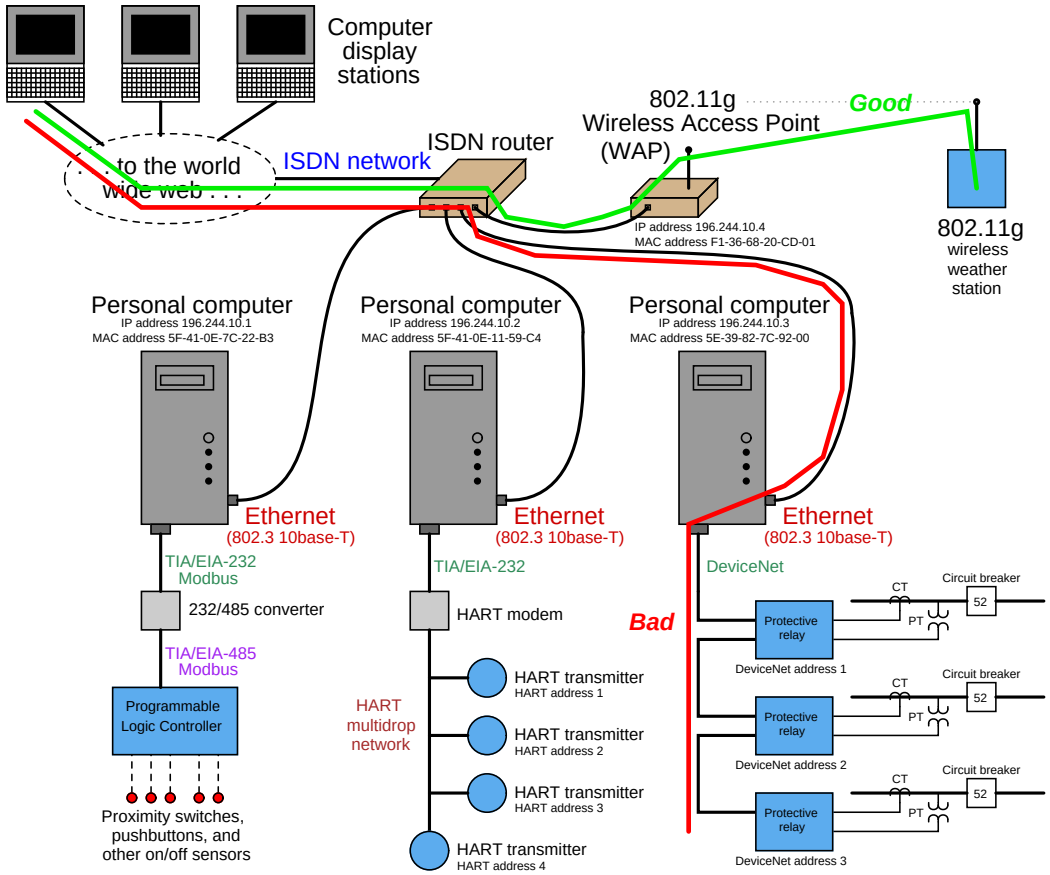
A method often useful for tracing the location of faults in complex systems is to identify where data is coming from (source), where it is going (destination), and all paths taken by the data in between. If we then plot those paths on a one-line diagram of the system, the intersection of paths often tells us where the problem lies.

For example, consider this system of networked devices in a data acquisition system:



Suppose operations personnel noticed they could no longer access any protective relay data from the left-most display station connected to the world-wide web (Internet), but they could still access live weather station data from that same display station. Applying the technique of tracing data paths may be helpful to us in locating the fault in this complex system, and also devising a good test to pinpoint the location.

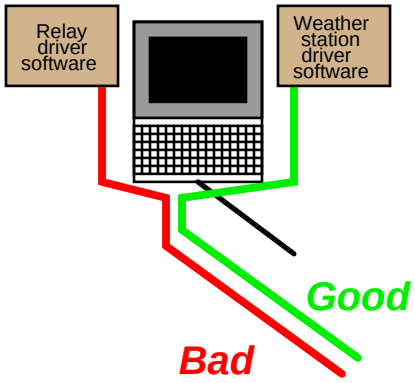
Here we see the same system with green and red lines overlaid showing good data paths and failed data paths:



Note how these two paths overlap in the display station, the world-wide-web, and the ISDN router. Since data from the weather station is getting through this part of the path just fine, yet protective relay data is not, the most likely location of the fault is in a part of the system where these two data paths are *not* common. In other words, it is unlikely that the problem lies within the display station, the Internet, or the ISDN router, because all those component are proven to work just fine for the weather station data. A more probable location of the fault is in an area of the "bad" data path that is *not* common to the "good" data path. In this particular case, it points to a problem from the ISDN router to the DeviceNet network.

A good test to do at this point is to try "pinging" the right-most personal computer from one of the other two personal computers connected directly to the ISDN router. This would be testing a data path from one PC to the other, thereby testing the integrity of the right-most PC and the cable connecting it to the ISDN router. If this test is successful, the problem likely lies farther beyond the PC (e.g. in the DeviceNet network) ; if this test is unsuccessful, the problem likely lies within that PC or within the cabling connecting it to the router.

Note the careful use of the words "likely" and "probable" in the hypotheses. Hypotheses are uncertain by their very nature, and are never "proven" in any final sense. Even though our initial approach of sketching and comparing data pathways suggests a problem between the ISDN router and the DeviceNet network connecting the protective relays together, it is still possible for the problem to lie somewhere closer to the display station. If, for example, the display station relied on dedicated software "drivers" to properly poll and interpret data from the protective relay network that were not used to do the same for weather station data, the problem could actually lie within the display station! A corrupted relay driver would prevent that station from properly displaying protective relay data and yet permit the display of weather station data. In order for our data path tracing procedure to encompass this possibility, we would need to show the pathways splitting within the display station, so they would no longer be common to each other:



As always, the assumptions we embed into our hypotheses can skew their validity. If we assume completely overlapping data paths for protective relay and weather data within the display station, we would not recognize the possibility of a driver problem.

34.7 Common diagnostic mistakes

Volumes could be written about poor diagnostic technique. The following mistakes are not intended to comprise a comprehensive list, but are merely warnings against errors that are all too common among students and within the profession.

34.7.1 Failing to gather data

Perhaps the most common mistake made by technicians attempting to diagnose a system problem is failing to gather data (i.e. taking measurements and performing simple system tests) during the troubleshooting process. Even a small amount of data gathered from a system may profoundly accelerate the process of diagnosis.

A colleague of mine has a very descriptive term for the poor habit of looking for faults before gathering data: *Easter-Egging*. The idea is that a technician goes about finding the problem in a system the same way they might go about searching for eggs hidden on Easter morning: randomly. With Easter egg hunting, the eggs could literally be hidden *anywhere*, and so there is no rational way to proceed on a search. In like manner, a technician who lacks information about the nature or source of a system problem is likely to hunt in random fashion for its source. Not only will this likely require significant time and effort, but it may very well fail entirely.

A much more efficient way to proceed is to gather new data with each and every step in the troubleshooting process. By "gathering data," I mean the following:

- Taking measurements with test equipment (multimeter, pressure gauges, etc.)
- Observing equipment indicator lights
- Stimulating the system and observing its response(s)
- Using your other senses (smell, hearing, touch) to gather clues
- Documenting new data in a notepad to help track and analyze the results of your measurements and tests

34.7.2 Failing to use relevant documentation

Diagrams are indispensable "maps" for solving problems in complex systems. A critical first step in diagnosing any system problem is to obtain correct diagrams of the system, so that you may see the pathways of power and signals in the system. Attempting to diagnose a system problem without consulting the relevant diagrams is like trying to find your way around a city without a map.

A corollary to the rule of obtaining relevant diagrams is to *use them* when reasoning through fault scenarios. All too often I see students locate the diagram for a system, glance at it, then set it aside and proceed to stumble through the rest of the diagnosis because they try to trace all the signal paths in the real world as they assess fault possibilities and devise tests. Diagrams are laid out in a clean and logical format for a reason: it is much easier to follow the flow of signals and power in a diagram than it is to follow the same flow through the convoluted paths of real-world wires and cables. If you reject the diagram in favor of tracing all pathways by looking at the real-world system, you are needlessly adding complexity to the problem: not only do you have to reason through the fault hypotheses and diagnostic tests, but you also must mentally "un-tangle" the signal paths as they are laid out in the real world (which can be a daunting task in itself!). *Do all your diagnostic thinking while looking at the diagram, and refer to the real-life system only when the time comes to execute a diagnostic test.* The result will be a much faster and less frustrating experience than if you try to trace everything in real life.

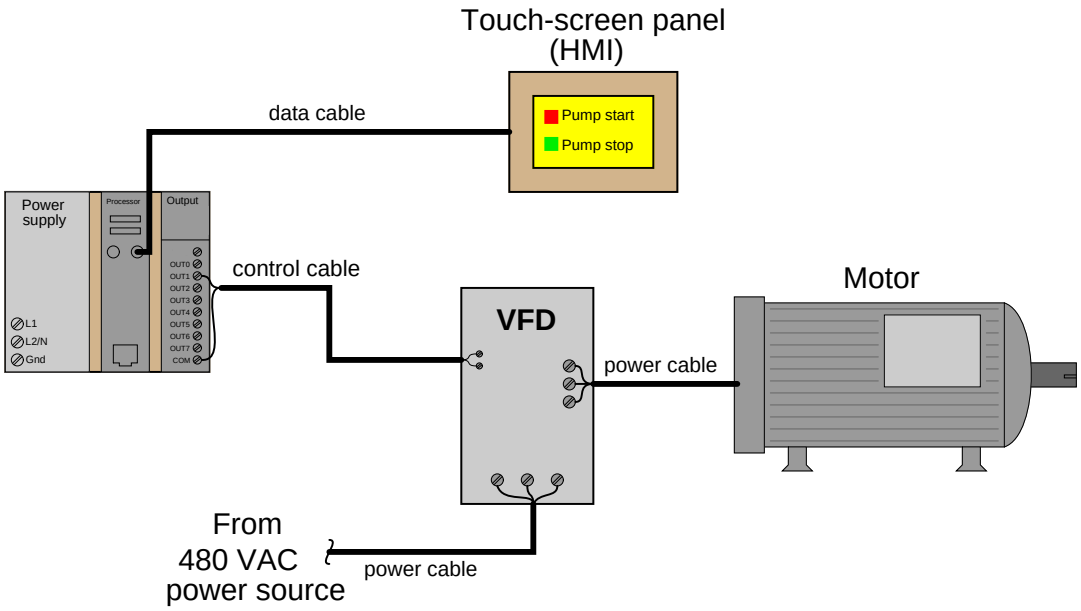
When diagnosing a problem in a system where one or more of the key components are unfamiliar to you, it is important to consult the relevant technical literature on those components. This is especially true if the component in question has been identified as suspect by your diagnostic test(s) and it is quite complex (e.g. loop controller, PLC, motor drive, data acquisition module, etc.). Just a few minutes' worth of reading the manual may save you hours of fruitless diagnosis.

This point also underscores the necessity of technical reading as a skill to be practiced and honed at every opportunity. Being able to quickly locate pertinent information in a dense technical document is key to fast and efficient troubleshooting!

34.7.3 Fixating on the first hypothesis

When diagnosing a faulted system, an efficient strategy is to brainstorm *multiple* hypotheses accounting for the symptoms, then devise tests to support or discredit those hypotheses in the fewest steps.

Let me illustrate by example. Suppose a pump motor is controlled by a PLC, the PLC sending a command signal to a variable-frequency motor drive (VFD) to command the motor to start and stop:



If the motor refuses to start when the "Pump start" icon is pressed on the HMI screen, a competent troubleshooter will begin to mentally list a range of problems that could prevent the motor from starting:

- Motor faulted
- VFD lacking power
- VFD not configured properly to receive signal from PLC
- PLC output defective, not sending signal to VFD
- PLC program halted or faulty
- HMI not sending signal to PLC
- . . . etc.

After brainstorming such a list, a competent troubleshooter will then devise a simple test to "divide the problem space in half." One such test[3] in this system would be to use a multimeter to measure the electrical signal from the PLC's output card to the VFD input terminals. If a signal appears when the "Pump start" icon on the HMI is pressed, it means everything in the HMI and PLC is working as it should, and that the problem must lie with the VFD or beyond. If no signal appears, it means the problem lies with the VFD, HMI, or associated cabling. Again, the wise strategy is to brainstorm multiple hypotheses explain why the motor won't start, then execute simple tests to eliminate most of those hypotheses so you may focus on those that are most likely.

By contrast, a novice might only think of one possibility – such as the VFD being improperly configured – and then immediately fixate on that hypothesis by inspecting the drive parameters looking for one that is improperly set. If the fault lies elsewhere, the novice could spend all day reviewing VFD parameters and never find the problem. Given the wide range of possible faults, fixating on any one fault from the start is very likely a waste of time. I have watched technicians and students alike waste hours of time trying to find a fault that was not where they were looking, simply because that was the first area they thought of to look toward. Only after squandering valuable time on one failed hypothesis will the novice then consider other possibilities and other tests.

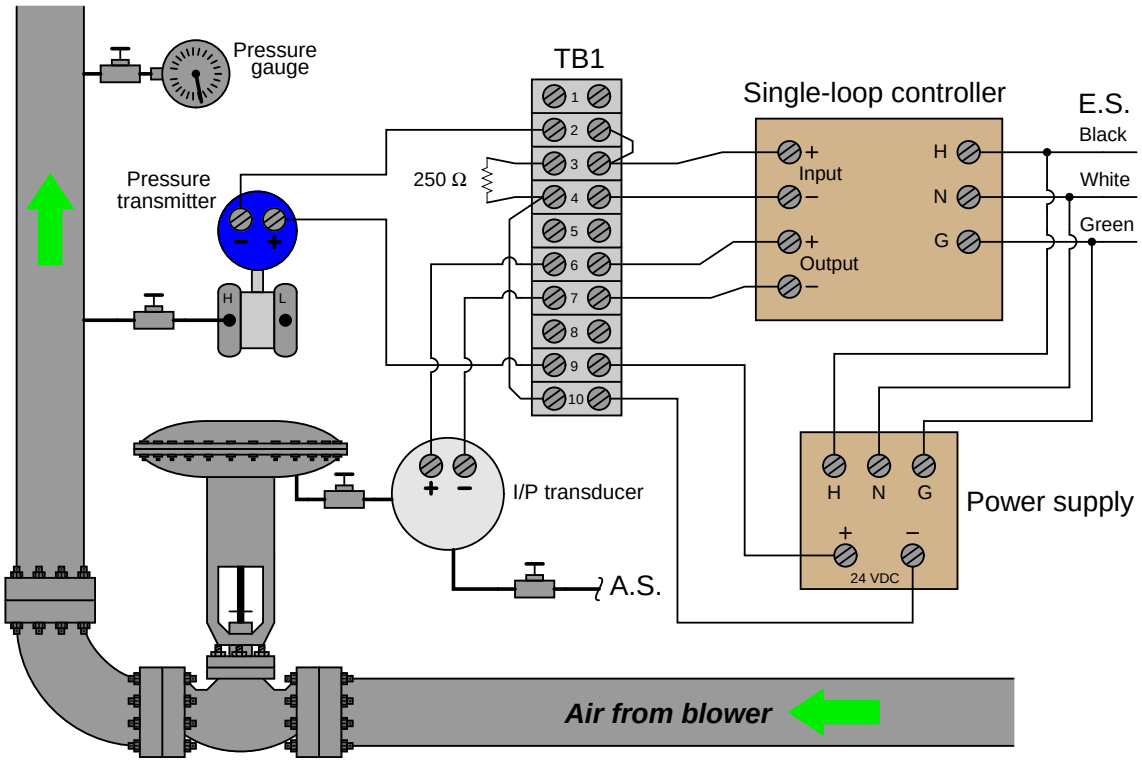
[3]Other possible tests include inspecting the LED status light on that PLC output card channel (a light indicates the HMI and PLC program are working correctly, and that the problem could lie within the output card or beyond to the motor) or measuring voltage at the drive output (voltage there indicates the problem must lie with the motor or the cable to the motor rather than further back).

34.7.4 Failing to build and test a new system in stages

Technicians must sometimes assemble new systems from components. A very common mistake is to assemble the system completely before attempting to test it for proper operation. This is almost always a grievous mistake.

The number of potential mistakes one can make when assembling a brand-new system is quite large. Given this large set of potential mistakes, the probability of making multiple mistakes when assembling the system is very high. Since diagnosis of a system with multiple faults is always more complicated than diagnosing a system with one fault, waiting for the entire system to be assembled before checking it invites multi-fault scenarios.

To illustrate, consider this pressure-control system, where an electronic pressure transmitter sends a 4-20 mA signal to a loop controller, which in turn drives a control valve with another 4-20 mA signal:



Imagine building this system, placing each component in the proper location, connecting all wires together, and testing it for proper operation. If one were to wait until the entire system were assembled before testing, the probability of having to diagnose multiple faults would be great.

A better strategy would be to assemble and test the system in stages. Consider this sequence of steps as a more practical alternative:

1. Install the I/P transducer, connecting air tubes to supply and valve diaphragm.
2. *Test the I/P and control valve operation using a loop calibrator in "source" mode to drive a 4-20 mA signal to the I/P.*
3. Install and wire power to the loop controller, ensuring it powers up properly.
4. Connect cabling between the I/P and the loop controller's output.
5. *Test the controller's ability in manual mode to "stroke" the control valve throughout its entire range.*
6. Connect wires between the loop controller's input and the 250 ohm resistor on the terminal block.
7. *Test the controller's ability to properly read an input signal by using a loop calibrator to drive 4-20 mA through the 250 ohm resistor.*
8. Install the pressure transmitter, connecting impulse line between it and the process line.
9. *Power the transmitter with a portable DC power supply (or loop calibrator set to the appropriate mode) and check its calibration by applying known pressures to the input tube.*
10. Connect wires between the permanent DC power supply, the transmitter, and the controller's input.
11. *Apply pressure to the transmitter input and check to see that it reads properly on the controller's digital display.*
12. *Test the controller's ability to monitor and control process pressure in manual mode.*
13. Perform manual-mode (open-loop) tests to verify process characteristics and obtain data needed for loop tuning (e.g. lag time, dead time, etc.).
14. Enter preliminary PID tuning parameter values.
15. *Test the controller's ability to monitor and control process pressure in automatic mode.*
16. Modify PID tuning parameter values and re-test in automatic mode until robust control is obtained.

Note how the pressure control instrumentation is constructed and then immediately tested as a series of sub-systems, rather than assembling the entire thing and testing only at the very end. Although the *built-test-build* sequence shown here may appear to be more time-intensive at first blush, it will actually save a lot of time and confusion over the *build-everything-then-test-last* method favored by novices.

34.8 Helpful "tricks" using a digital multimeter (DMM)

The digital multimeter (DMM) is quite possibly the most useful tool in the instrument technician's collection[4]. This one piece of test equipment, properly wielded, yields valuable insight into the status and operation of many electrical and electronic systems. Not only is a good-quality multimeter capable of precisely indicating electrical voltage, current, and resistance, but it is also useful for more advanced tests. The subject of this section is how to use a digital multimeter for some of these advanced tests[5].

For all these tests, I suggest the use of a top-quality field multimeter. I am personally a great fan of *Fluke* brand meters, having used this particular brand for nearly my whole professional career. The ability of these multimeters to accurately measure true RMS amplitude, discriminate between AC and DC signals, measure AC signals over a wide frequency range, and survive abuse both mechanical and electrical, is outstanding.

34.8.1 Recording unattended measurements

Many modern multimeters have a feature that records the highest and lowest measurements sensed during the duration of a test. On Fluke brand multimeters, this is called the *Min/Max* function. This feature is extremely useful when diagnosing intermittent problems, where the relevant voltages or currents indicating or causing the problem are not persistent, but rather come and go. Many times I have used this feature to monitor a signal with an intermittent "glitch," while I attended to other tasks.

The most basic high-low capture function on a multimeter only tells you what the highest and lowest measured readings were during the test interval (and that only within the meter's scan time – it is possible for a very brief transient signal to go undetected by the meter if its duration is less than the meter's scan time). More advanced multimeters actually log the *time* when an event occurs, which is obviously a more useful feature. If your tool budget can support a digital multimeter with "logging" capability, spend the extra money and take the time to learn how this feature works!

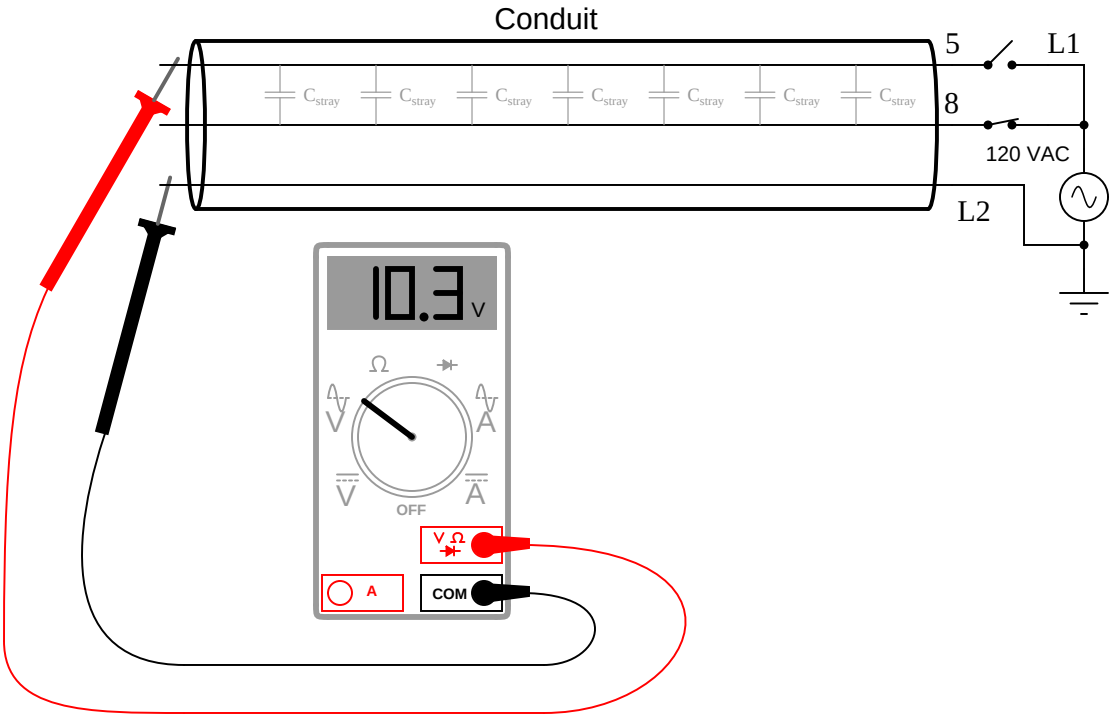
[4] As a child, I often watched episodes of the American science-fiction television show *Star Trek*, in which the characters made frequent use of a diagnostic tool called a *tricorder*. Week after week the protagonists of this show would avoid trouble and solve problems using this nifty device. The *sonic screwdriver* was a similar tool in the British science-fiction television show *Doctor Who*. Little did I realize while growing up that my career would make just as frequent use of another diagnostic tool: the electrical multimeter.

[5] I honestly considered naming this section "Stupid Multimeter Tricks," but changed my mind when I realized how confusing this could be for some of my readers not familiar with colloquial American English.

34.8.2 Avoiding "phantom" voltage readings

My first "trick" is not a feature of a high-quality DMM so much as it is a solution to a common problem *caused* by the use of a high-quality DMM. Most digital multimeters exhibit very high input impedance in their voltage-measuring modes. This is commendable, as an ideal voltmeter should have infinite input impedance (so as to not "load" the voltage signal it measures). However, in industrial applications, this high input impedance may cause the meter to register the presence of voltage where none should rightfully appear.

Consider the case of testing for the absence of AC voltage on an isolated power conductor that happens to lie near other (energized) AC power conductors within a long run of conduit:

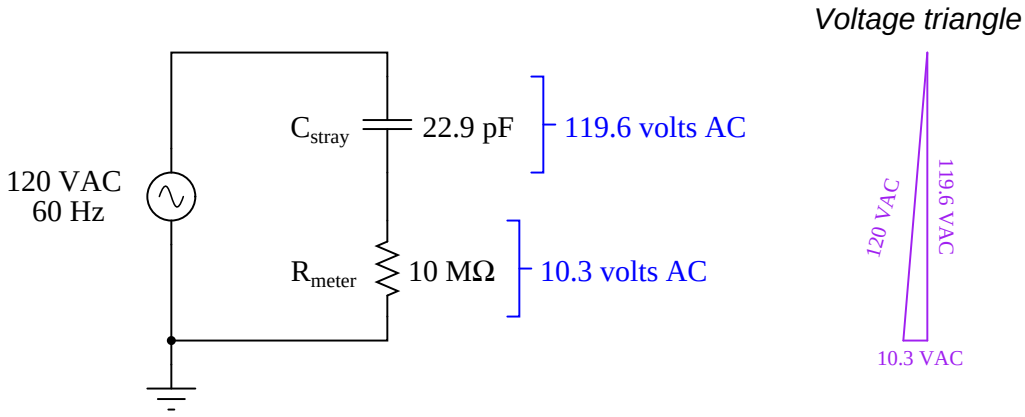


With the power switch feeding wire 5 in the open state, there should be no AC voltage measured between wire 5 and neutral (L2), yet the voltmeter registers slightly over 10 volts AC. This "phantom voltage" is due to capacitive coupling between wire 5 and wire 8 (still energized) throughout the length of their mutual paths within the conduit.

Such phantom voltages may be very misleading if the technician encounters them while troubleshooting a faulty electrical system. Phantom voltages give the impression of connection (or at least high-resistance connection) where no continuity actually exists. The example shown, where the phantom voltage is 10.3 volts compared to the source voltage value of 120 volts, is actually quite modest. With increased stray capacitance between the conductors (longer wire runs in close proximity, and/or more than one energized "neighboring" wire), phantom voltage magnitude begins

to approach that of the source voltage[6].

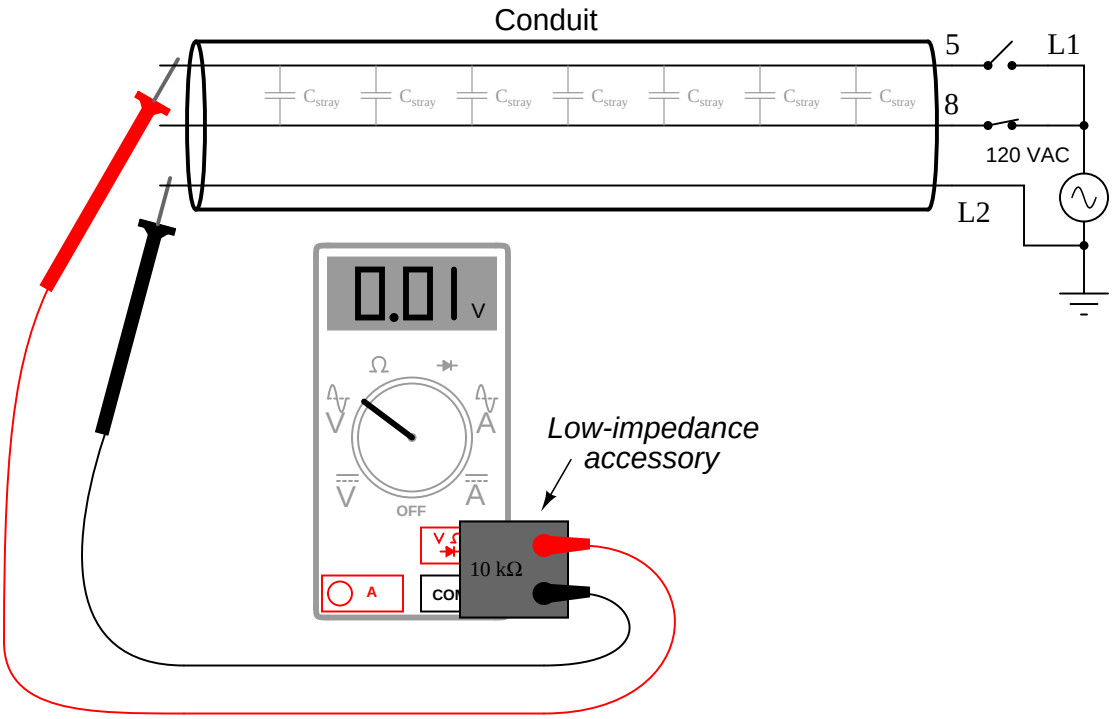
The equivalent circuit is shown here, with the DMM modeled as a 10 MΩ resistance:



An analog voltmeter would never have registered 10.3 volts under the same conditions, due to its substantially lower input impedance. Thus, "phantom voltage" readings are a product of modern test equipment more than anything else.

[6] I have personally measured "phantom" voltages in excess of 100 volts AC, in systems where the source voltage was 120 volts AC.

The obvious solution to this problem is to use a different voltmeter – one with a much lesser input impedance. But what is a technician to do if their only voltmeter is a high-impedance DMM? Connect a modest resistance in parallel with the meter input terminals, of course! Fluke happens to market just this type of accessory[7], the SV225 "Stray Voltage Adapter" for the purpose of eliminating stray voltage readings on a high-impedance DMM:



With the voltmeter's input impedance artificially decreased by the application of this accessory, the capacitive coupling is insufficient to produce any substantial voltage dropped across the voltmeter's input terminals, thus eliminating. The technician may now proceed to test for the presence of AC control signal (or power) voltages with confidence.

[7]Before there was such an accessory available, I used a 20 kΩ high-power resistor network connected in parallel with my DMM's input terminals, which I fabricated myself. It was ugly and cumbersome, but it worked well. When I made this, I took great care in selecting resistors with power ratings high enough that accidental contact with a truly "live" AC power source (up to 600 volts) would not cause damage to them. A pre-manufactured device such as the Fluke SV225, however, is a much better option.

34.8.3 Non-contact AC voltage detection

While the last multimeter "trick" was the elimination of a parasitic effect, this trick is the exploitation of that same effect: "phantom voltage" readings obtained through capacitive coupling of a high-impedance voltmeter to a conductor energized with AC voltage (with respect to ground). You may use a high-impedance AC voltmeter to perform qualitative measurements of ground-referenced AC power voltage by setting the meter to the most sensitive AC range possible, grounding one test lead, and simply touching the other test lead to the insulation of the conductor under test. The presence of voltage (usually in the range of millivolts AC) upon close proximity to the energized conductor will indicate the energization of that conductor.

This trick is useful for determining whether or not particular AC power or control wires are energized in a location where the only access you have to those wires is their insulating sheaths. An example of where you might encounter this situation is where you have removed the cover from a conduit elbow or other fitting to gain access to a wire bundle, and you find those wires labeled for easy identification, but the wires do not terminate to any exposed metal terminals for you to contact with your multimeter's probe tips. In this case, you may firmly connect one probe to the metal conduit fitting body, while individually touching the other probe tip to the desired conductors (one at a time), watching the meter's indication in AC millivolts.

Several significant caveats limit the utility of this "trick:"

- The impossibility of quantitative measurement
- The potential for "false negative" readings (failure to detect a voltage that is present)
- The potential for "false positive" readings (detection of a "phantom voltage" from an adjacent conductor)
- The exclusive applicability to AC voltages of significant magnitude ($\geq$ 100 VAC)

Being a qualitative test only, the millivoltage indication displayed by the high-impedance voltmeter tells you nothing about the actual magnitude of AC voltage between the conductor and ground. Although the meter's input impedance is quite constant, the parasitic capacitance formed by the surface area of the test probe tip and the thickness (and dielectric constant) of the conductor insulation is quite variable. However, in conditions where the validity of the measurement may be established (e.g. cases where you can touch the probe tip to a conductor known to be energized in order to establish a "baseline" millivoltage signal), the technique is useful for quickly checking the energization status of conductors where ohmic (metal-to-metal) contact is impossible.

For the same reason of wildly variable parasitic capacitance, this technique should *never* be used to establish the de-energization of a conductor for safety purposes. The only time you should trust a voltmeter's non-indication of line voltage is when that same meter is validated against a known source of similar voltage in close proximity, and when the test is performed with direct metal-to-metal (probe tip to wire) contact. A non-indicating voltmeter *may* indicate the absence of dangerous voltage, or it may indicate an insensitive meter.

34.8.4 Detecting AC power harmonics

The presence of *harmonic* voltages[8] in an AC power system may cause many elusive problems. Power-quality instruments exist for the purpose of measuring harmonic content in a power system, but a surprisingly good qualitative check for harmonics may be performed using a multimeter with a frequency-measuring function.

Setting a multimeter to read AC voltage (or AC current, if that is the quantity of interest) and then activating the "frequency" measurement function should produce a measurement of exactly 60.0 Hz in a properly functioning power system (50.0 Hz in Europe and some other parts of the world). The only way the meter should ever read anything significantly different from the base frequency is if there is significant harmonic content in the circuit. For example, if you set your multimeter to read frequency of AC voltage, then obtained a measurement of 60 Hz that intermittently jumped up to some higher value (say 78 Hz) and then back down to 60 Hz, it would suggest your meter was detecting harmonic voltages of sufficient amplitude to make it difficult for your meter to "lock on" to the fundamental frequency.

It is very important to note that this is a crude test of power system harmonics, and that measurements of "solid" base frequency do not guarantee the absence of harmonics. Certainly, if your multimeter produces unstable readings when set to measure frequency, it suggests the presence of strong harmonics in the circuit. However, the absence of such instability does not necessarily mean the circuit is free of harmonics. In other words, a stable reading for frequency is *inconclusive*: the circuit might be harmonic-free, or the harmonics may be weak enough that your multimeter ignores them and only displays the fundamental circuit frequency.

[8] These are AC voltages having frequencies that are integer-multiples of the fundamental powerline frequency. In the United States, where 60 Hz is standard, harmonic frequencies would be whole-number multiples of 60: 120 Hz, 180 Hz, 240 Hz, 300 Hz, etc.

34.8.5 Identifying noise in DC signal paths

An aggravating source of trouble in analog electronic circuits is the presence of AC "noise" voltage superimposed on DC signals. Such "noise" is immediately evident when the signal is displayed on an oscilloscope screen, but how many technicians carry a portable oscilloscope with them for troubleshooting?

A high-quality multimeter exhibiting good discrimination between AC and DC voltage measurement is very useful as a qualitative noise-detection instrument. Setting the multimeter to read AC voltage, and connecting it to an signal source where pure (unchanging) DC voltage is expected, should yield a reading of nearly zero millivolts. If noise is superimposed on this DC signal, it will reveal itself as an AC voltage, which your meter will display.

Not only is the AC voltage capability of a high-quality (discriminating) multimeter useful in detecting the presence of "noise" voltage superimposed on analog DC signals, it may also give clues as to the source of the noise. By activating the frequency-measuring function of the multimeter while measuring AC voltage (or AC millivoltage), you will be able to track the frequency of the noise to see its value and stability.

Once on a job I was diagnosing a problem in an analog power control system, where the control device was acting strangely. Suspecting that noise on the measurement signal line might be causing the problem, I set my Fluke multimeter to measure AC volts, and read a noise voltage of several tenths of a volt (superimposed on a DC signal a few volts in magnitude). This told me the noise *was* indeed a significant problem. Pressing the "Hz" button on my multimeter, I measured a noise frequency of 360 Hz, which happens to be the "ripple" frequency of a six-pulse (three-phase) AC-to-DC rectifier operating on a base frequency of 60 Hz. This told me where the likely source of the noise was, which led me to the physical location of the problem (a bad shield on a cable run near the rectified power output wiring).

34.8.6 Generating test voltages

Modern digital multimeters are fantastically capable measurement tools, but did you know they are also capable of *generating* simple test signals? Although this is not the design purpose of the resistance and diode-check functions of a multimeter, the meter does output a low DC voltage in each of these settings.

This is useful when qualitatively testing certain instruments such as electronic indicators, recorders, controllers, data acquisition modules, and alarm relays, all designed to input a DC voltage signal from a 250 ohm resistor conducting the 4-20 mA electronic transmitter signal. By setting a multimeter to either the resistance (Ω) or diode check function and then connecting the test leads to the input terminals of the instrument, the instrument's response may be noted.

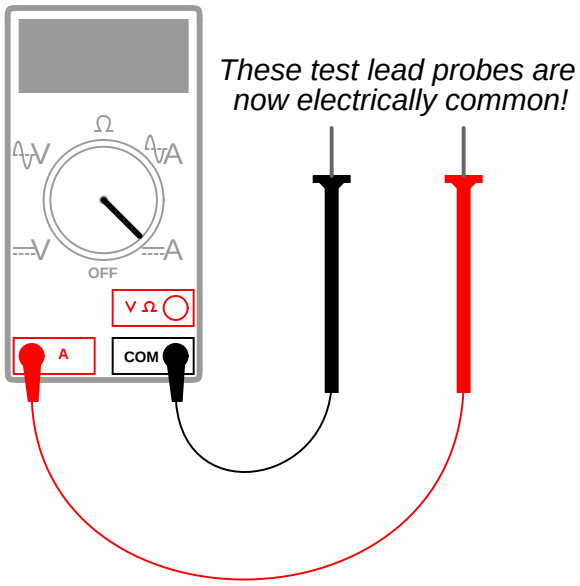
Of course, this is a *qualitative* test only, since multimeters are not designed to output any precise amount of voltage in either the resistance or diode-check modes. However, for testing the basic response of a process indicator, recorder, controller, data acquisition channel, DCS input, or any other DC-signal-receiving devices, it is convenient and useful. In every multimeter I have ever tried this with, the diode-check function outputs *more* voltage than the resistance measurement function[9]. This gives you two levels of "test signal" generation: a low level (resistance) and a high level (diode check). If you are interested in using your multimeter to generate test voltages, I recommend you take the time to connect your multimeter to a high-impedance voltmeter (such as another digital multimeter set to measure DC volts) and note just how much voltage your meter outputs in each mode. Knowing this will allow you to perform tests that are more quantitative than qualitative.

[9] There is a design reason for this. Most digital multimeters are designed to be used on semiconductor circuits, where the minimum "turn-on" voltage of a silicon PN junction is approximately 500 to 700 millivolts. The diode-check function must output more than that, in order to force a PN junction into forward conduction. However, it is useful to be able to check ohmic resistance in a circuit *without* activating any PN junctions, and so the resistance measurement function typically uses test voltages *less than 500 millivolts*.

34.8.7 Using the meter as a temporary jumper

Often in the course of diagnosing problems in electrical and electronic systems, there is a need to temporarily connect two or more points in a circuit together to force a response. This is called "jumpering," and the wires used to make these temporary connections are called *jumper wires*.

More than once I have found myself in a position where I needed to make a temporary "jumper" connection between two points in a circuit, but I did not have any wires with me to make that connection. In such cases, I learned that I could use my multimeter test leads while plugged into the *current-sensing* jacks of the meter. Most digital multimeters have a separate jack for the red test lead, internally connected to a low-resistance *shunt* leading to the common (black) test lead jack. With the red test lead plugged into this jack, the two test leads are effectively common to one another, and act as a single length of wire.



Touching the meter's test leads to two points in a circuit will now "jumper" those two points together, any current flowing through the shunt resistance of the multimeter. If desired, the meter may be turned on to monitor how much current goes through the "jumper" if this is diagnostically relevant.

An additional benefit to using a multimeter in the current-measuring mode as a test jumper is that this setting is usually current-protected by a fuse inside the meter. Applying jumper wires to a live circuit may harbor some danger if significant potential and current-sourcing capability exist between those two points: the moment a jumper wire bridges those points, a dangerous current may develop within the wire. Using the multimeter in this manner gives you a *fused* jumper wire: an added degree of safety in your diagnostic procedure.

References

Adler, Mortimer, "How to Mark a Book", *The McGraw-Hill Reader*, McGraw-Hill Book Company, New York, NY, 1982.